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(54) **SYSTEM AND METHOD FOR ENHANCING NETWORK RELIABILITY AND COUNTERACTING CYBERSECURITY BREACH VIA SOFTWARE UPDATE**

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(21) Appl. No.: **19/213,742**

(22) Filed: **May 20, 2025**

Related U.S. Application Data

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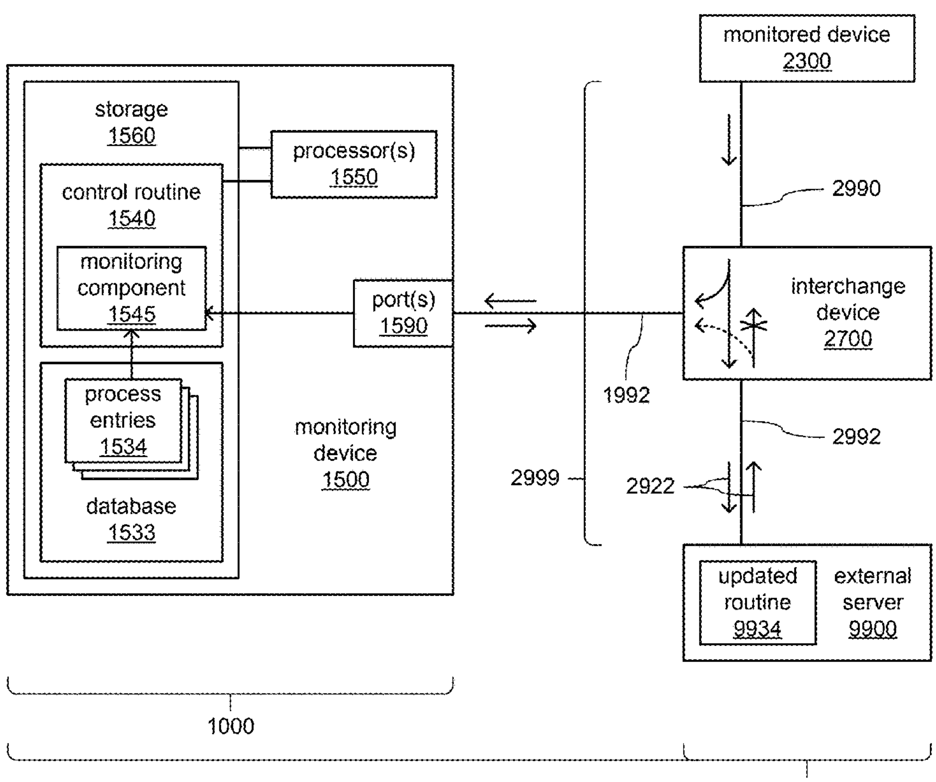
Publication Classification

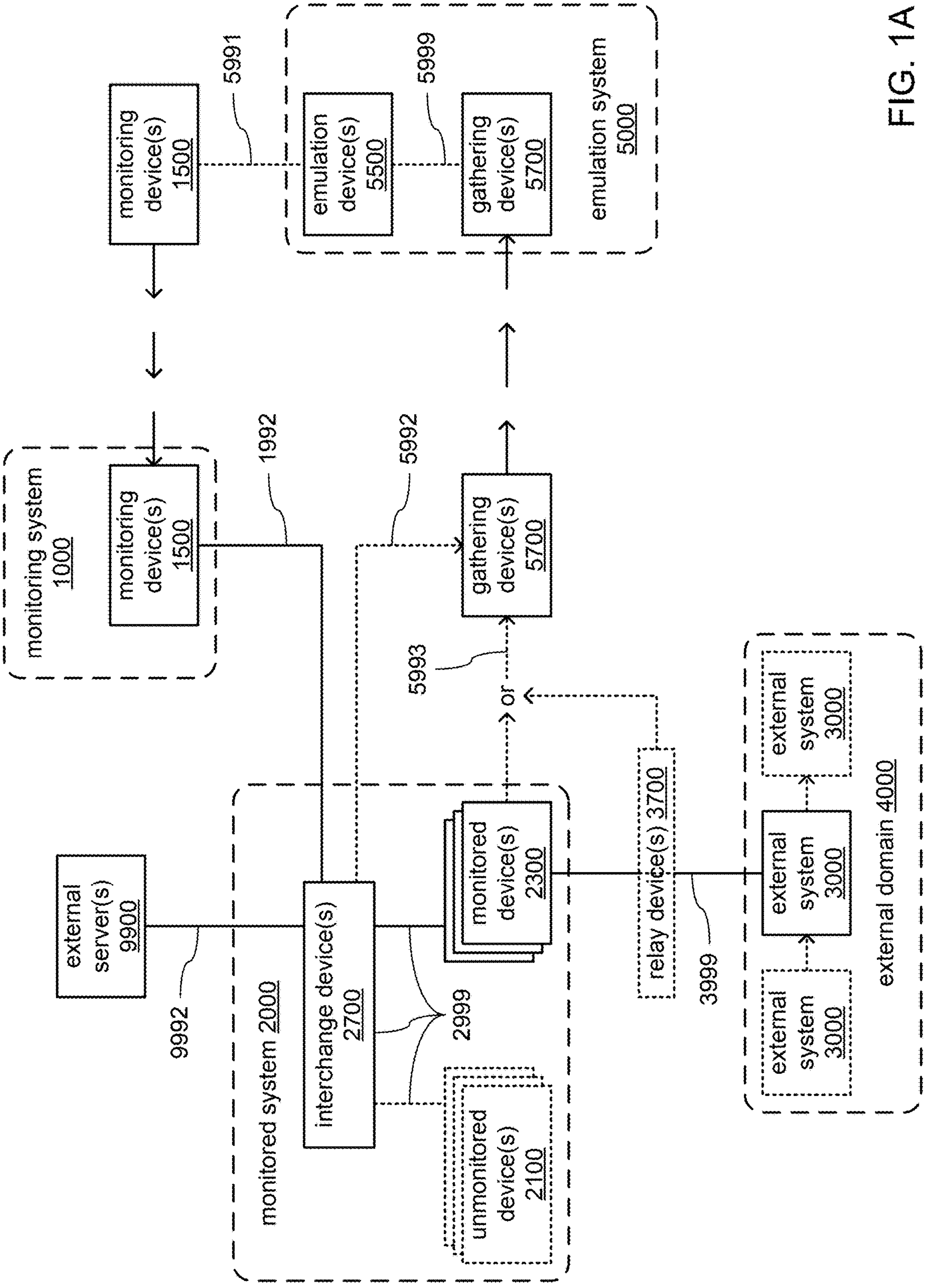
(51) **Int. Cl.**
G06F 8/65 (2018.01)
G06N 20/00 (2019.01)

(52) **U.S. Cl.**
CPC ***G06F 8/65*** (2013.01); ***G06N 20/00*** (2019.01)

(57) **ABSTRACT**

A monitoring system including a processor to: place the monitoring system into an operating mode to use a model of a software update process to analyze observed transmissions of operational commands or information associated with the process; receive, from one or more interchange devices, indications of observed transmissions of operational commands or information among multiple monitored devices of a monitored system, wherein at least one of the monitored devices is configured to provide an updated routine in the process; and compare the received indications of observed transmissions of operational commands or information associated with the process to indications in the model of expected transmissions of operational commands or information associated with the process to determine whether a particular transmitted operational command is a proper operational command associated with the process or to determine whether particular transmitted operational information is proper operational information associated with the process.

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5991
emulation device(s) 5500
5999
gathering device(s) 5700
emulation system 5000
monitoring system 1000
monitoring device(s) 1500
1992
5992
gathering device(s) 5700
5993
or
external server(s) 9900
9992
monitored system 2000
interchange device(s) 2700
2999
monitored device(s) 2300
unmonitored device(s) 2100
relay device(s) 3700
3999
external system 3000
external system 3000
external system 3000
external domain 4000

FIG. 1A

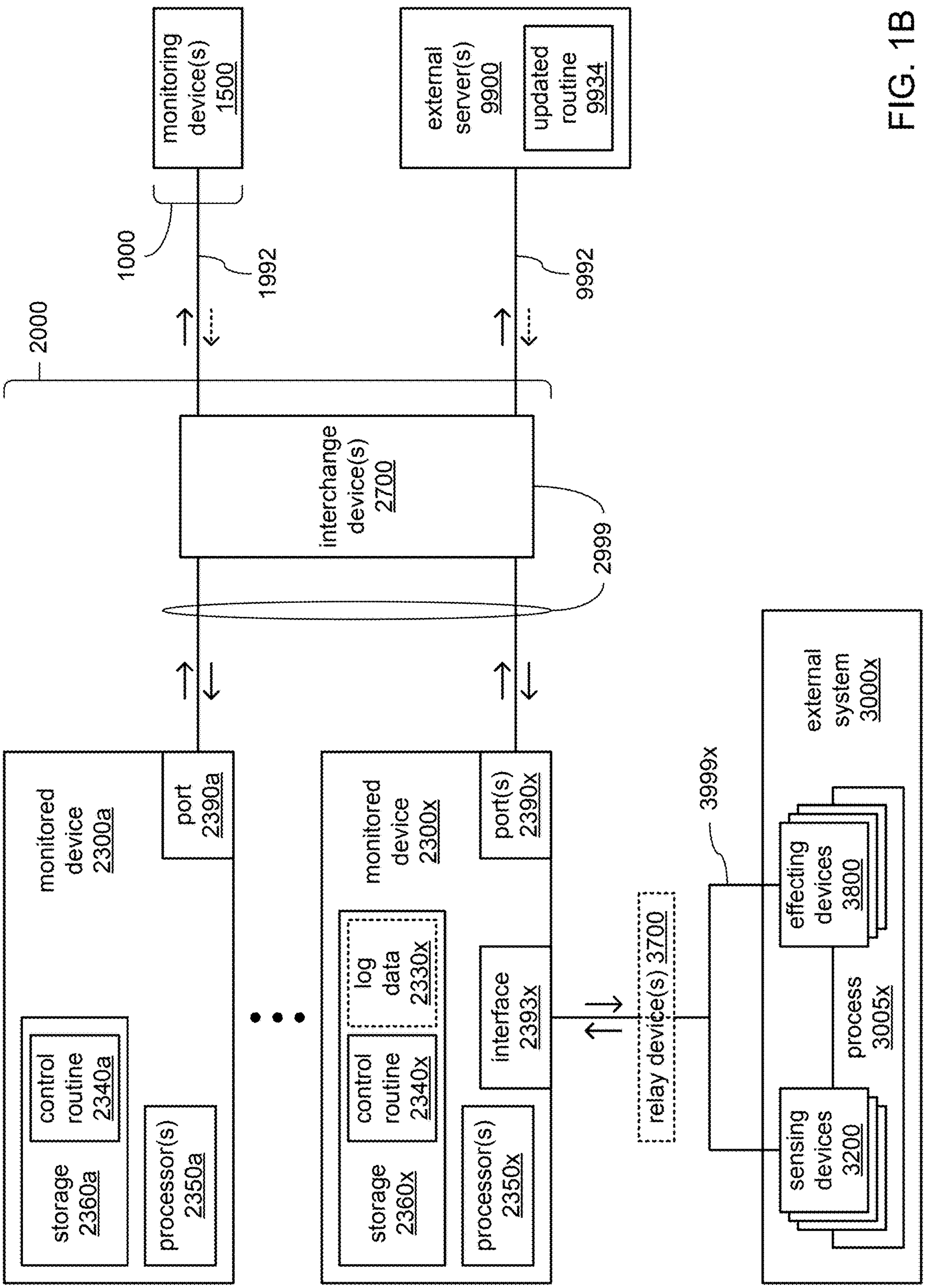
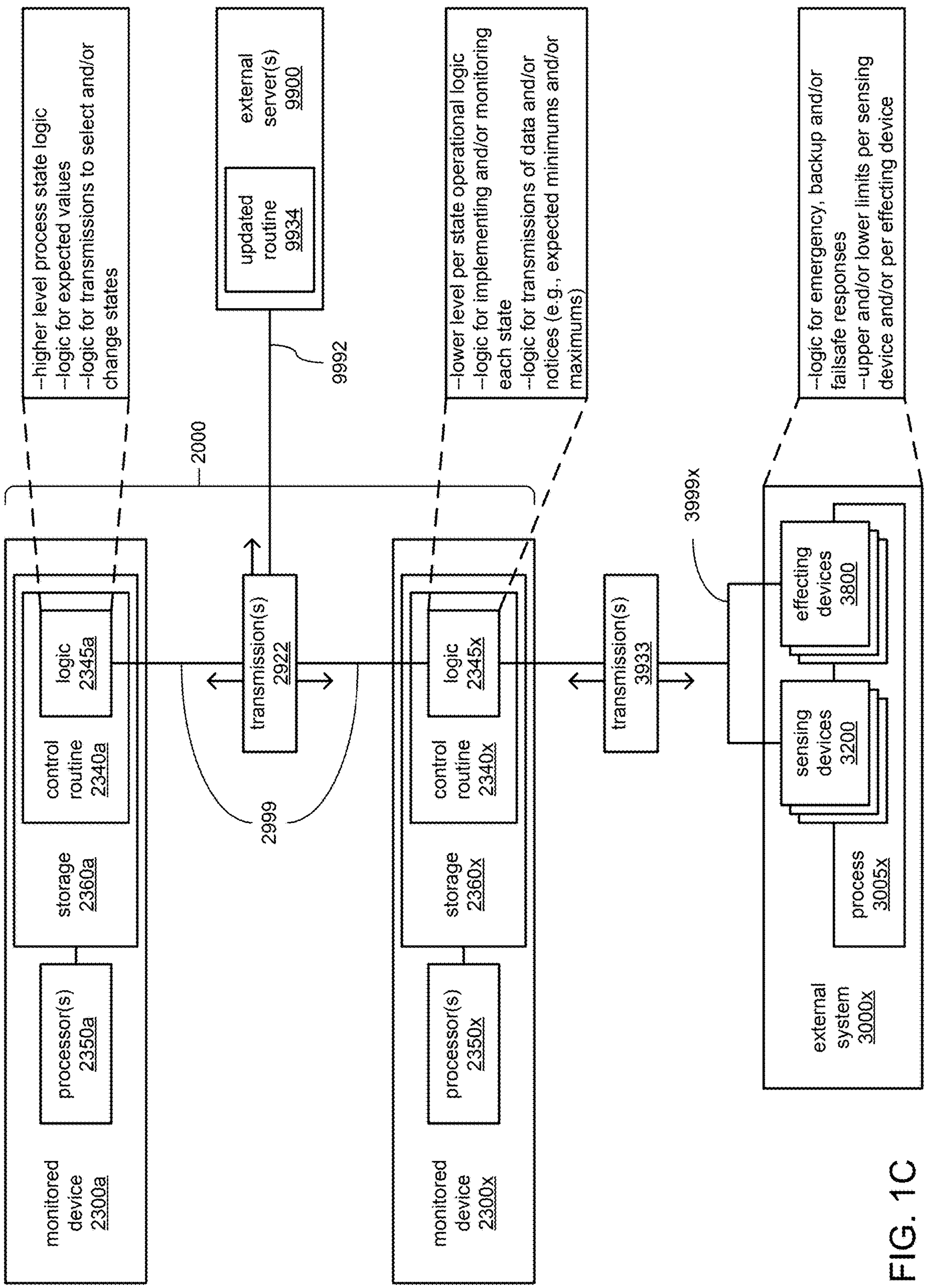

monitoring device(s) 1500
external server(s) 9900
updated routine 9934
1000
1992
9992
2000
interchange device(s) 2700
2999
monitored device 2300a
port 2390a
storage 2360a
control routine 2340a
processor(s) 2350a
monitored device 2300x
port(s) 2390x
storage 2360x
control routine 2340x
log data 2330x
interface 2393x
processor(s) 2350x
relay device(s) 3700
3999x
external system 3000x
effecting devices 3800
process 3005x
sensing devices 3200

FIG. 1B



monitored device 2300a
processor(s) 2350a
storage 2360a
control routine 2340a
logic 2345a
--higher level process state logic
--logic for expected values
--logic for transmissions to select and/or change states
2000
2999
transmission(s) 2922
external server(s) 9900
updated routine 9934
9992
monitored device 2300x
processor(s) 2350x
storage 2360x
control routine 2340x
logic 2345x
--lower level per state operational logic
--logic for implementing and/or monitoring each state
--logic for transmissions of data and/or notices (e.g., expected minimums and/or maximums)
transmission(s) 3933
3999x
external system 3000x
process 3005x
sensing devices 3200
effecting devices 3800
--logic for emergency, backup and/or failsafe responses
--upper and/or lower limits per sensing device and/or per effecting device

FIG. 1C

FIG. 1D

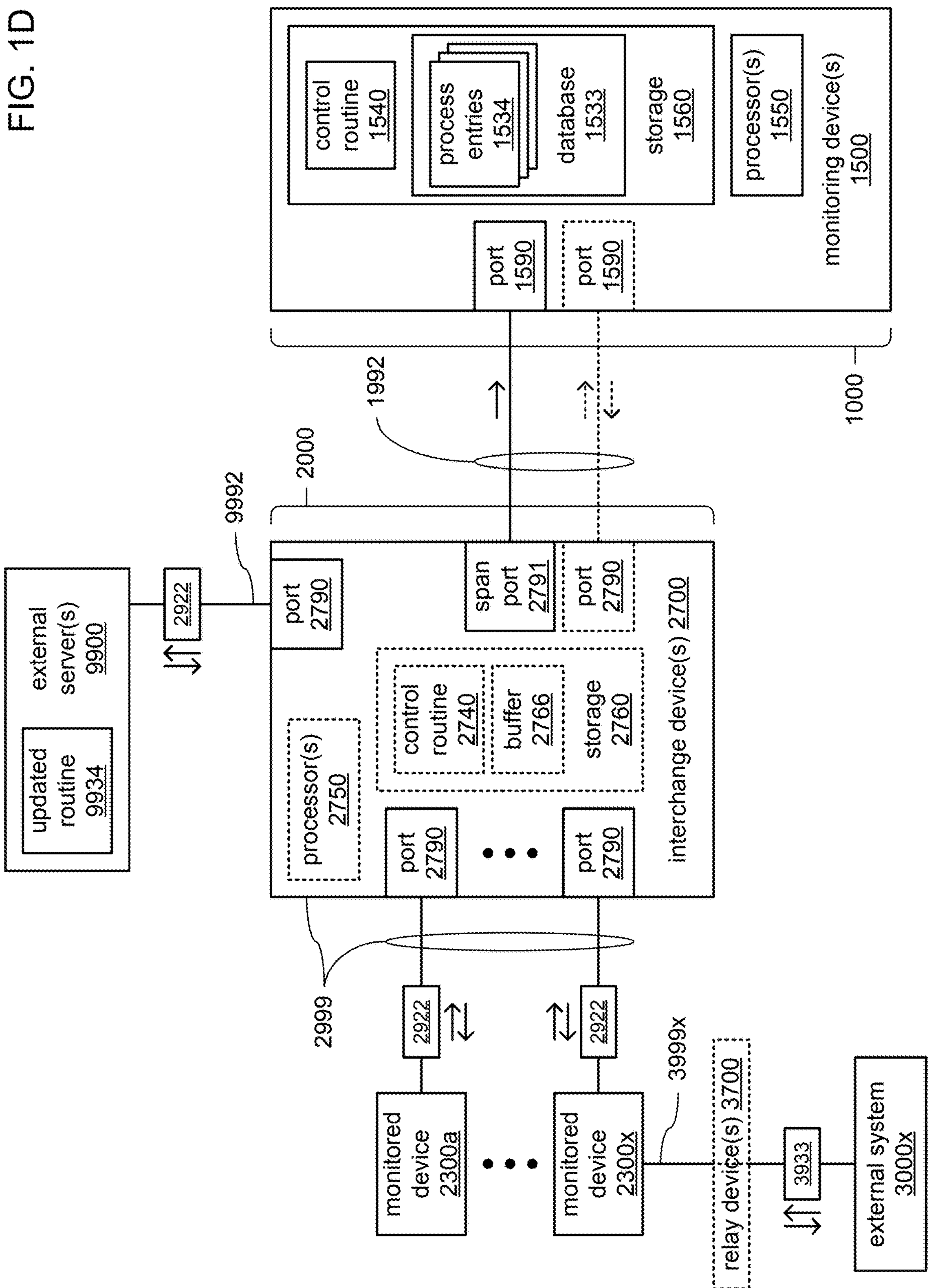
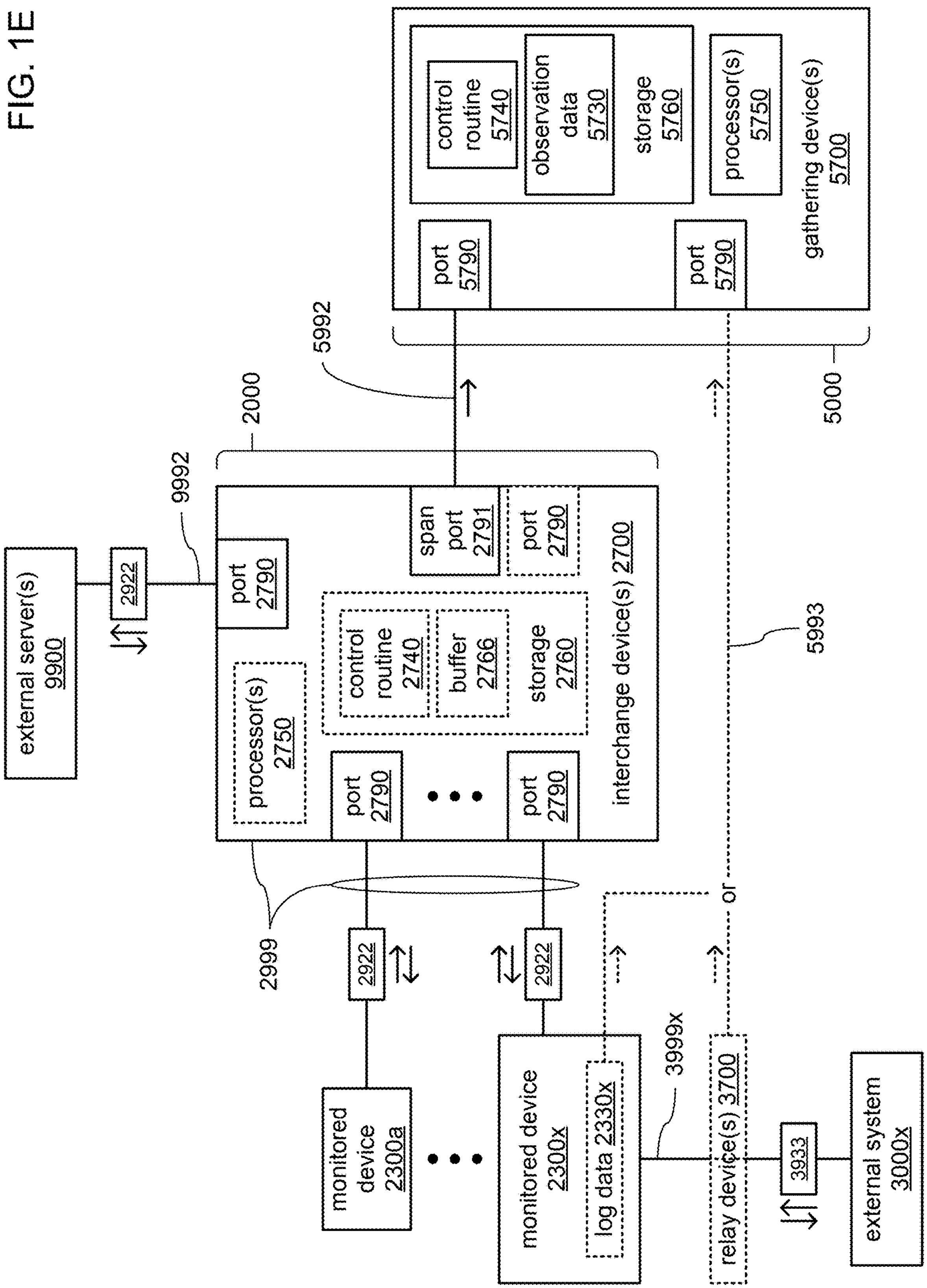

external server(s) 9900
updated routine 9934
2922
9992
2000
port 2790
span port 2791
port 2790
processor(s) 2750
control routine 2740
buffer 2766
storage 2760
port 2790
port 2790
interchange device(s) 2700
2999
2922
2922
monitored device 2300a
monitored device 2300x
3999x
relay device(s) 3700
3933
external system 3000x
1992
1000
port 1590
port 1590
control routine 1540
process entries 1534
database 1533
storage 1560
processor(s) 1550
monitoring device(s) 1500

FIG. 1E



external server(s) 9900
2922
9992
2000
port 2790
processor(s) 2750
control routine 2740
buffer 2766
storage 2760
span port 2791
port 2790
port 2790
port 2790
interchange device(s) 2700
5992
5993
5000
port 5790
port 5790
control routine 5740
observation data 5730
storage 5760
processor(s) 5750
gathering device(s) 5700
2999
2922
2922
or
monitored device 2300a
monitored device 2300x
log data 2330x
3999x
relay device(s) 3700
3933
external system 3000x

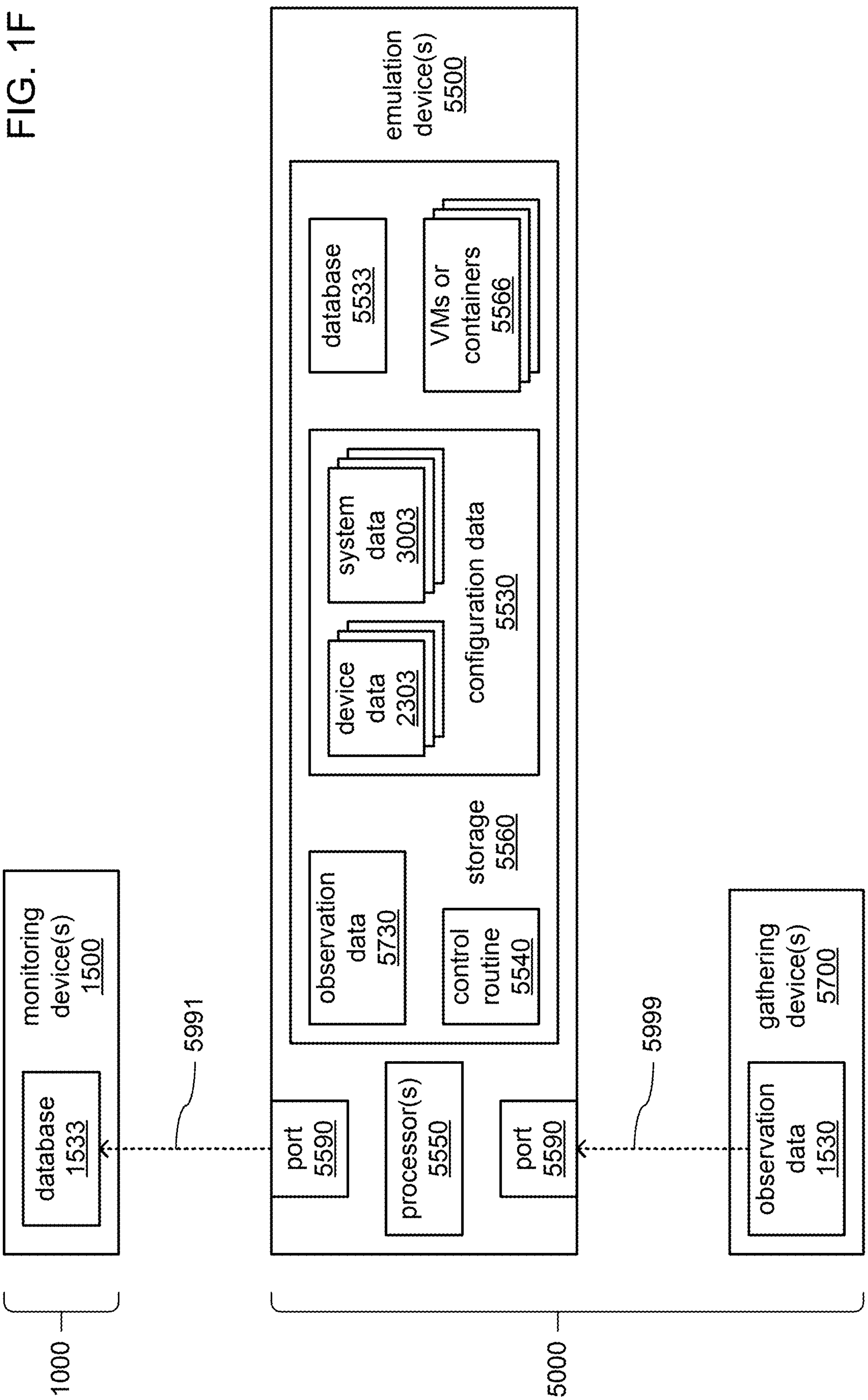

1000
monitoring device(s) 1500
database 1533
5991
5000
emulation device(s) 5500
database 5533
VMs or containers 5566
system data 3003
device data 2303
configuration data 5530
storage 5560
observation data 5730
control routine 5540
port 5590
processor(s) 5550
port 5590
5999
gathering device(s) 5700
observation data 1530

FIG. 1F

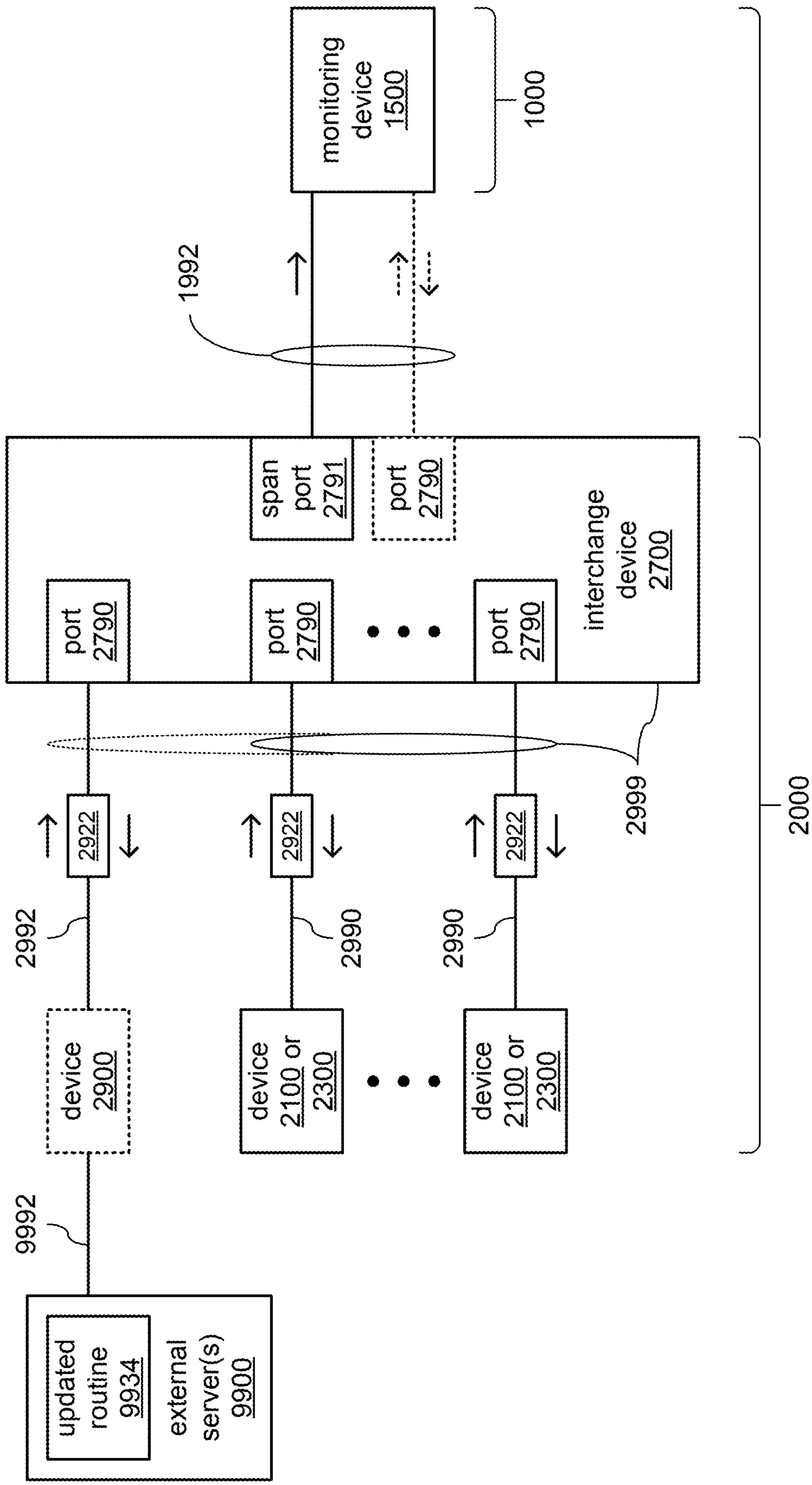

monitoring device 1500
1000
1992
span port 2791
port 2790
interchange device 2700
port 2790
port 2790
port 2790
2999
2922
2922
2922
2000
2992
2990
2990
device 2900
device 2100 or 2300
device 2100 or 2300
9992
updated routine 9934
external server(s) 9900

FIG. 2A

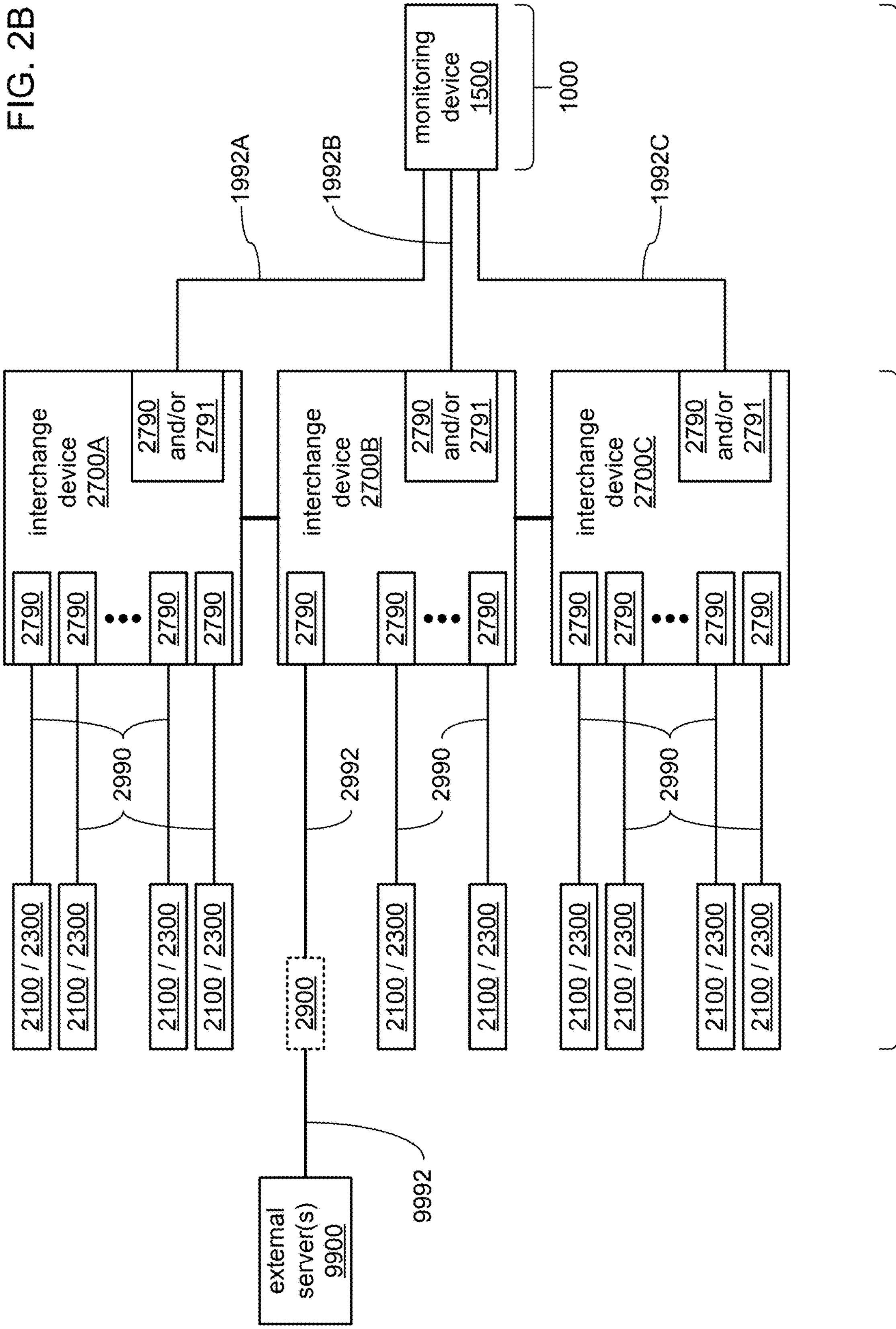

monitoring device 1500
1000
1992A
1992B
1992C
interchange device 2700A
interchange device 2700B
interchange device 2700C
2790 and/or 2791
2790
2990
2992
2900
2100 / 2300
9992
external server(s) 9900
2000

FIG. 2B

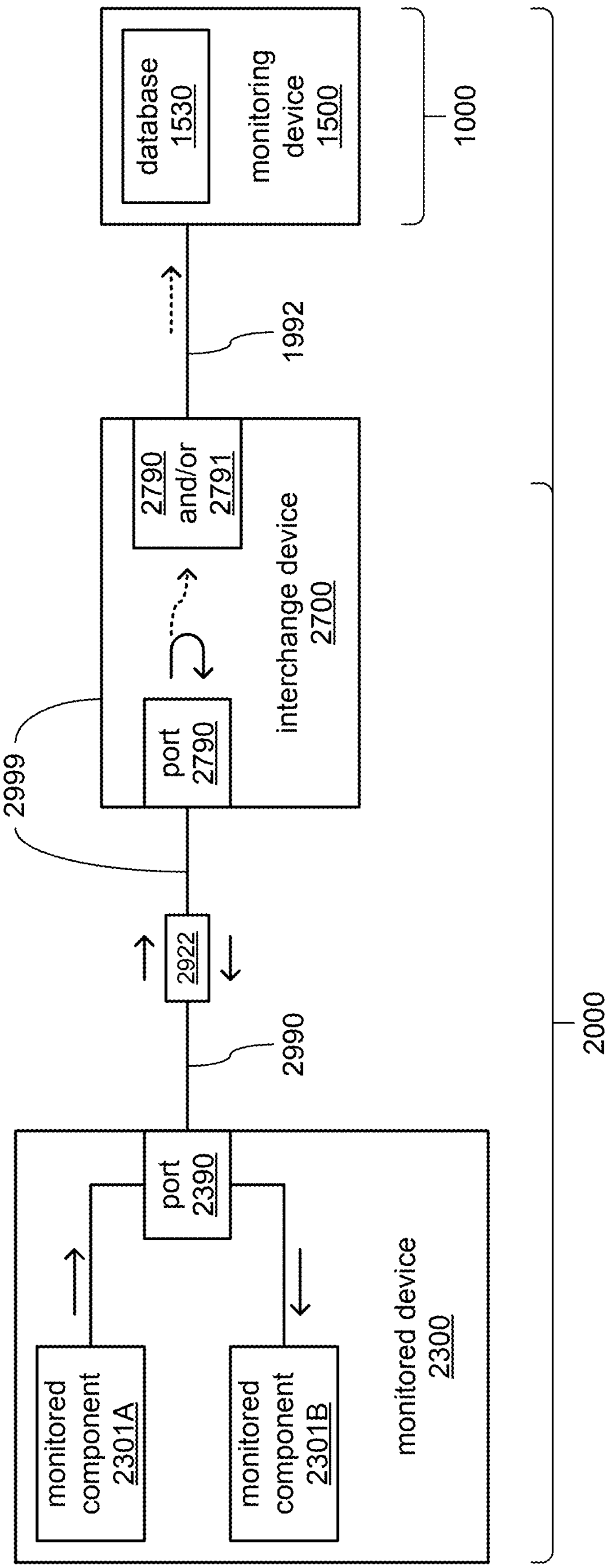

database
1530
monitoring
device
1500
1000
1992
2790
and/or
2791
interchange device
2700
port
2790
2999
2922
2000
2990
port
2390
monitored
component
2301A
monitored
component
2301B
monitored device
2300

FIG. 2C

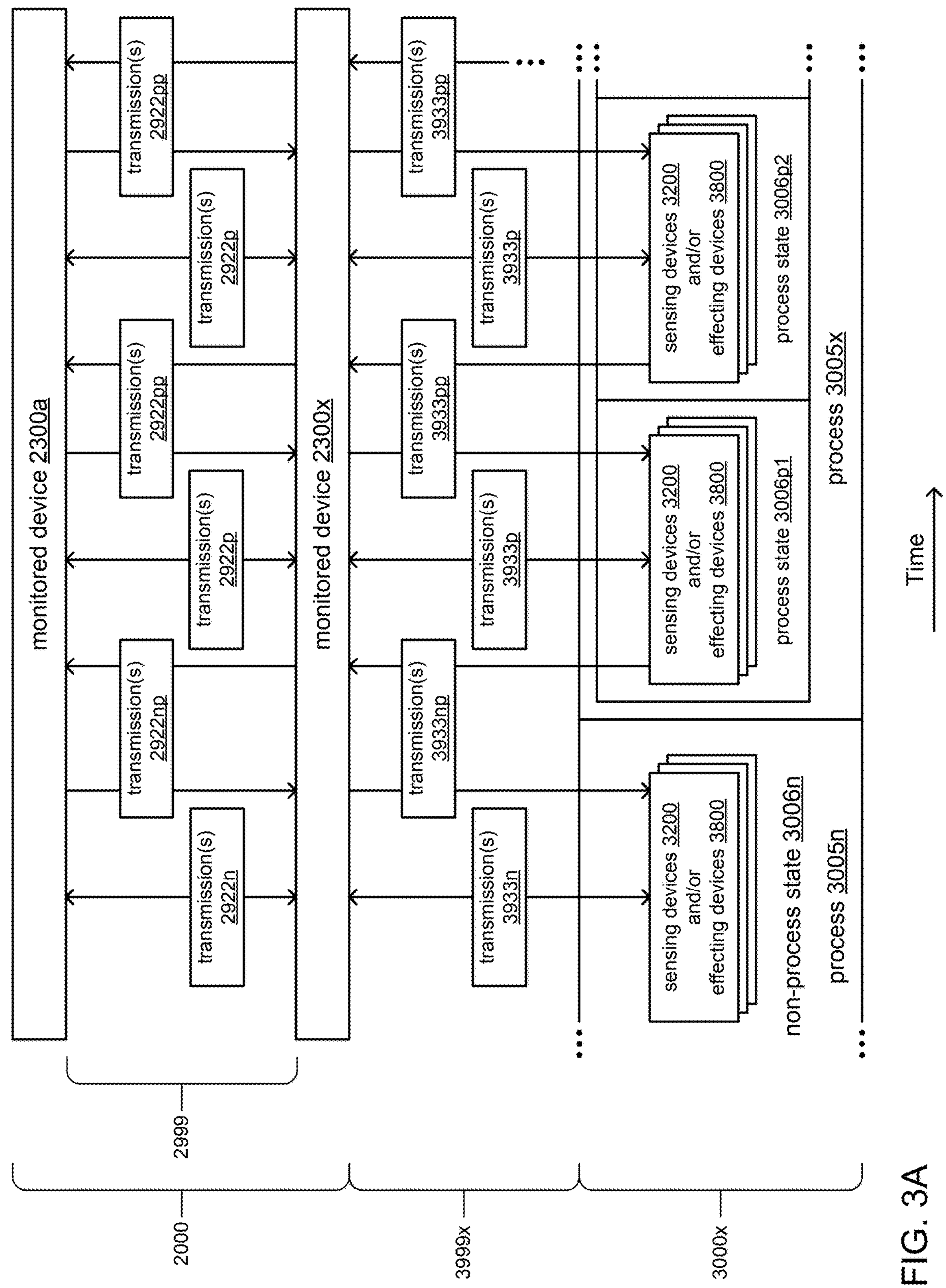

monitored device 2300a
transmission(s) 2922n
transmission(s) 2922np
transmission(s) 2922p
transmission(s) 2922pp
transmission(s) 2922p
transmission(s) 2922pp
monitored device 2300x
transmission(s) 3933n
transmission(s) 3933np
transmission(s) 3933p
transmission(s) 3933pp
transmission(s) 3933p
transmission(s) 3933pp
sensing devices 3200 and/or effecting devices 3800
non-process state 3006n
process 3005n
sensing devices 3200 and/or effecting devices 3800
process state 3006p1
sensing devices 3200 and/or effecting devices 3800
process state 3006p2
process 3005x
Time
2999
2000
3999x
3000x

FIG. 3A

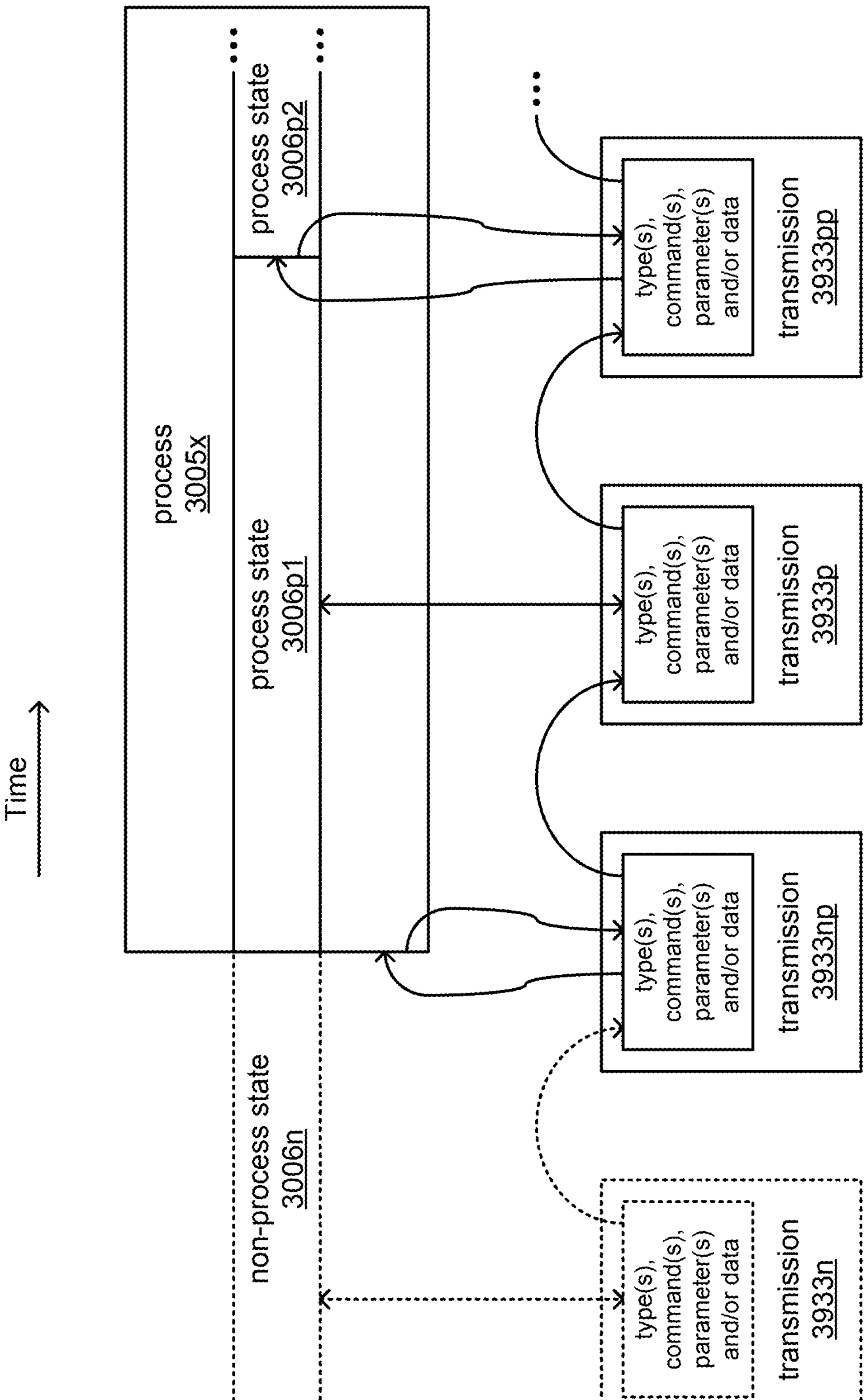
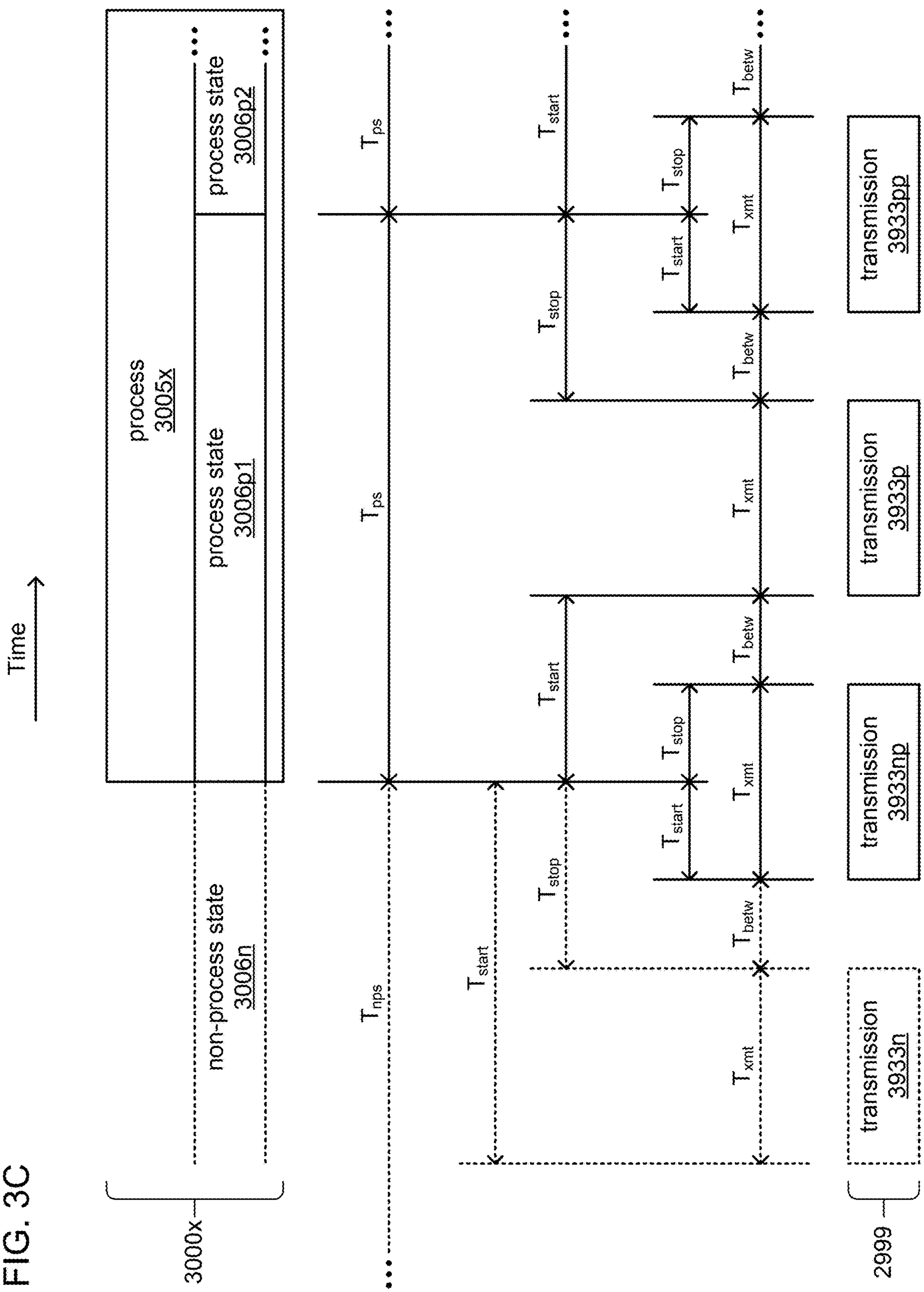

Time
process
3005x
process state
3006p1
process state
3006p2
non-process state
3006n
type(s), command(s), parameter(s) and/or data
transmission
3933n
transmission
3933np
transmission
3933p
transmission
3933pp
3000x
2999

FIG. 3B

FIG. 3C


Time
process
3005x
process state
3006p1
process state
3006p2
non-process state
3006n
3000x
T_{ps}
T_{nps}
T_{start}
T_{stop}
T_{xmt}
T_{betw}
transmission
3933n
transmission
3933np
transmission
3933p
transmission
3933pp
2999

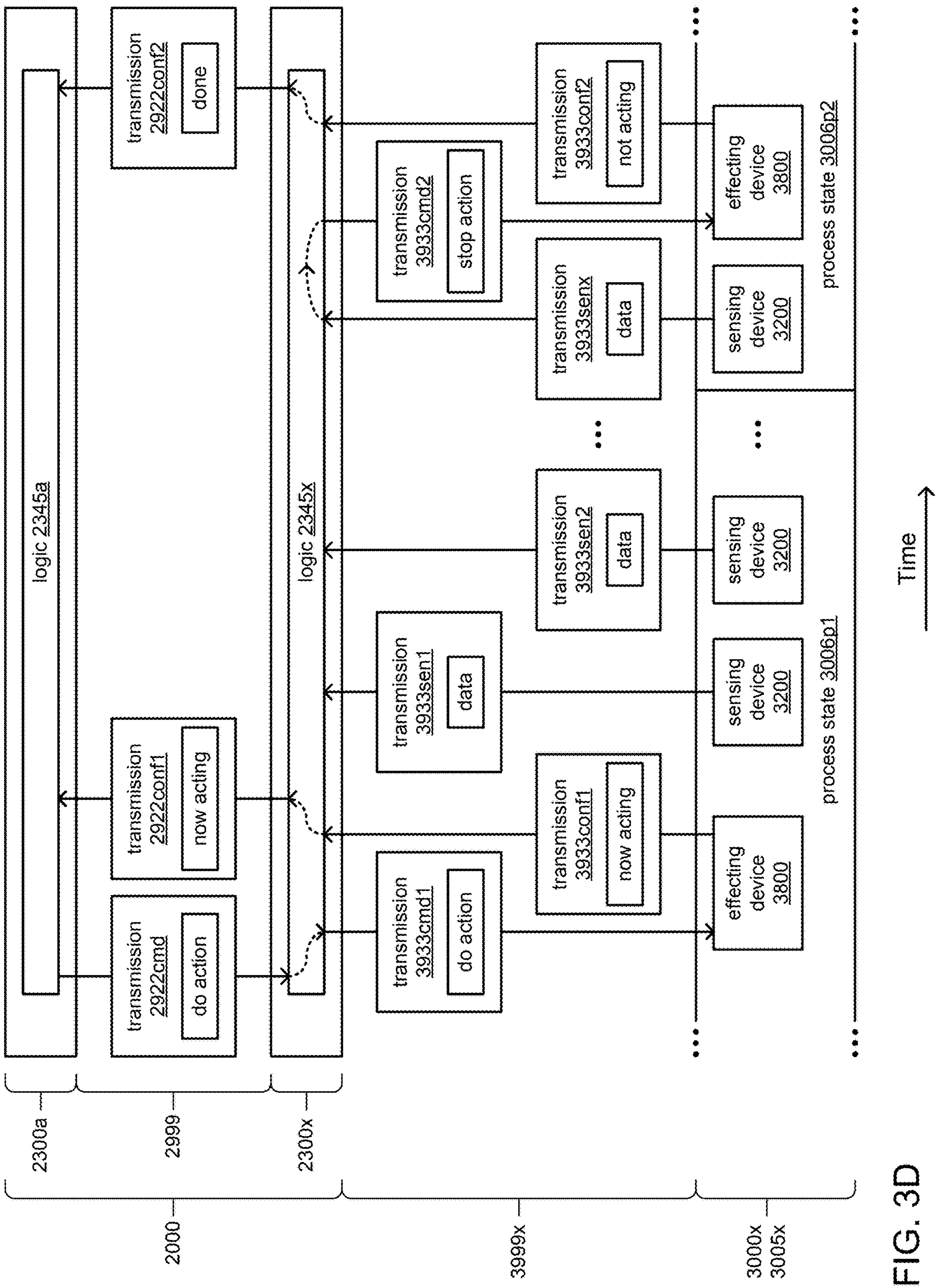

2000
2300a
2999
2300x
3999x
3000x
3005x
logic 2345a
logic 2345x
transmission 2922cmd
do action
transmission 2922conf1
now acting
transmission 2922conf2
done
transmission 3933cmd1
do action
transmission 3933conf1
now acting
transmission 3933sen1
data
transmission 3933sen2
data
transmission 3933senx
data
transmission 3933cmd2
stop action
transmission 3933conf2
not acting
effecting device 3800
sensing device 3200
sensing device 3200
sensing device 3200
effecting device 3800
process state 3006p1
process state 3006p2
Time

FIG. 3D

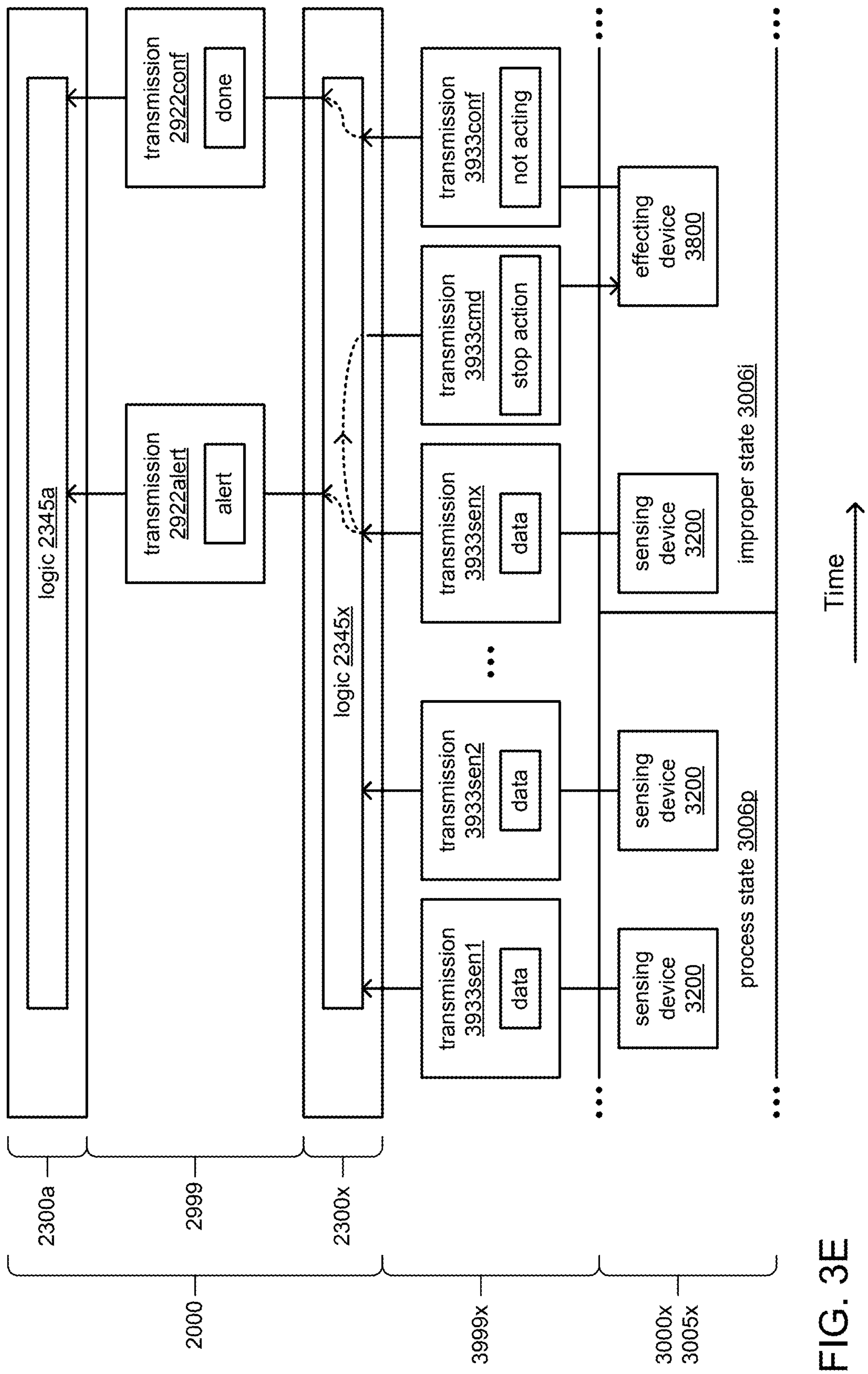

logic 2345a
transmission 2922alert
alert
transmission 2922conf
done
logic 2345x
transmission 3933sen1
data
transmission 3933sen2
data
transmission 3933senx
data
transmission 3933cmd
stop action
transmission 3933conf
not acting
sensing device 3200
sensing device 3200
sensing device 3200
effecting device 3800
process state 3006p
improper state 3006i
Time
2300a
2999
2300x
2000
3999x
3000x
3005x

FIG. 3E

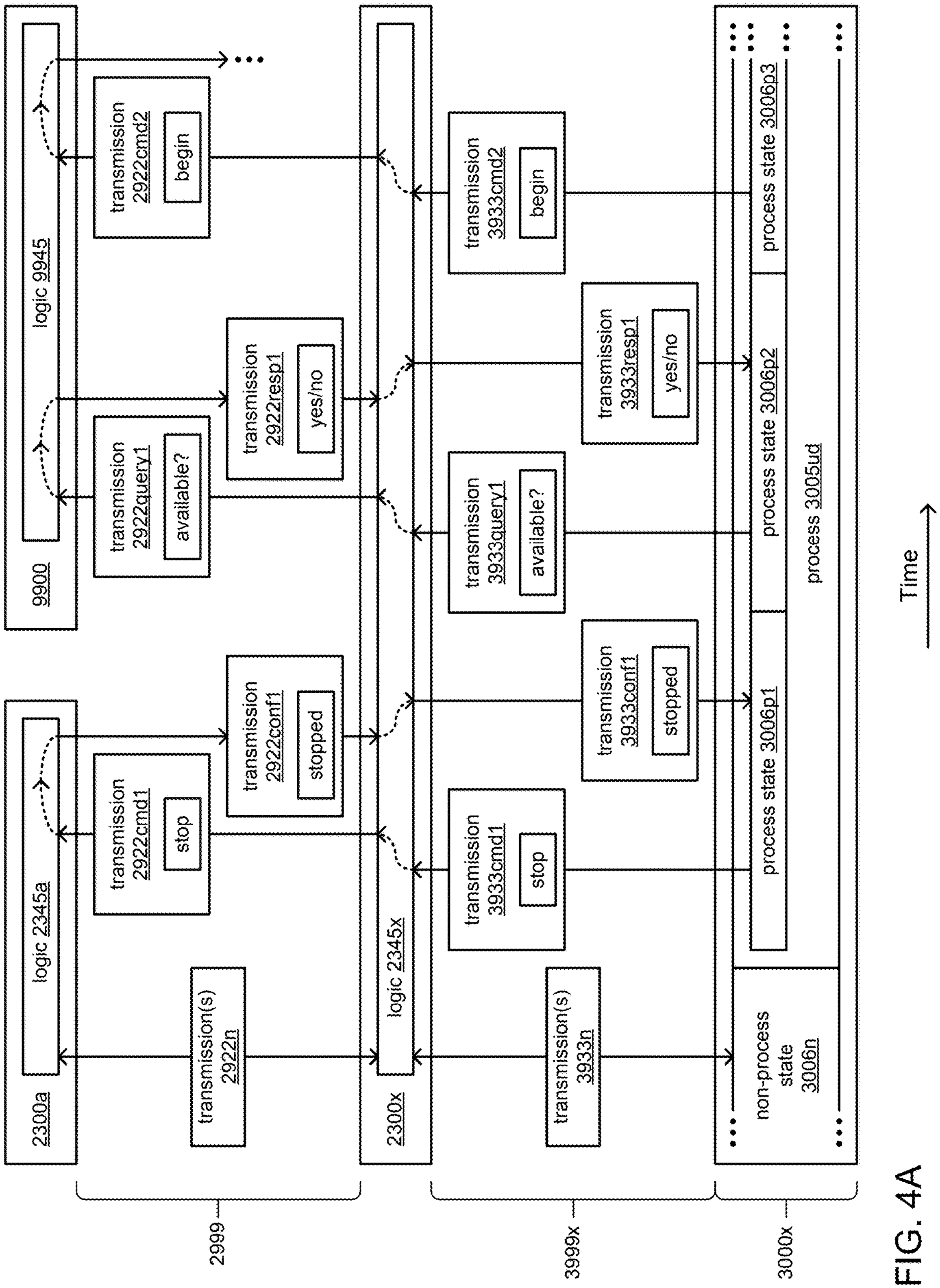

2300a
logic 2345a
9900
logic 9945
transmission(s) 2922n
transmission 2922cmd1
stop
transmission 2922conf1
stopped
transmission 2922query1
available?
transmission 2922resp1
yes/no
transmission 2922cmd2
begin
2300x
logic 2345x
transmission(s) 3933n
transmission 3933cmd1
stop
transmission 3933conf1
stopped
transmission 3933query1
available?
transmission 3933resp1
yes/no
transmission 3933cmd2
begin
non-process state 3006n
process state 3006p1
process state 3006p2
process state 3006p3
process 3005ud
Time
2999
3999x
3000x

FIG. 4A

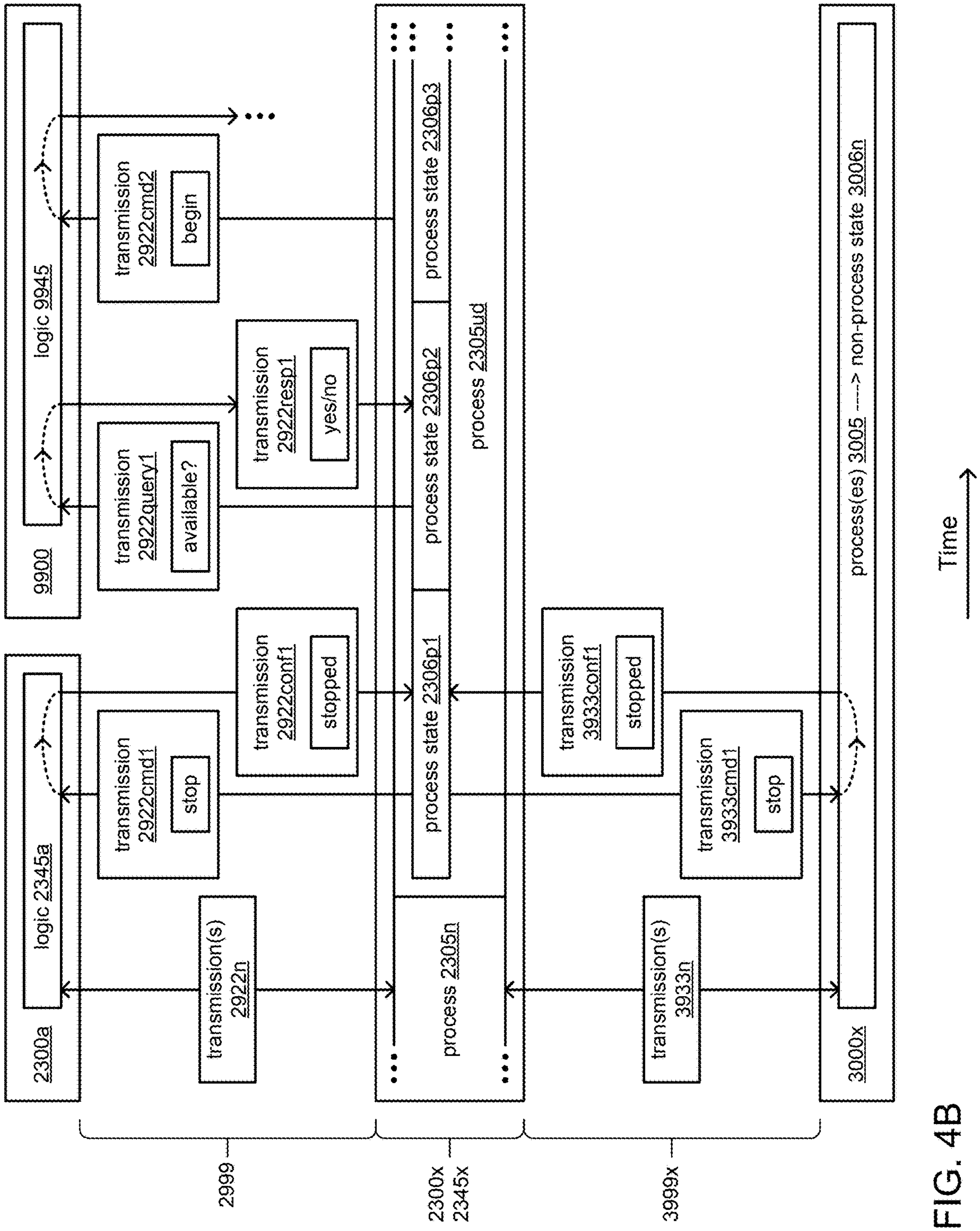

2300a
logic 2345a
transmission(s) 2922n
transmission 2922cmd1
stop
transmission 2922conf1
stopped
9900
logic 9945
transmission 2922query1
available?
transmission 2922resp1
yes/no
transmission 2922cmd2
begin
process 2305n
process state 2306p1
process state 2306p2
process state 2306p3
process 2305ud
transmission(s) 3933n
transmission 3933cmd1
stop
transmission 3933conf1
stopped
3000x
process(es) 3005 ----> non-process state 3006n
Time
2999
2300x
2345x
3999x

FIG. 4B

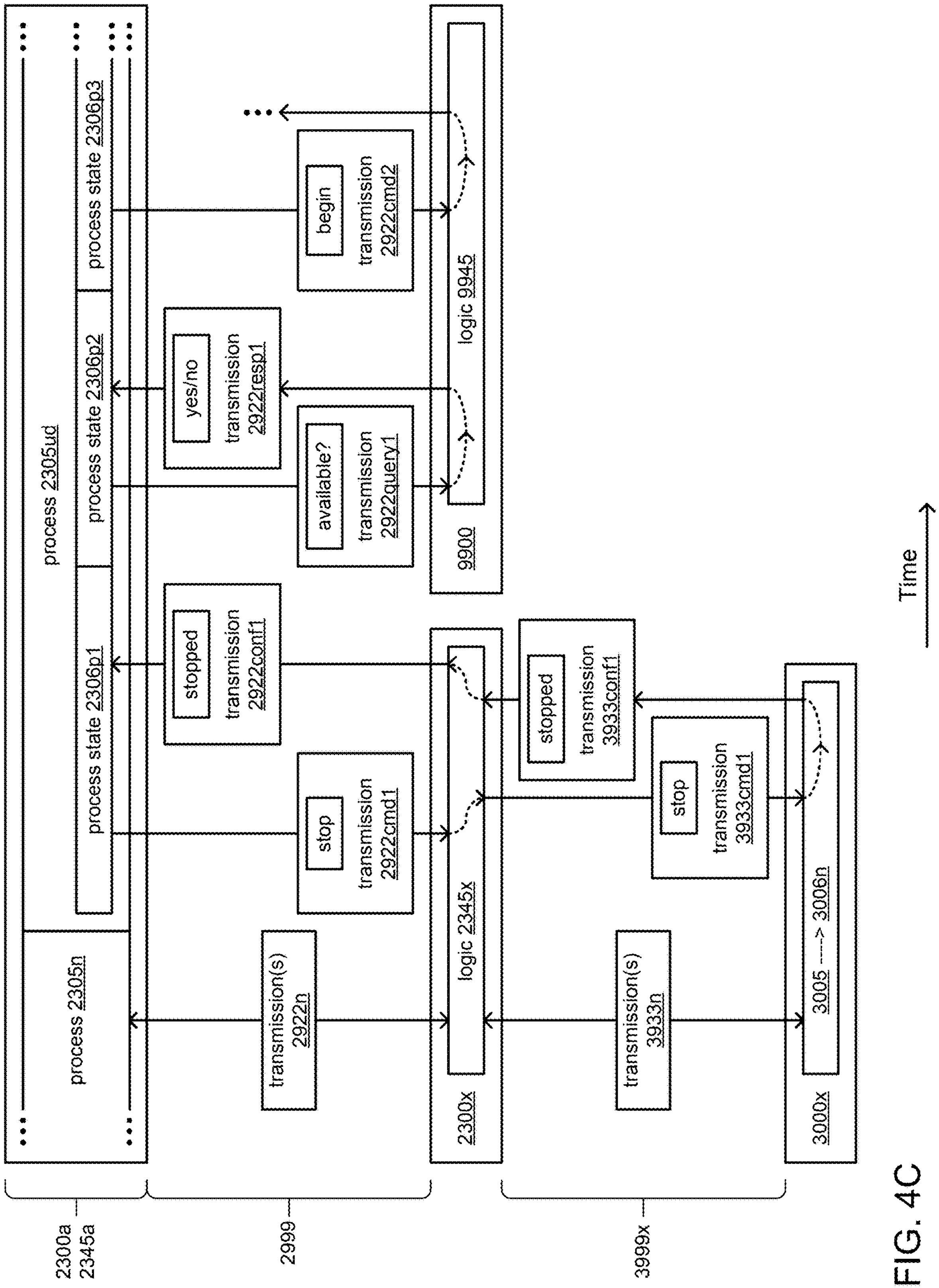

process 2305ud
process state 2306p1
process state 2306p2
process state 2306p3
process 2305n
stop
transmission 2922cmd1
stopped
transmission 2922conf1
available?
transmission 2922query1
yes/no
transmission 2922resp1
begin
transmission 2922cmd2
transmission(s) 2922n
logic 2345x
2300x
logic 9945
9900
stop
transmission 3933cmd1
stopped
transmission 3933conf1
transmission(s) 3933n
3005 ----> 3006n
3000x
2300a
2345a
2999
3999x
Time

FIG. 4C

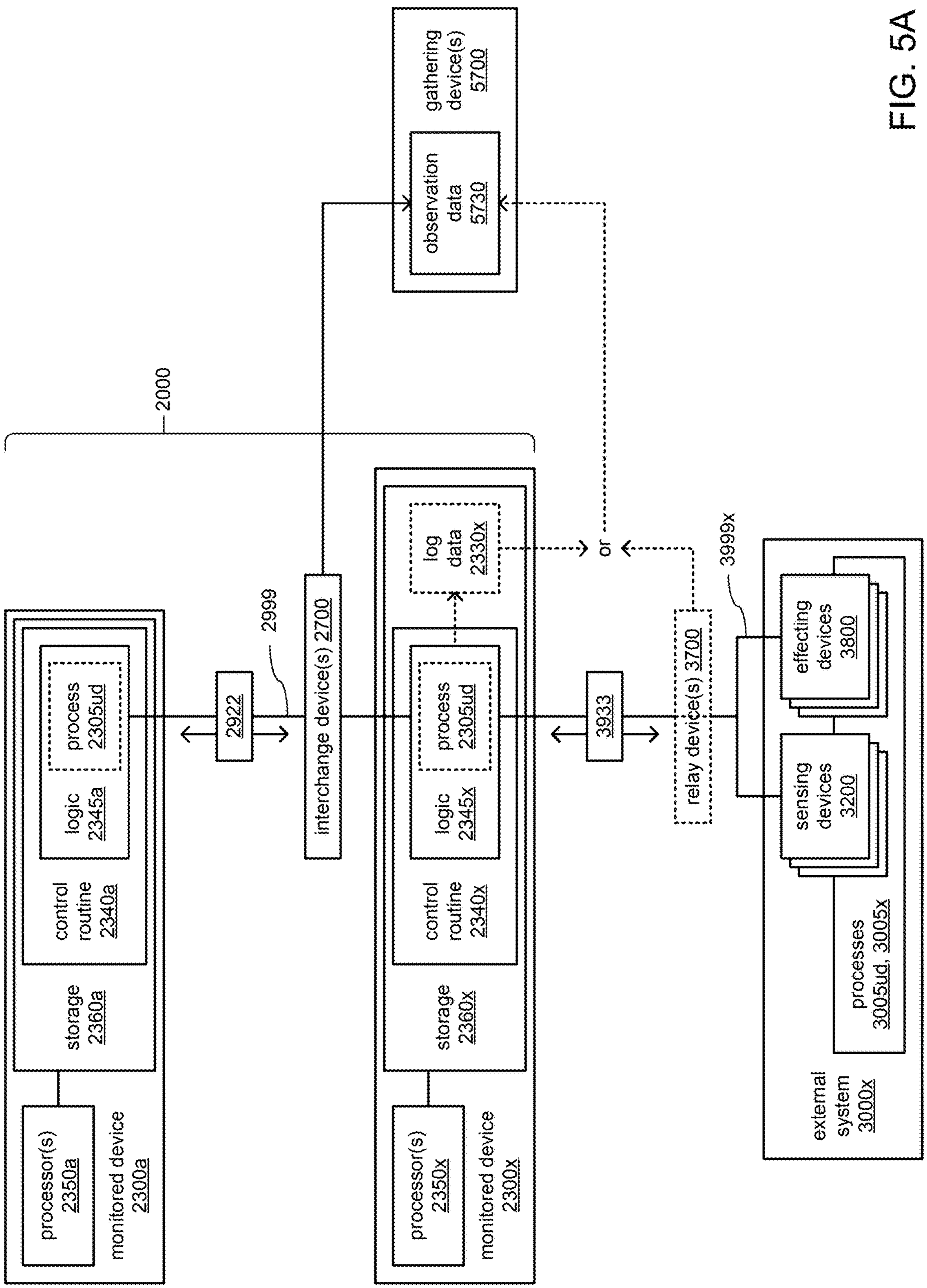

2000
monitored device 2300a
processor(s) 2350a
storage 2360a
control routine 2340a
logic 2345a
process 2305ud
2922
2999
interchange device(s) 2700
monitored device 2300x
processor(s) 2350x
storage 2360x
control routine 2340x
logic 2345x
process 2305ud
log data 2330x
3933
relay device(s) 3700
or
3999x
external system 3000x
processes 3005ud, 3005x
sensing devices 3200
effecting devices 3800
gathering device(s) 5700
observation data 5730

FIG. 5A

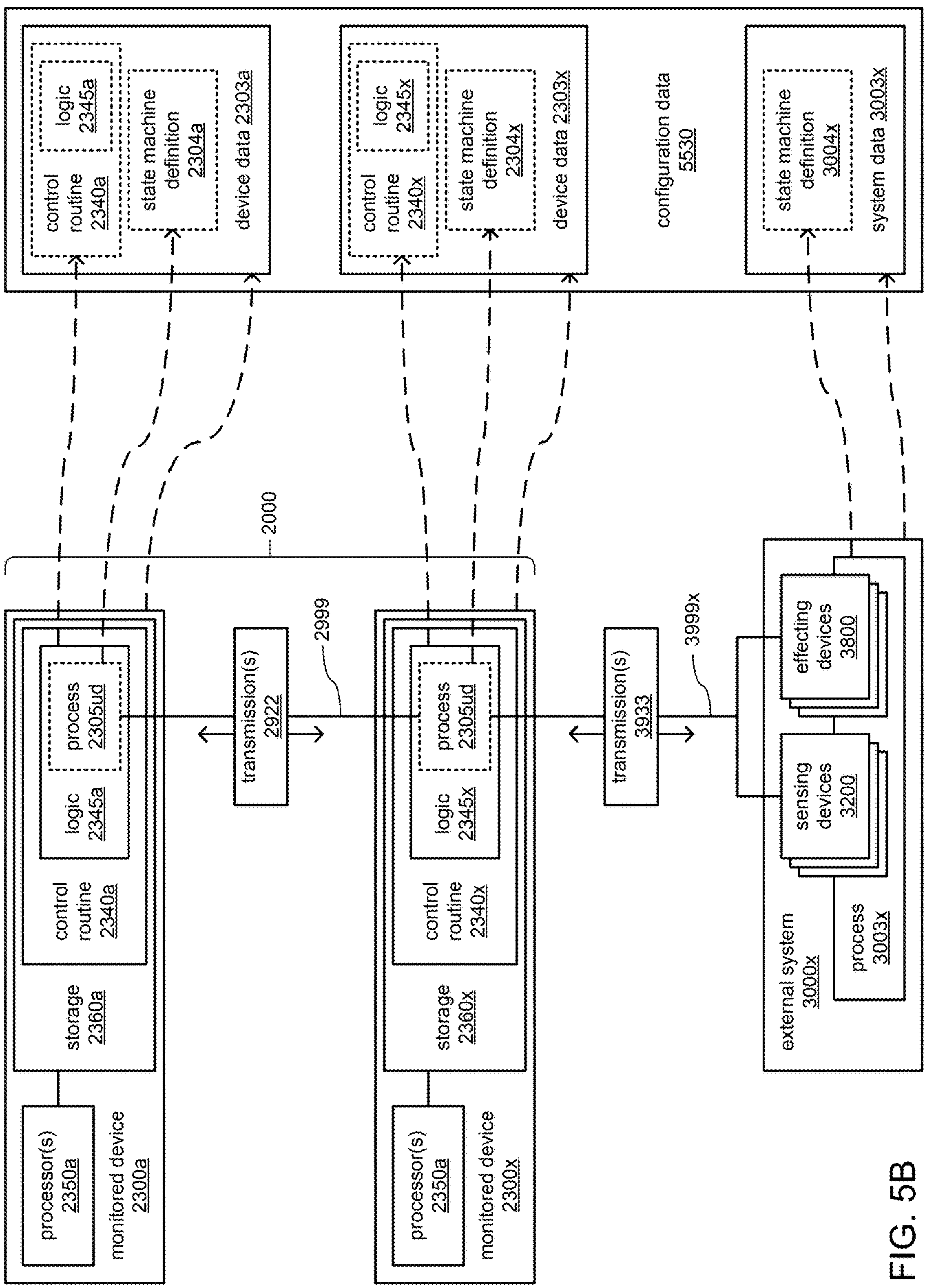

control routine 2340a
logic 2345a
state machine definition 2304a
device data 2303a
control routine 2340x
logic 2345x
state machine definition 2304x
device data 2303x
configuration data 5530
state machine definition 3004x
system data 3003x
2000
processor(s) 2350a
monitored device 2300a
storage 2360a
control routine 2340a
logic 2345a
process 2305ud
transmission(s) 2922
2999
processor(s) 2350a
monitored device 2300x
storage 2360x
control routine 2340x
logic 2345x
process 2305ud
transmission(s) 3933
3999x
external system 3000x
process 3003x
sensing devices 3200
effecting devices 3800

FIG. 5B

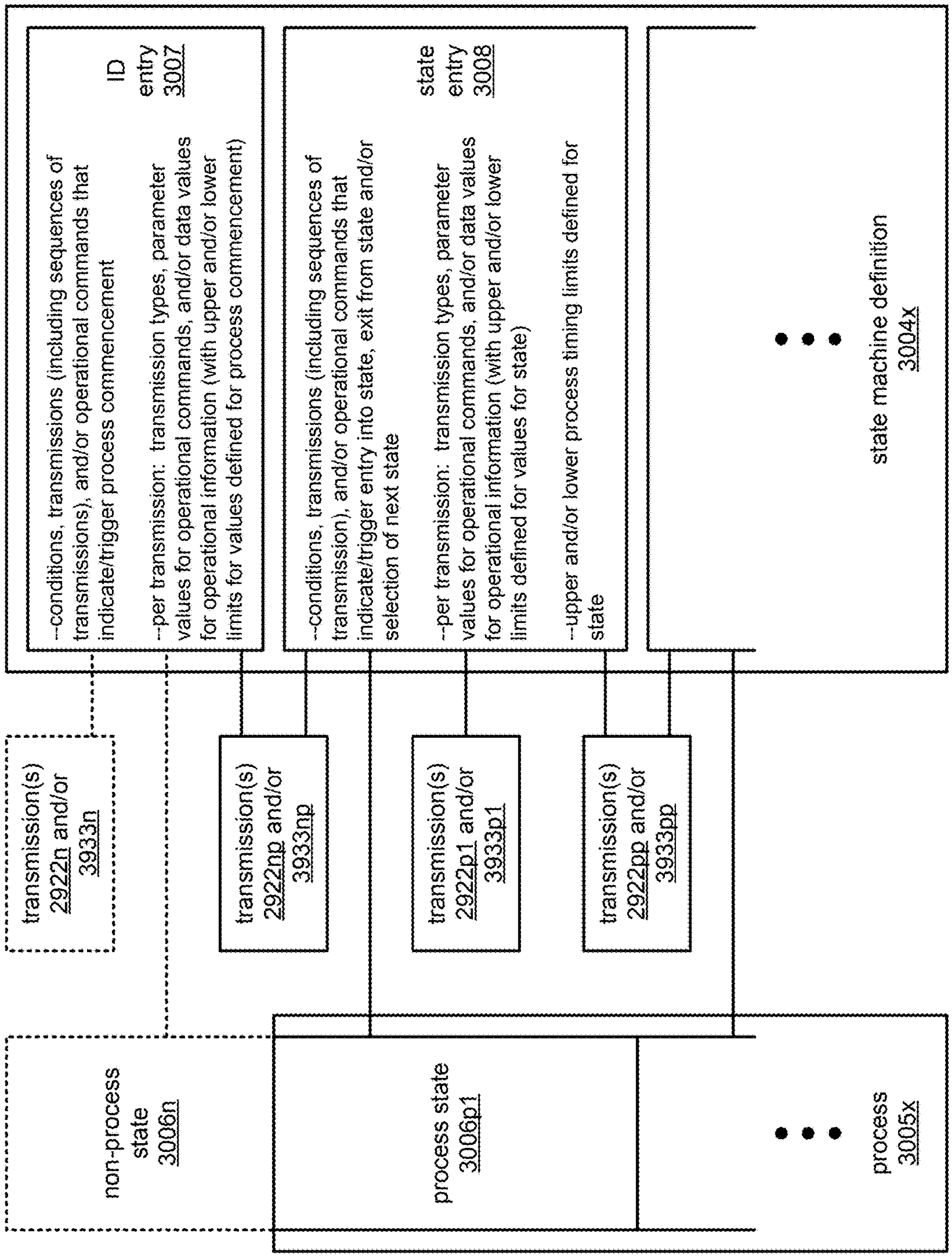

--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/trigger process commencement
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits for values defined for process commencement)
ID entry 3007
--conditions, transmissions (including sequences of transmission), and/or operational commands that indicate/trigger entry into state, exit from state and/or selection of next state
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits defined for values for state)
--upper and/or lower process timing limits defined for state
state entry 3008
state machine definition 3004x
transmission(s) 2922n and/or 3933n
transmission(s) 2922np and/or 3933np
transmission(s) 2922p1 and/or 3933p1
transmission(s) 2922pp and/or 3933pp
non-process state 3006n
process state 3006p1
process 3005x

FIG. 5C

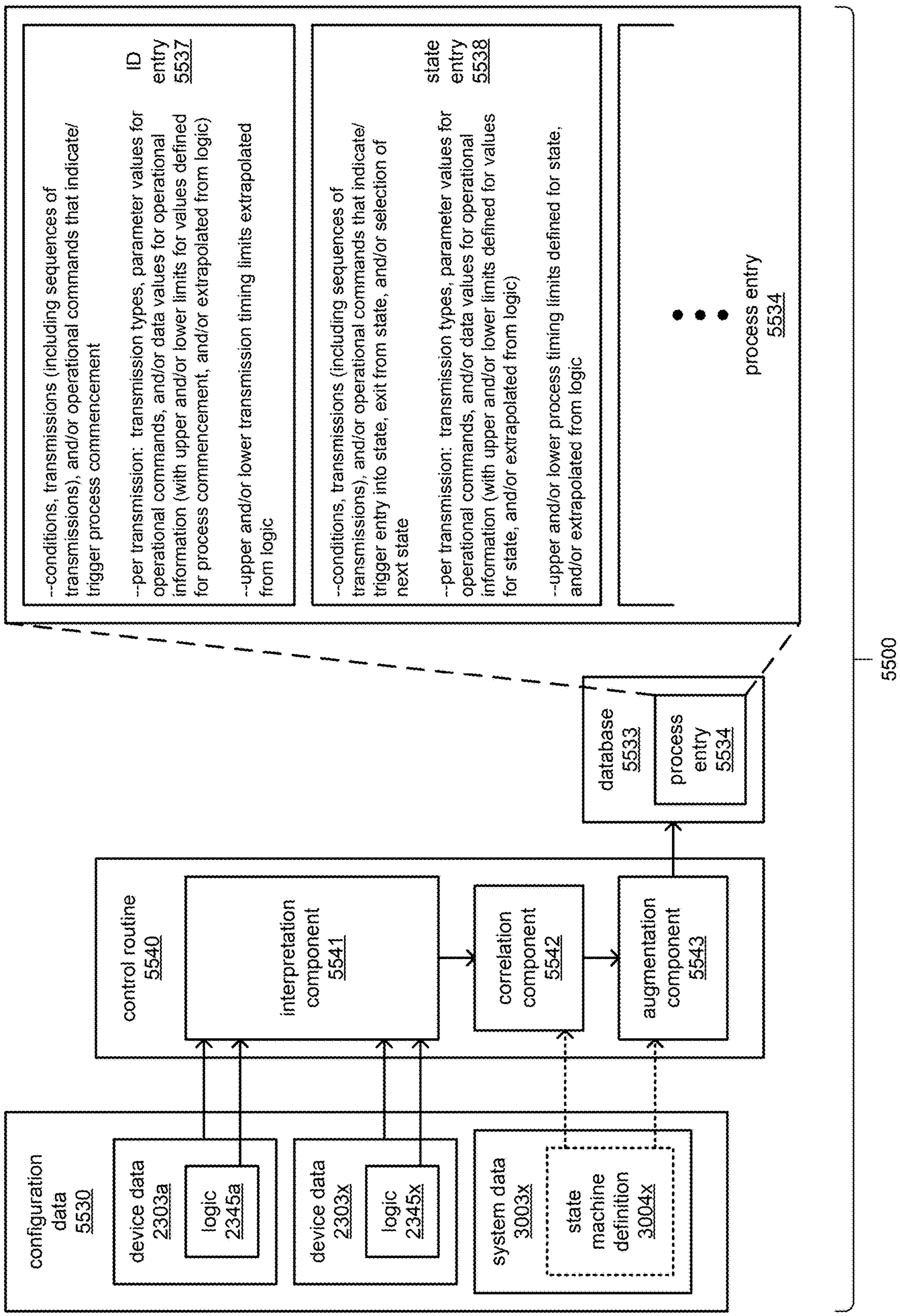

configuration data 5530
device data 2303a
logic 2345a
device data 2303x
logic 2345x
system data 3003x
state machine definition 3004x
control routine 5540
interpretation component 5541
correlation component 5542
augmentation component 5543
database 5533
process entry 5534
--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/ trigger process commencement
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits for values defined for process commencement, and/or extrapolated from logic)
--upper and/or lower transmission timing limits extrapolated from logic
ID entry 5537
--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/ trigger entry into state, exit from state, and/or selection of next state
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits defined for values for state, and/or extrapolated from logic)
--upper and/or lower process timing limits defined for state, and/or extrapolated from logic
state entry 5538
process entry 5534
5500

FIG. 6A

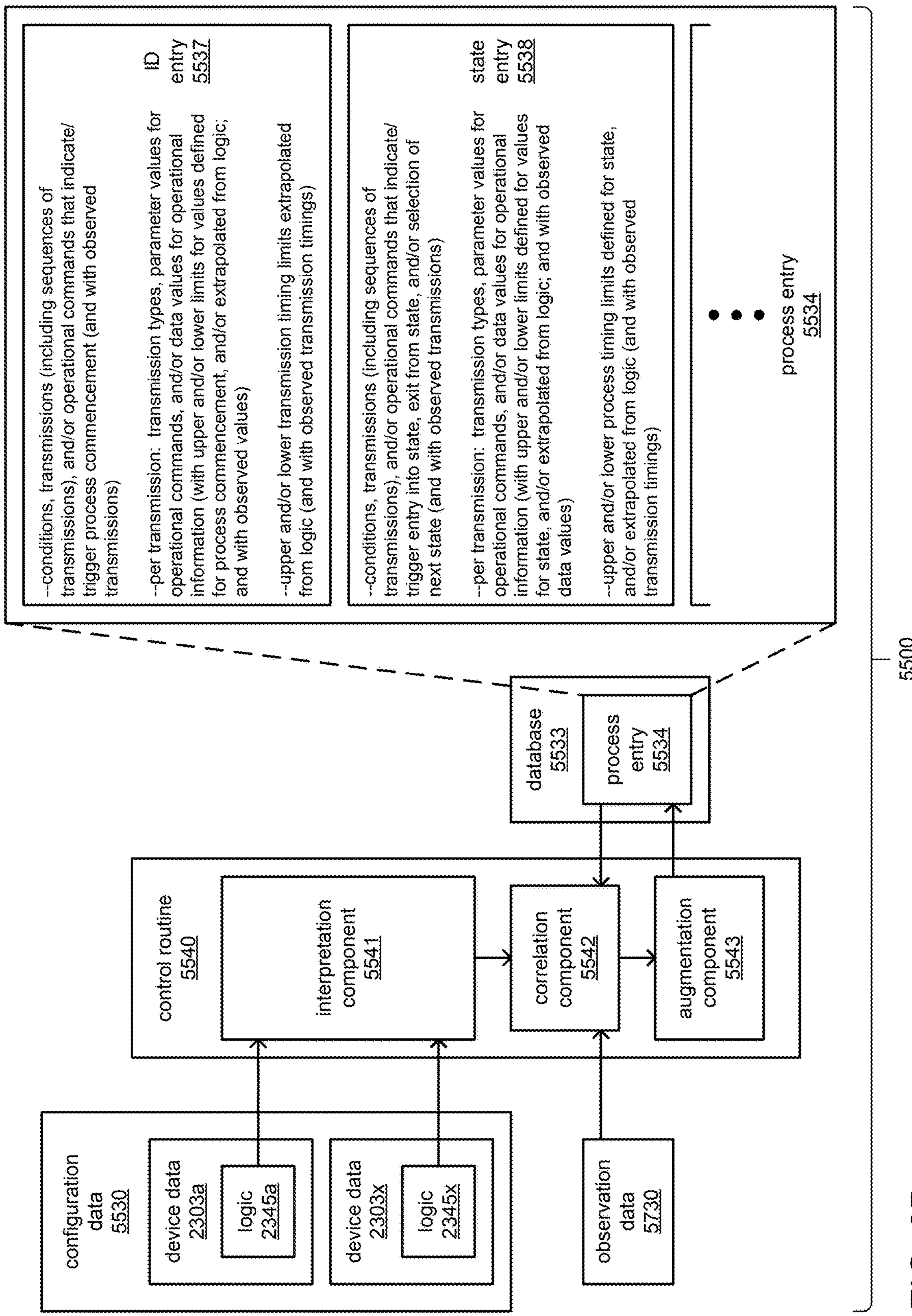

configuration data 5530
device data 2303a
logic 2345a
device data 2303x
logic 2345x
observation data 5730
control routine 5540
interpretation component 5541
correlation component 5542
augmentation component 5543
database 5533
process entry 5534
--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/ trigger process commencement (and with observed transmissions)
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits for values defined for process commencement, and/or extrapolated from logic; and with observed values)
--upper and/or lower transmission timing limits extrapolated from logic (and with observed transmission timings)
ID entry 5537
--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/ trigger entry into state, exit from state, and/or selection of next state (and with observed transmissions)
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits defined for values for state, and/or extrapolated from logic; and with observed data values)
--upper and/or lower process timing limits defined for state, and/or extrapolated from logic (and with observed transmission timings)
state entry 5538
process entry 5534
5500

FIG. 6B

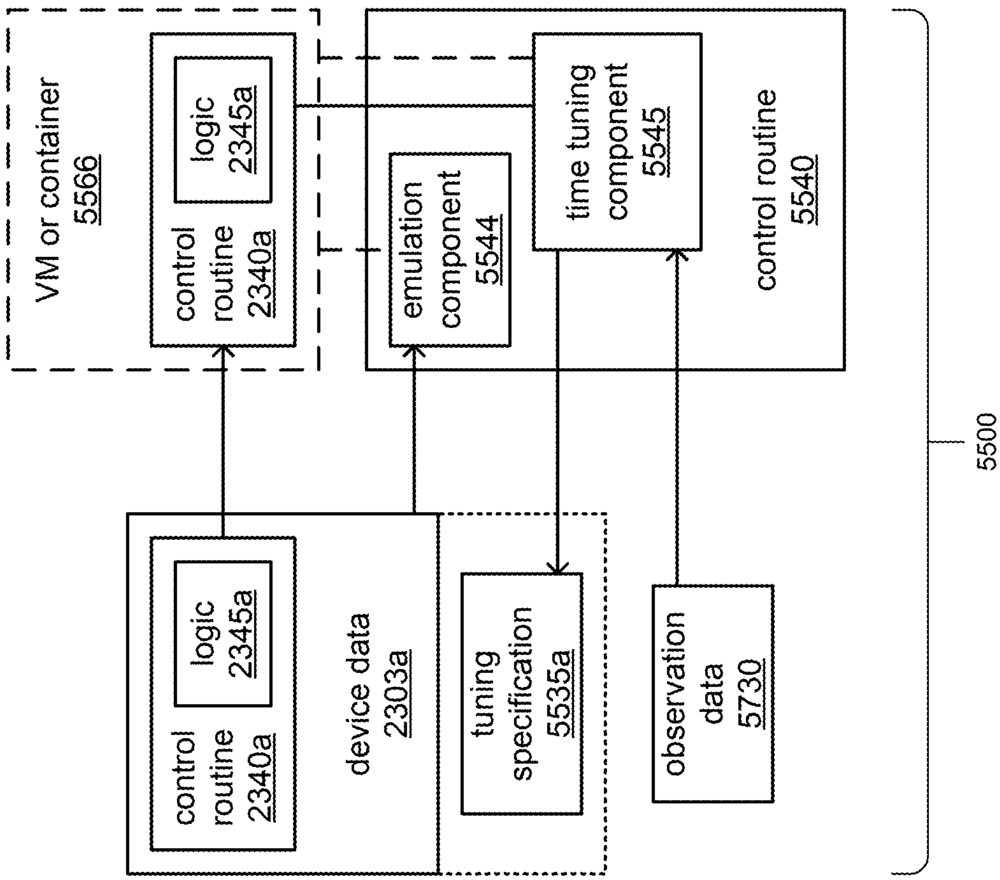

VM or container 5566
control routine 2340a
logic 2345a
emulation component 5544
time tuning component 5545
control routine 5540
control routine 2340a
logic 2345a
device data 2303a
tuning specification 5535a
observation data 5730
5500

FIG. 6C

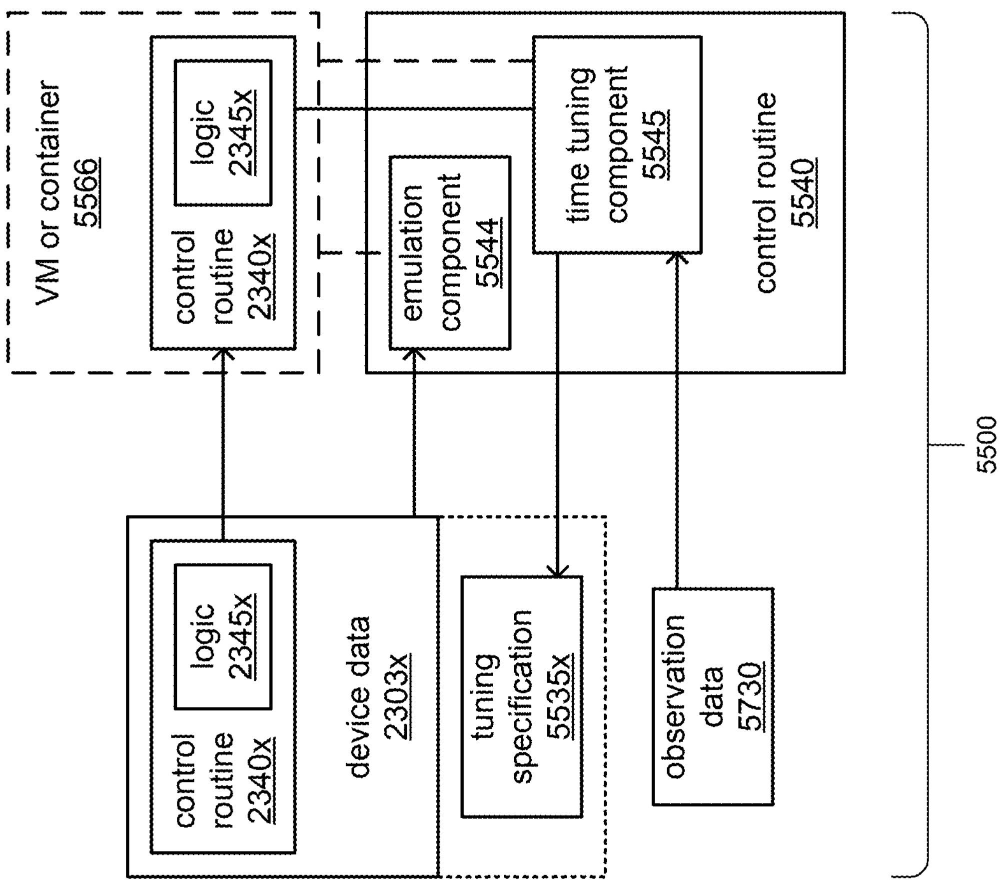

VM or container
5566
control
routine
2340x
logic
2345x
emulation
component
5544
time tuning
component
5545
control routine
5540
control
routine
2340x
logic
2345x
device data
2303x
tuning
specification
5535x
observation
data
5730
5500

FIG. 6D

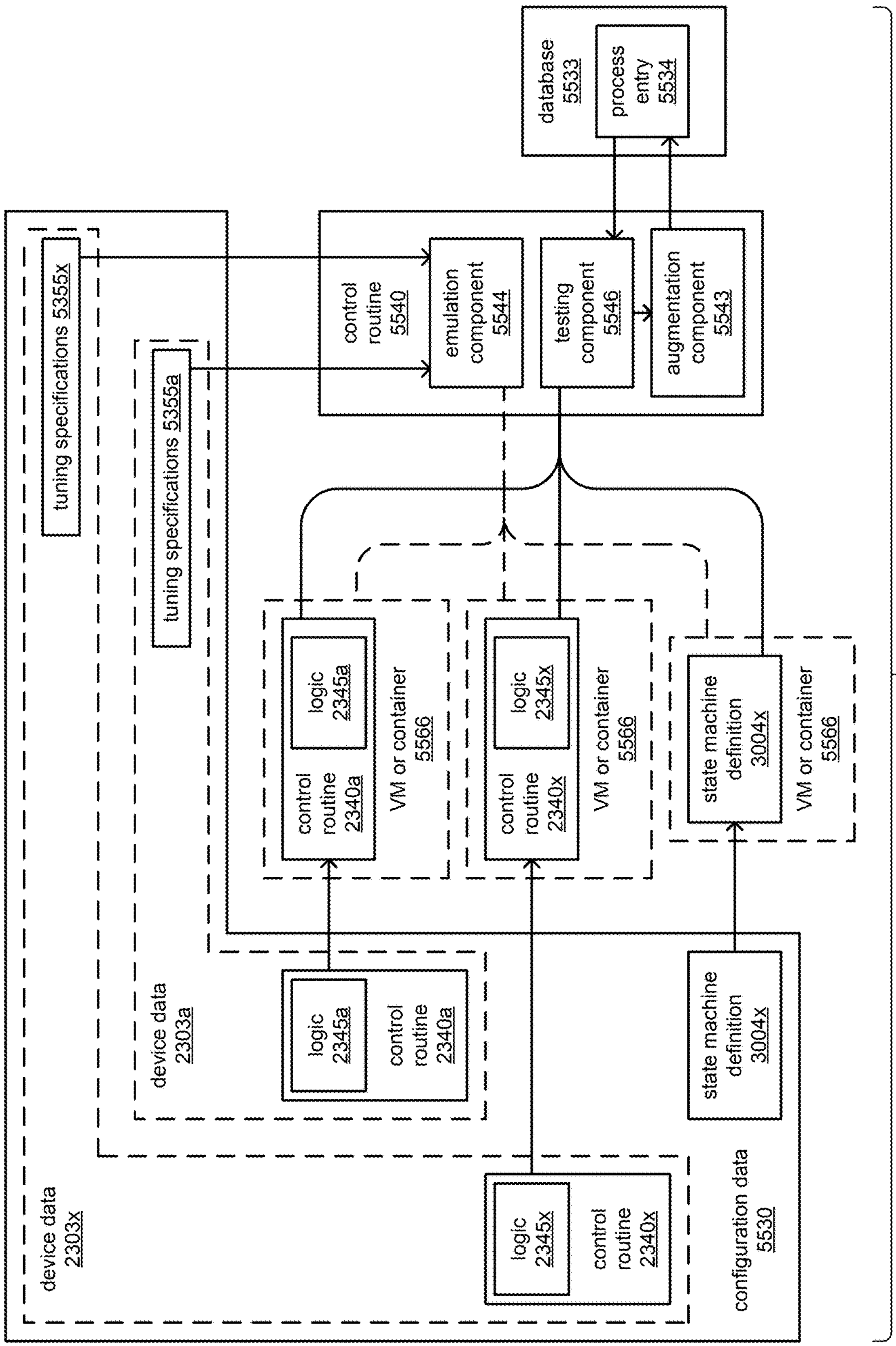
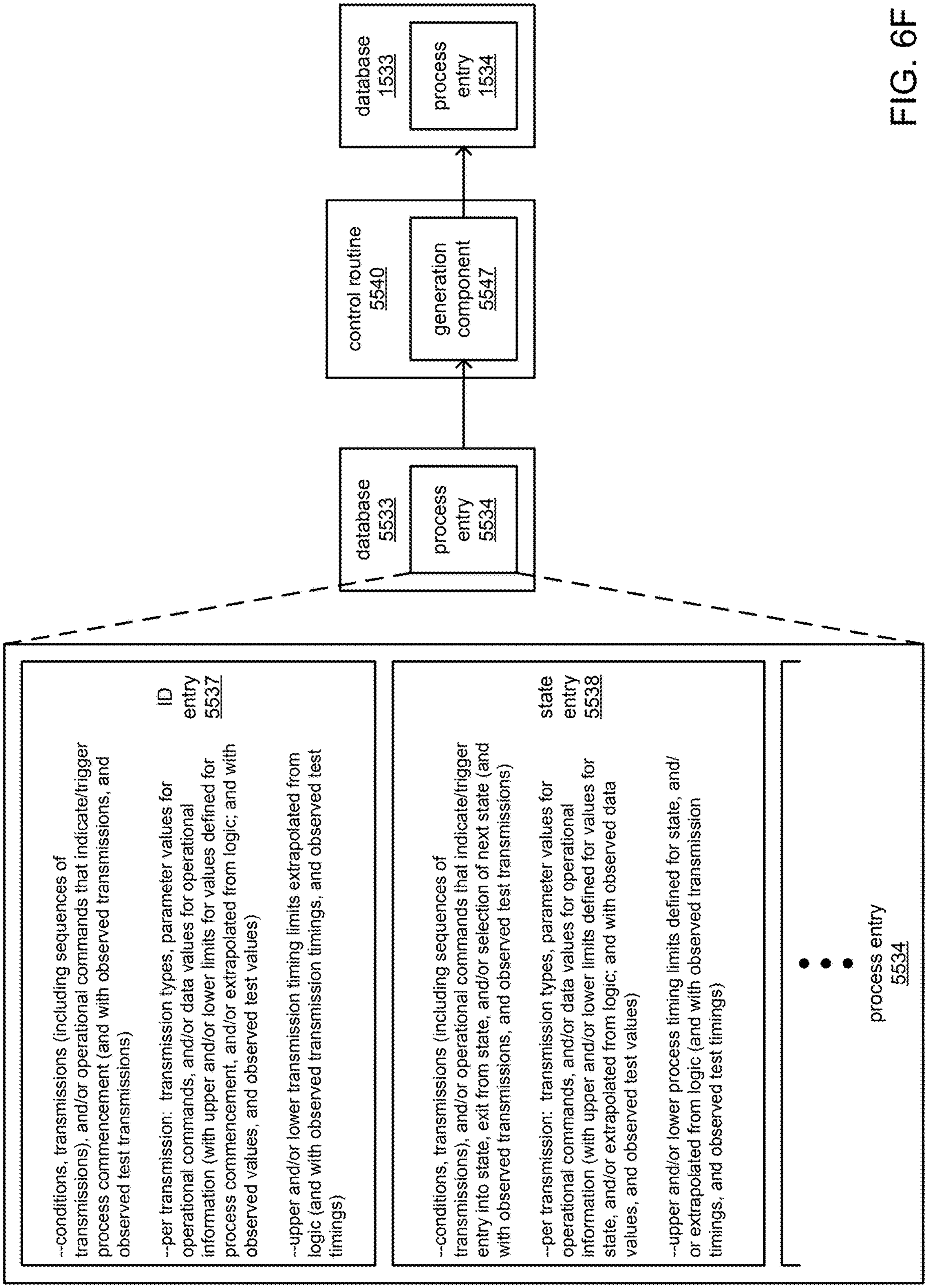

database 5533
process entry 5534
control routine 5540
emulation component 5544
testing component 5546
augmentation component 5543
tuning specifications 5355x
tuning specifications 5355a
control routine 2340a
logic 2345a
VM or container 5566
control routine 2340x
logic 2345x
VM or container 5566
state machine definition 3004x
VM or container 5566
device data 2303a
logic 2345a
control routine 2340a
state machine definition 3004x
device data 2303x
logic 2345x
control routine 2340x
configuration data 5530
5500

FIG. 6E



database
5533
process
entry
5534
control routine
5540
generation
component
5547
database
5533
process
entry
5534
--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/trigger process commencement (and with observed transmissions, and observed test transmissions)
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits for values defined for process commencement, and/or extrapolated from logic; and with observed values, and observed test values)
--upper and/or lower transmission timing limits extrapolated from logic (and with observed transmission timings, and observed test timings)
ID
entry
5537
--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/trigger entry into state, exit from state, and/or selection of next state (and with observed transmissions, and observed test transmissions)
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits defined for values for state, and/or extrapolated from logic; and with observed data values, and observed test values)
--upper and/or lower process timing limits defined for state, and/or extrapolated from logic (and with observed transmission timings, and observed test timings)
state
entry
5538
process entry
5534

FIG. 6F

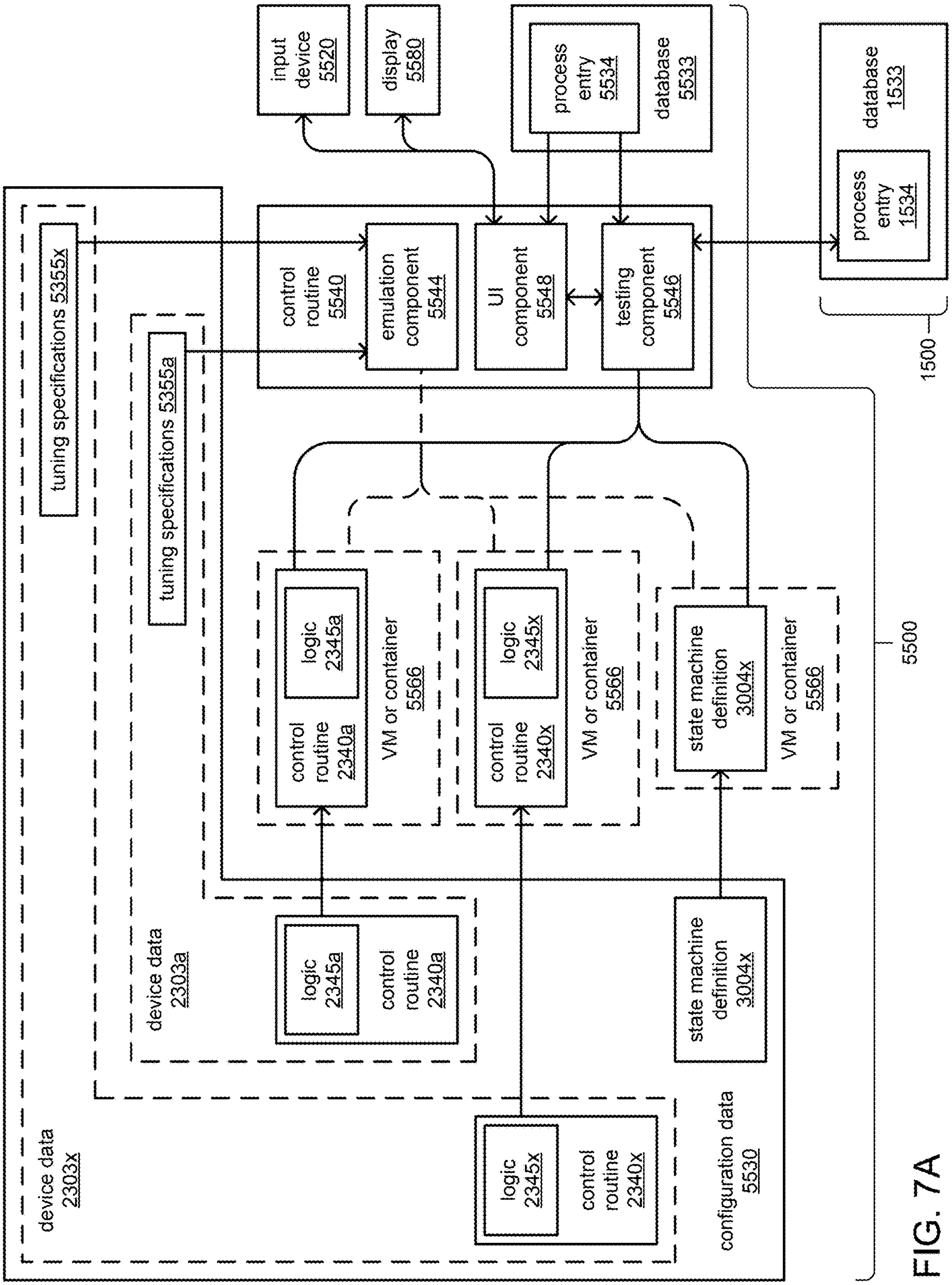
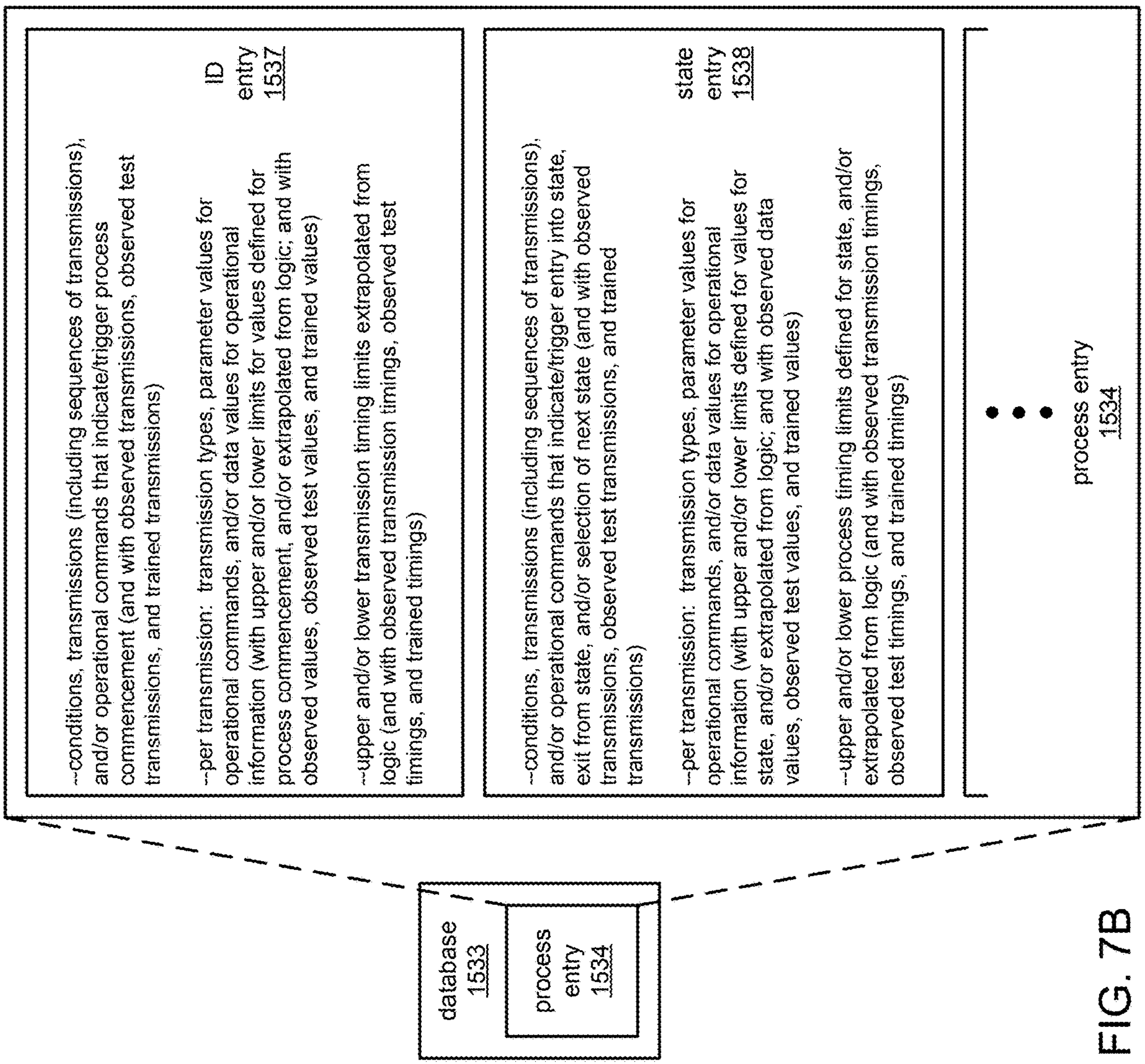

input device 5520
display 5580
process entry 5534
database 5533
control routine 5540
emulation component 5544
UI component 5548
testing component 5546
process entry 1534
database 1533
1500
tuning specifications 5355x
tuning specifications 5355a
control routine 2340a
logic 2345a
VM or container 5566
control routine 2340x
logic 2345x
VM or container 5566
state machine definition 3004x
VM or container 5566
5500
device data 2303a
logic 2345a
control routine 2340a
state machine definition 3004x
device data 2303x
logic 2345x
control routine 2340x
configuration data 5530

FIG. 7A



database
1533
process
entry
1534
--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/trigger process commencement (and with observed transmissions, observed test transmissions, and trained transmissions)
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits for values defined for process commencement, and/or extrapolated from logic; and with observed values, observed test values, and trained values)
--upper and/or lower transmission timing limits extrapolated from logic (and with observed transmission timings, observed test timings, and trained timings)
ID
entry
1537
--conditions, transmissions (including sequences of transmissions), and/or operational commands that indicate/trigger entry into state, exit from state, and/or selection of next state (and with observed transmissions, observed test transmissions, and trained transmissions)
--per transmission: transmission types, parameter values for operational commands, and/or data values for operational information (with upper and/or lower limits defined for values for state, and/or extrapolated from logic; and with observed data values, observed test values, and trained values)
--upper and/or lower process timing limits defined for state, and/or extrapolated from logic (and with observed transmission timings, observed test timings, and trained timings)
state
entry
1538
process entry
1534

FIG. 7B

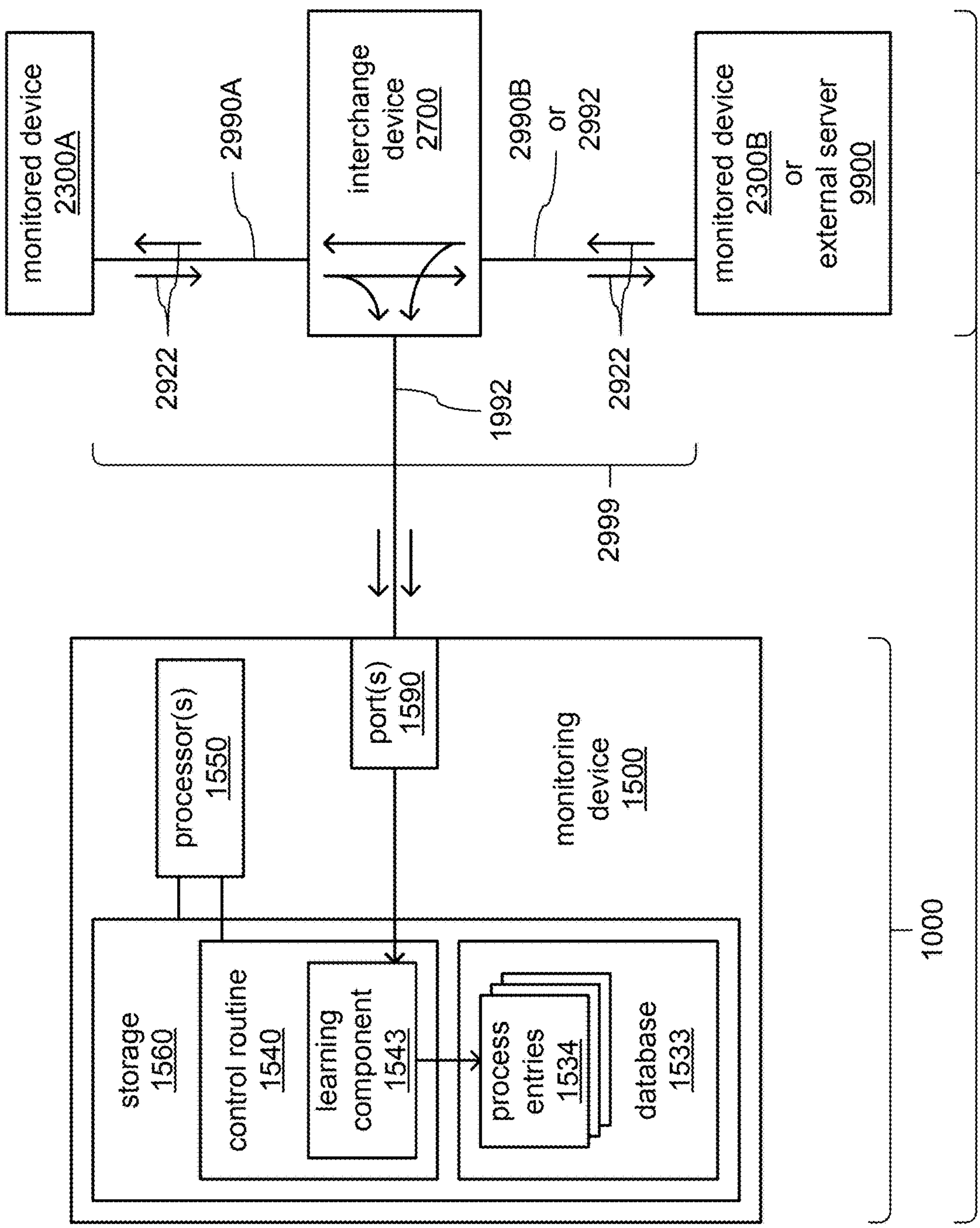

monitored device 2300A
2990A
interchange device 2700
2990B or 2992
monitored device 2300B or external server 9900
2000
2922
1992
2922
2999
processor(s) 1550
port(s) 1590
monitoring device 1500
storage 1560
control routine 1540
learning component 1543
process entries 1534
database 1533
1000

FIG. 8A

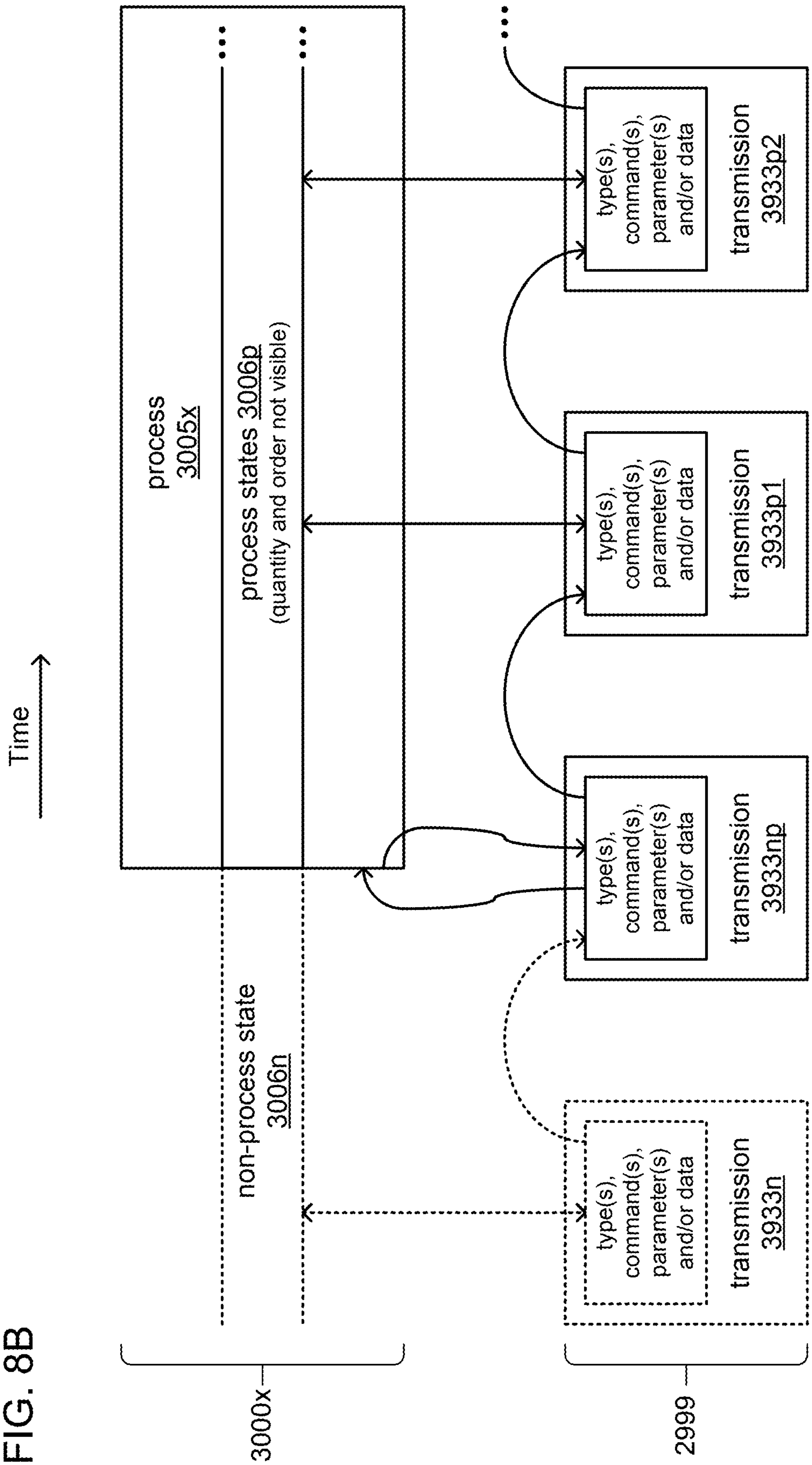
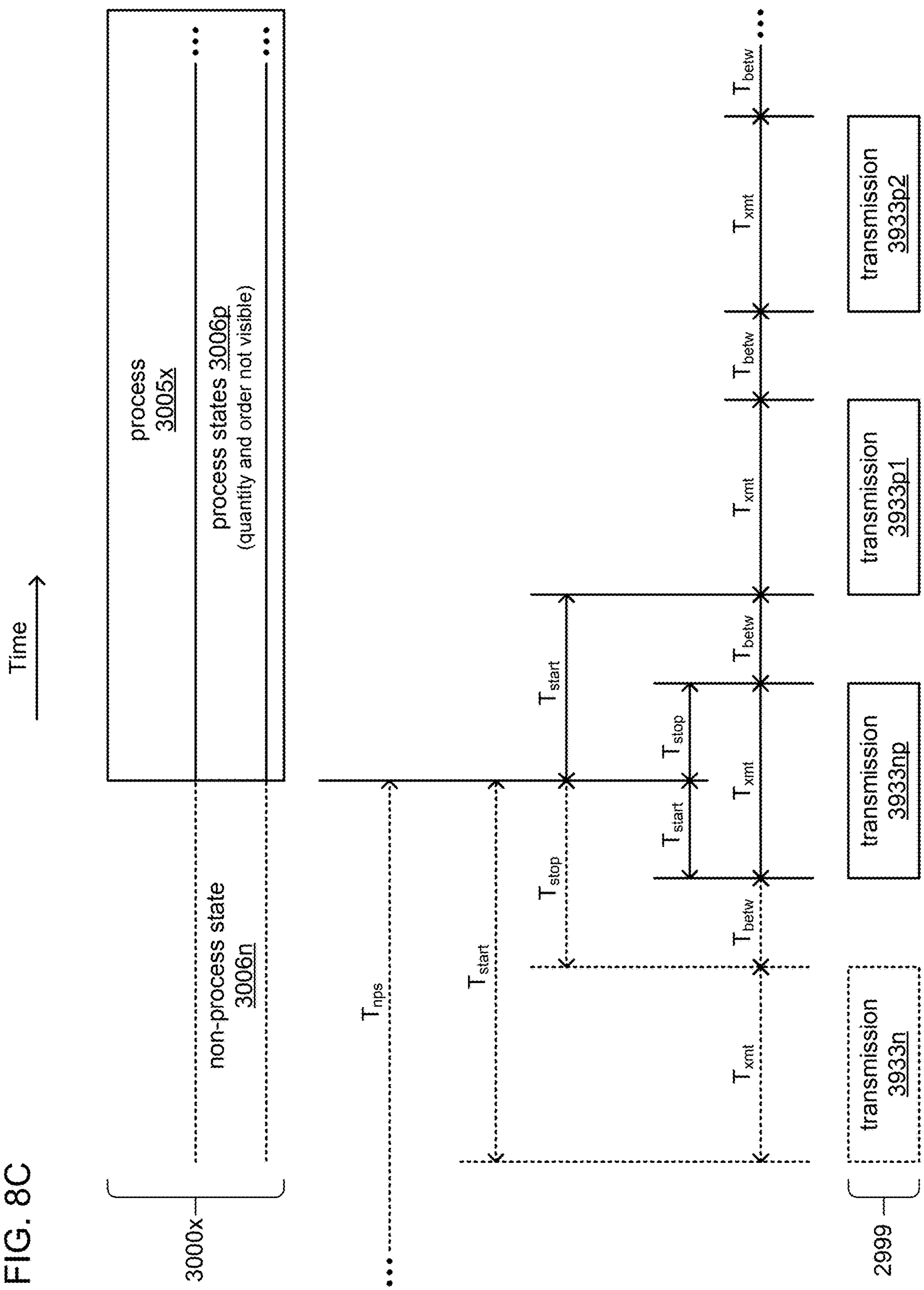
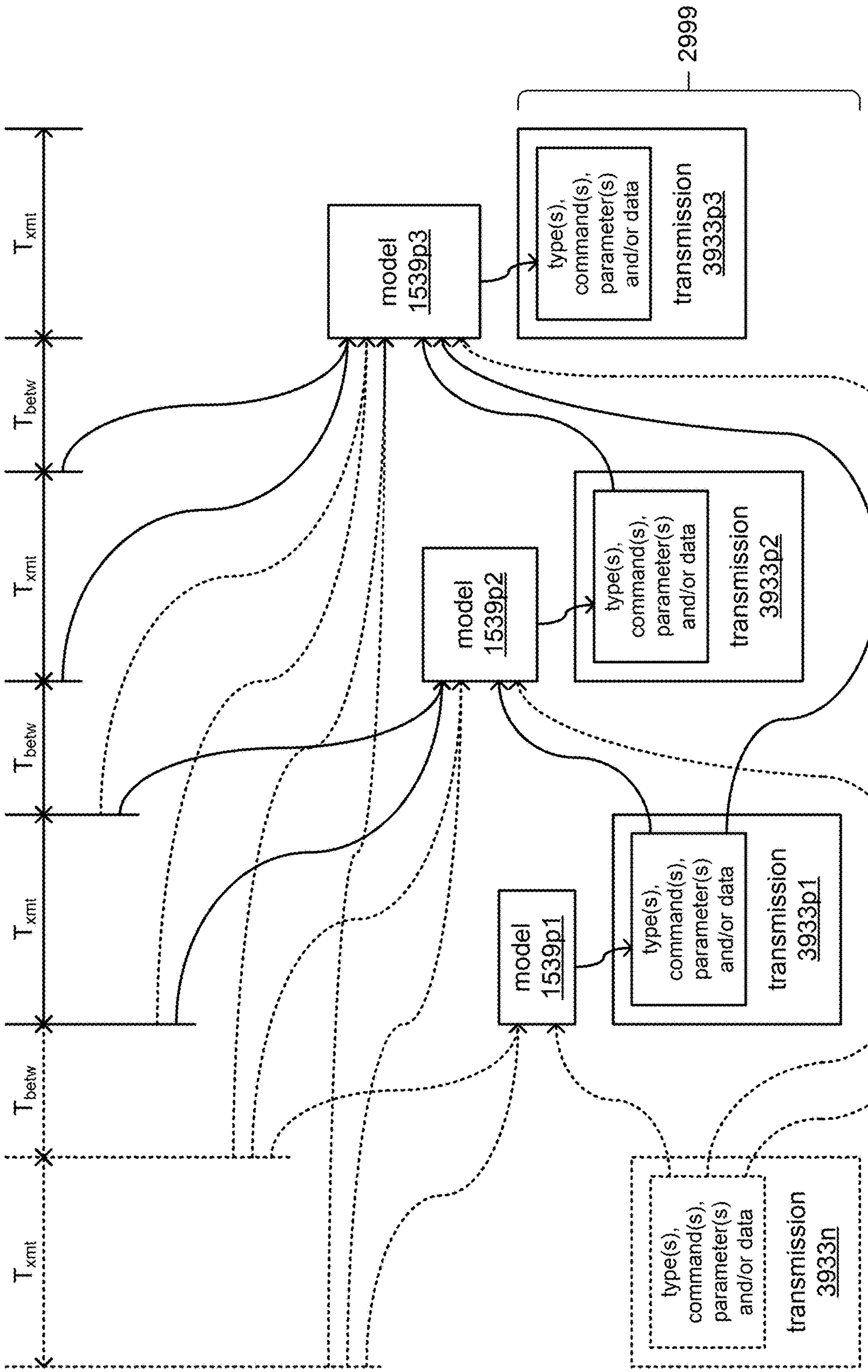

Time
process 3005x
process states 3006p
(quantity and order not visible)
non-process state 3006n
type(s), command(s), parameter(s) and/or data
transmission 3933p2
transmission 3933p1
transmission 3933np
transmission 3933n
3000x
2999

FIG. 8B

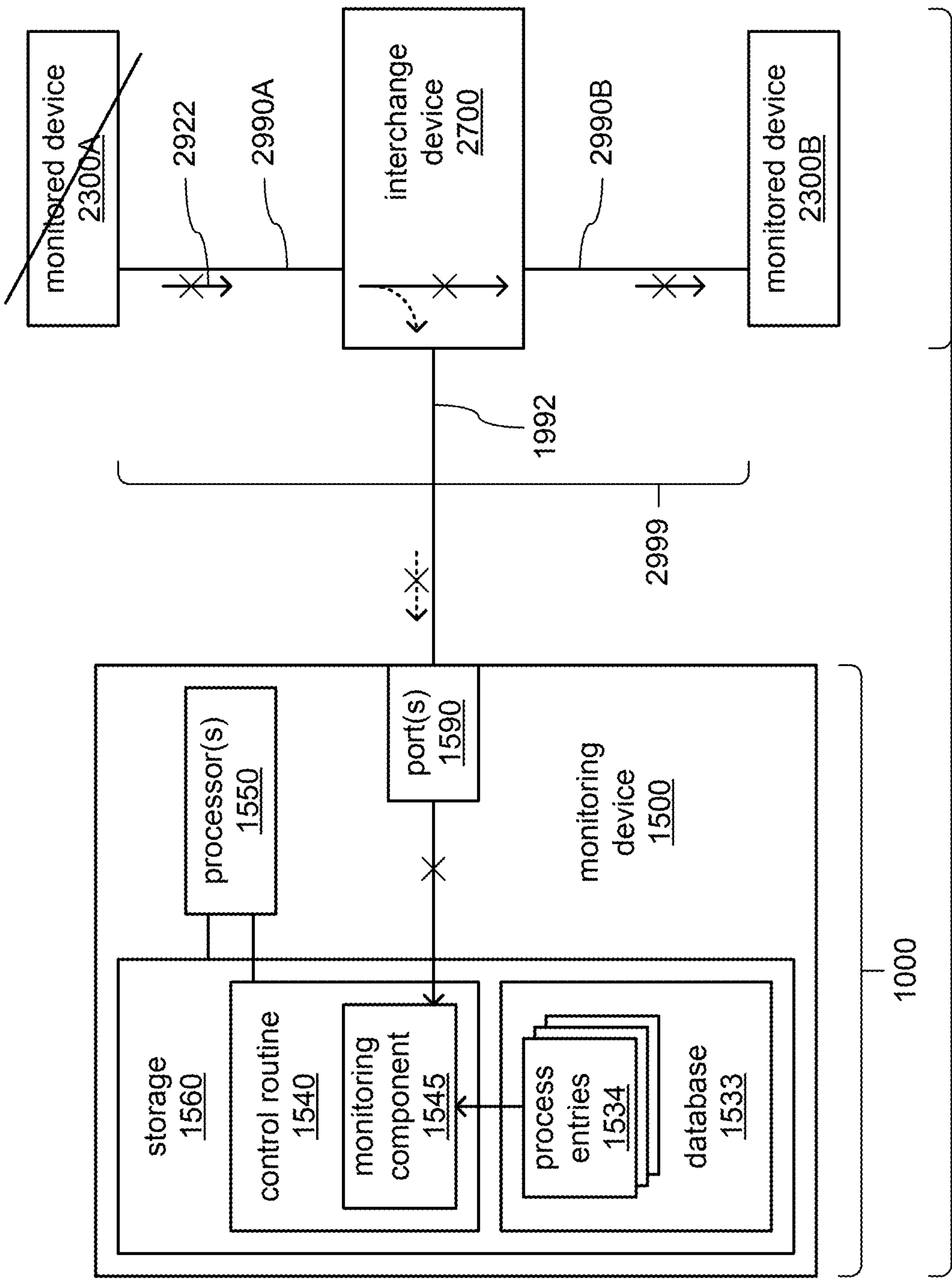
FIG. 8C


Time
process
3005x
process states 3006p
(quantity and order not visible)
non-process state
3006n
3000x
T_{nps}
T_{start}
T_{stop}
T_{start}
T_{stop}
T_{start}
T_{xmt}
T_{betw}
T_{xmt}
T_{betw}
T_{xmt}
T_{betw}
T_{xmt}
T_{betw}
transmission
3933n
transmission
3933np
transmission
3933p1
transmission
3933p2
2999



T xmt
T betw
T xmt
T betw
T xmt
T betw
T xmt
model 1539p3
type(s), commard(s), parameter(s) and/or data
transmission 3933p3
2999
model 1539p2
type(s), commard(s), parameter(s) and/or data
transmission 3933p2
model 1539p1
type(s), commard(s), parameter(s) and/or data
transmission 3933p1
type(s), commard(s), parameter(s) and/or data
transmission 3933n

FIG. 8D



monitored device
2300A
2922
2990A
interchange
device
2700
2990B
monitored device
2300B
2000
1992
2999
port(s)
1590
processor(s)
1550
monitoring
device
1500
storage
1560
control routine
1540
monitoring
component
1545
process
entries
1534
database
1533
1000

FIG. 9A

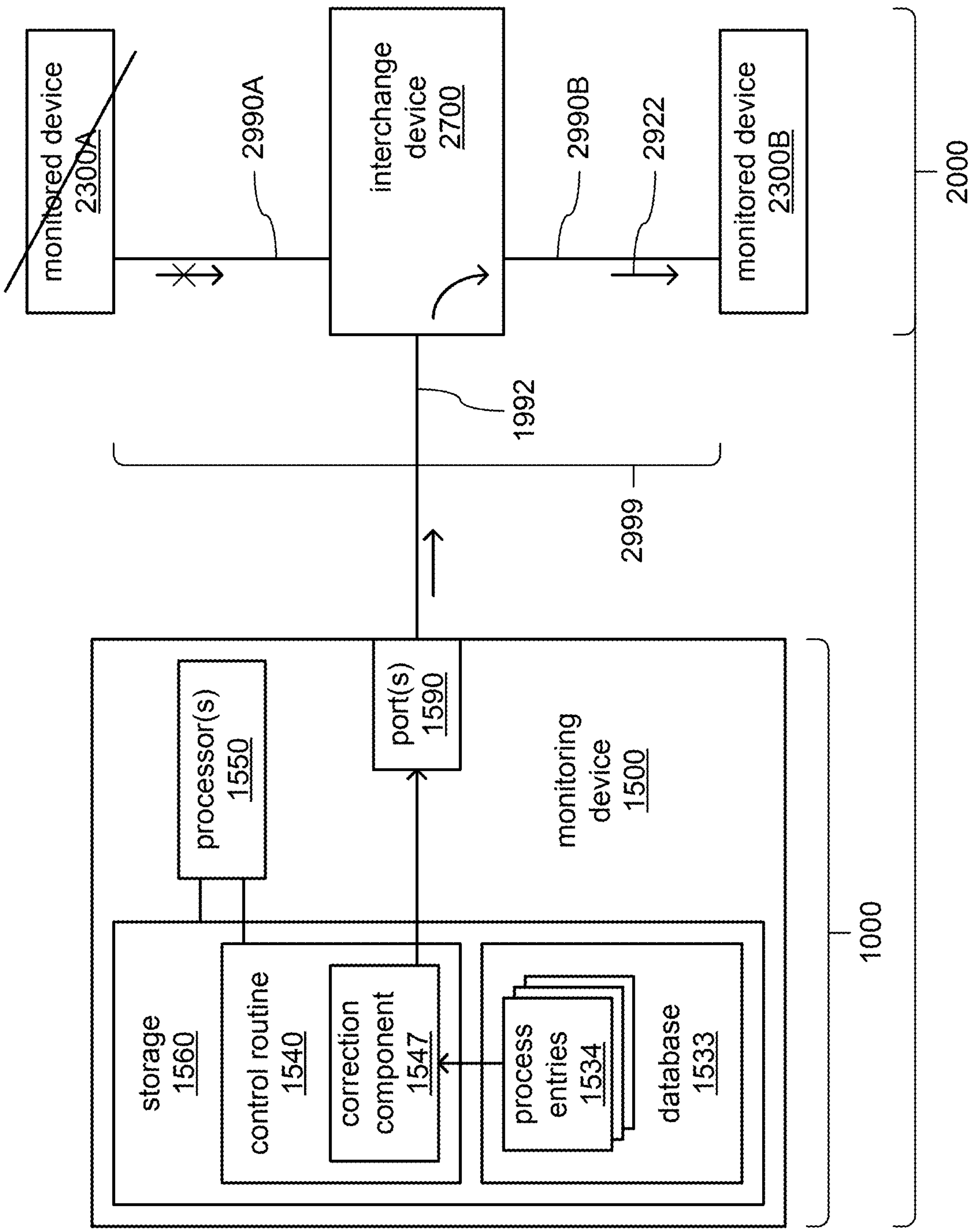

monitored device 2300A
2990A
interchange device 2700
2990B
2922
monitored device 2300B
2000
1992
2999
port(s) 1590
processor(s) 1550
monitoring device 1500
storage 1560
control routine 1540
correction component 1547
process entries 1534
database 1533
1000

FIG. 9B

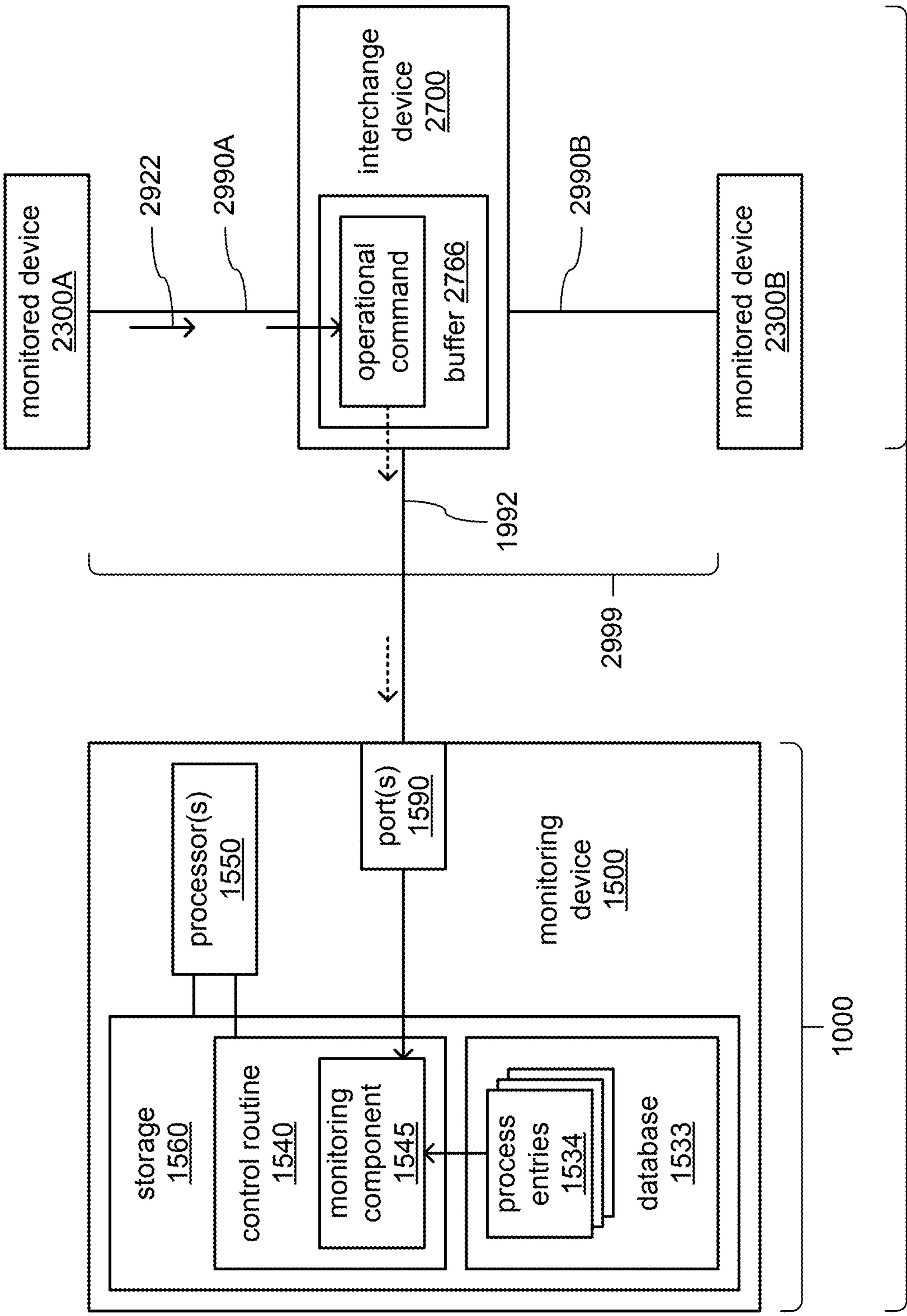

monitored device 2300A
2922
2990A
interchange device 2700
operational command
buffer 2766
2990B
monitored device 2300B
2000
1992
2999
port(s) 1590
processor(s) 1550
monitoring device 1500
storage 1560
control routine 1540
monitoring component 1545
process entries 1534
database 1533
1000

FIG. 10A

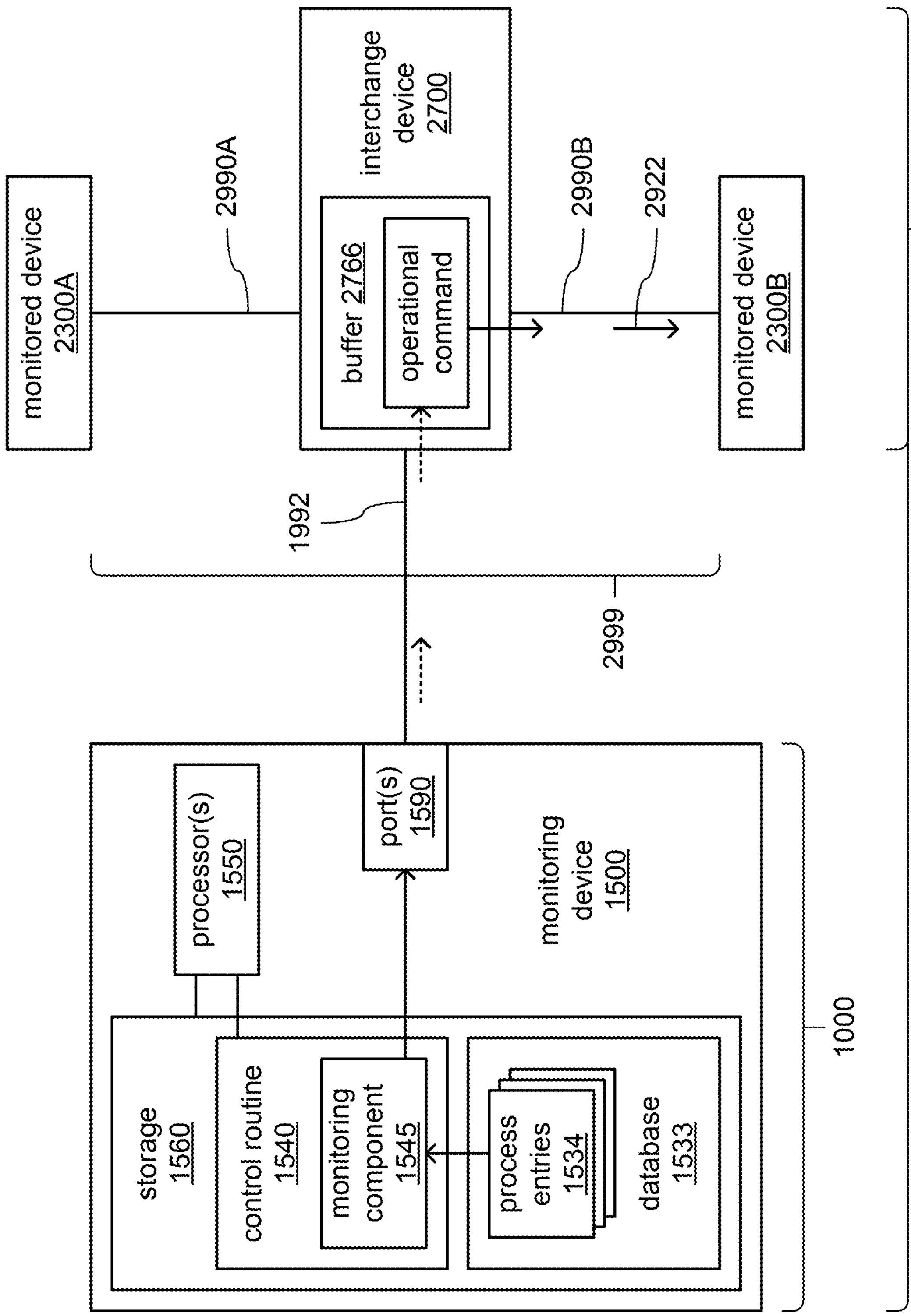

monitored device 2300A
2990A
interchange device 2700
buffer 2766
operational command
2990B
2922
monitored device 2300B
1992
2999
2000
port(s) 1590
processor(s) 1550
monitoring device 1500
storage 1560
control routine 1540
monitoring component 1545
process entries 1534
database 1533
1000

FIG. 10B

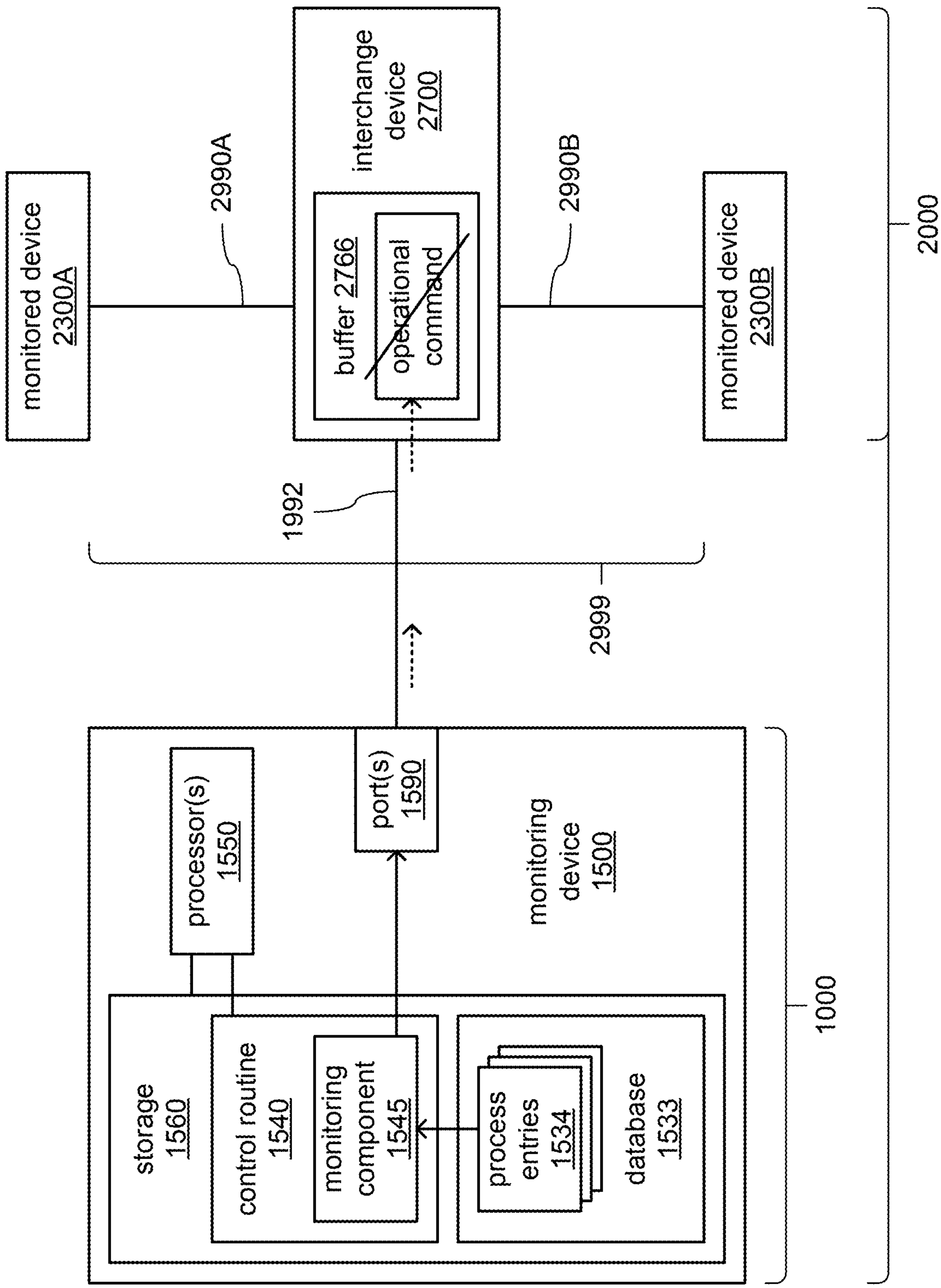

monitored device 2300A
2990A
interchange device 2700
buffer 2766
operational command
2990B
monitored device 2300B
2000
1992
2999
port(s) 1590
processor(s) 1550
monitoring device 1500
storage 1560
control routine 1540
monitoring component 1545
process entries 1534
database 1533
1000

FIG. 10C

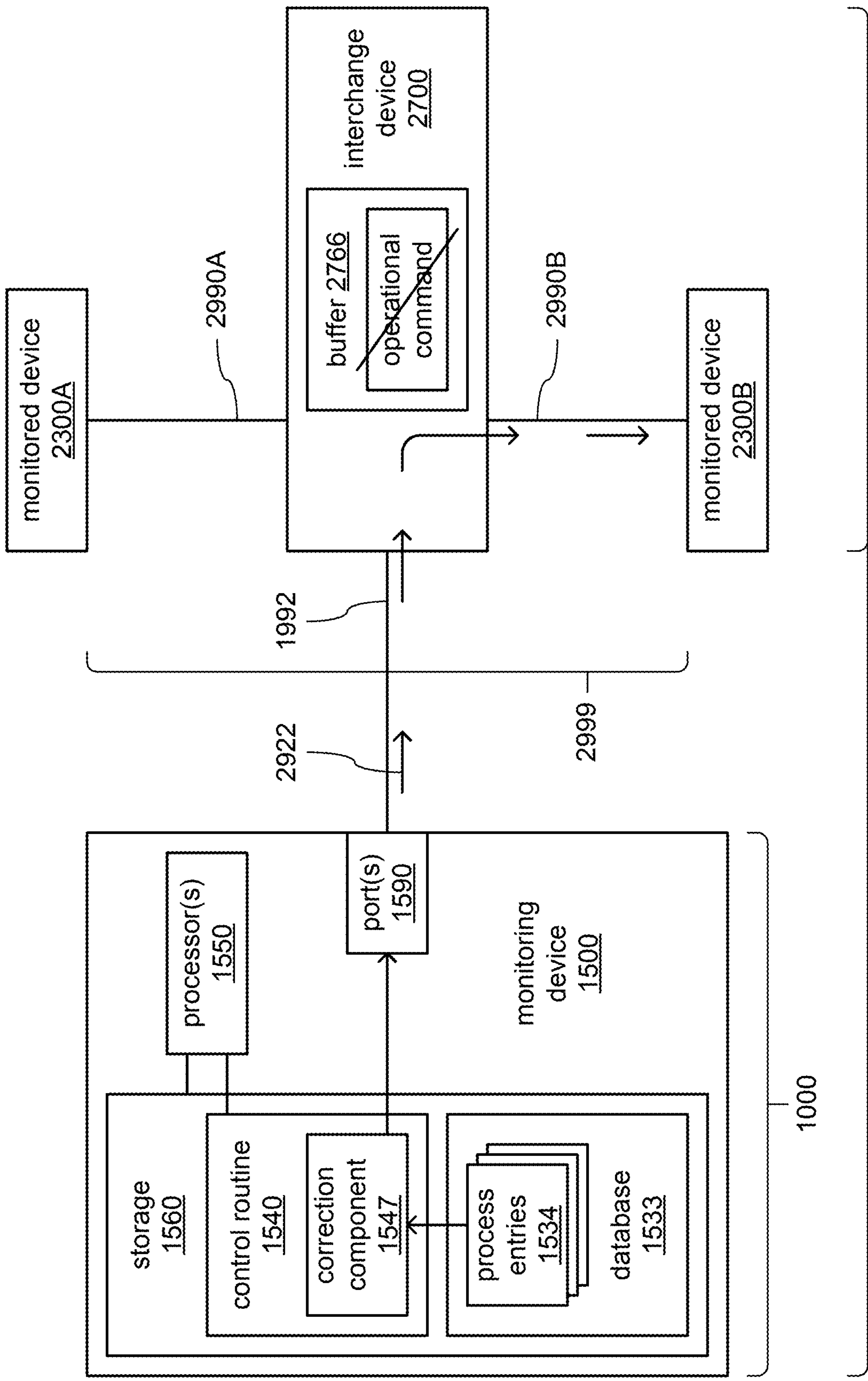

monitored device 2300A
2990A
interchange device 2700
buffer 2766
operational command
2990B
monitored device 2300B
2000
1992
2999
2922
port(s) 1590
processor(s) 1550
monitoring device 1500
storage 1560
control routine 1540
correction component 1547
process entries 1534
database 1533
1000

FIG. 10D

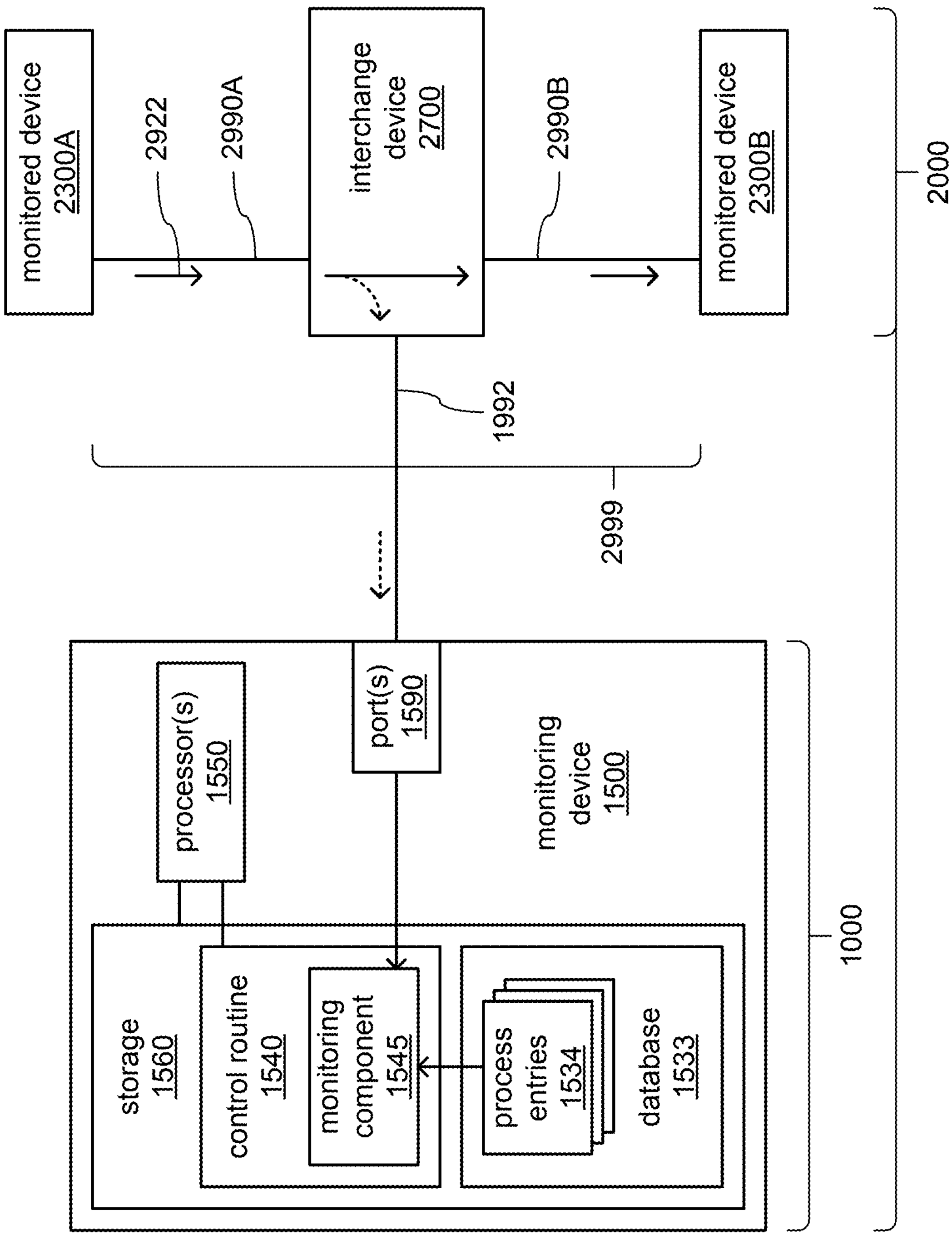

monitored device
2300A
2922
2990A
interchange
device
2700
2990B
monitored device
2300B
2000
1992
2999
processor(s)
1550
port(s)
1590
monitoring
device
1500
storage
1560
control routine
1540
monitoring
component
1545
process
entries
1534
database
1533
1000

FIG. 11A

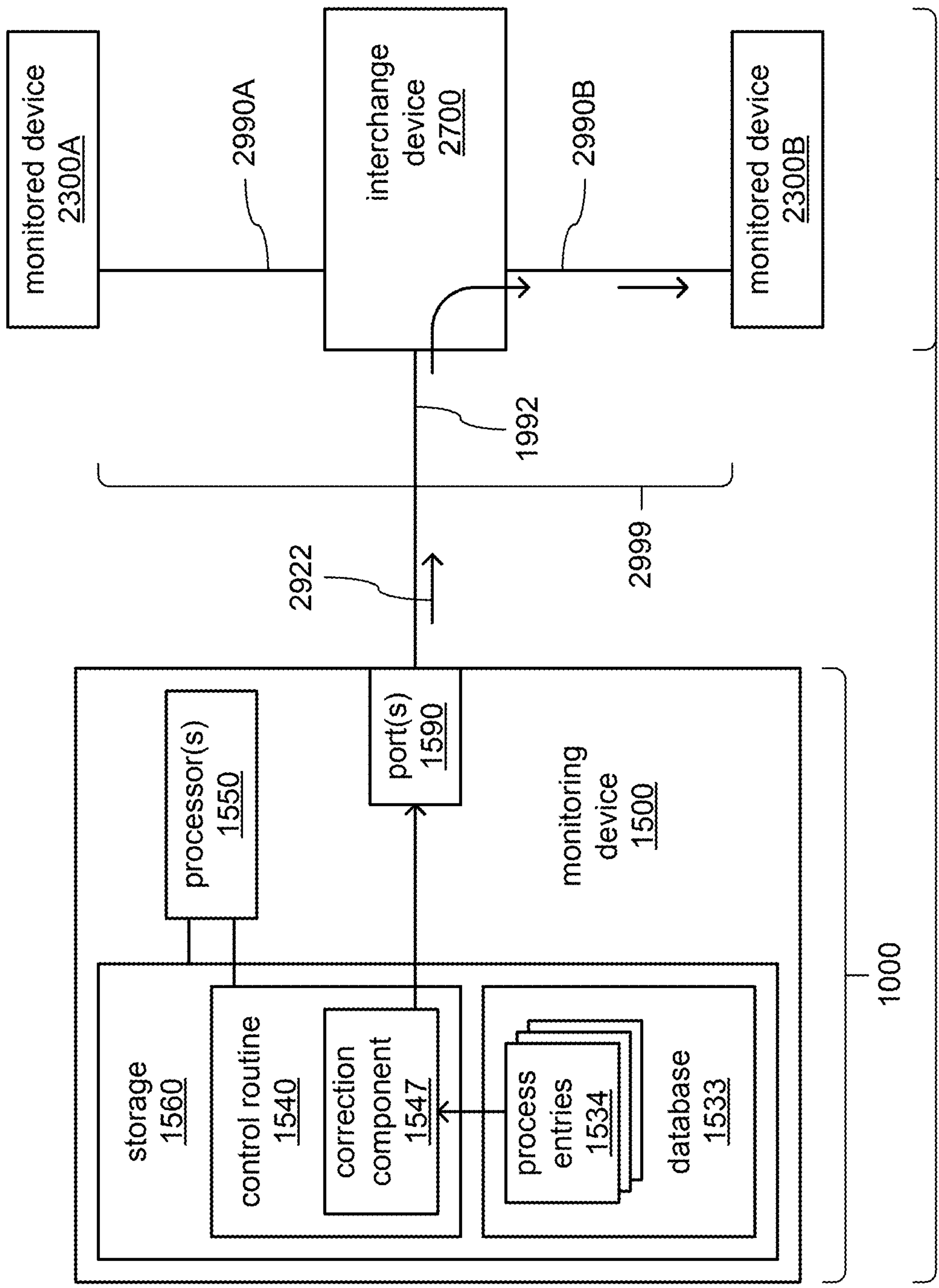

monitored device
2300A
2990A
interchange
device
2700
2990B
monitored device
2300B
2000
1992
2999
2922
processor(s)
1550
port(s)
1590
monitoring
device
1500
storage
1560
control routine
1540
correction
component
1547
process
entries
1534
database
1533
1000

FIG. 11B

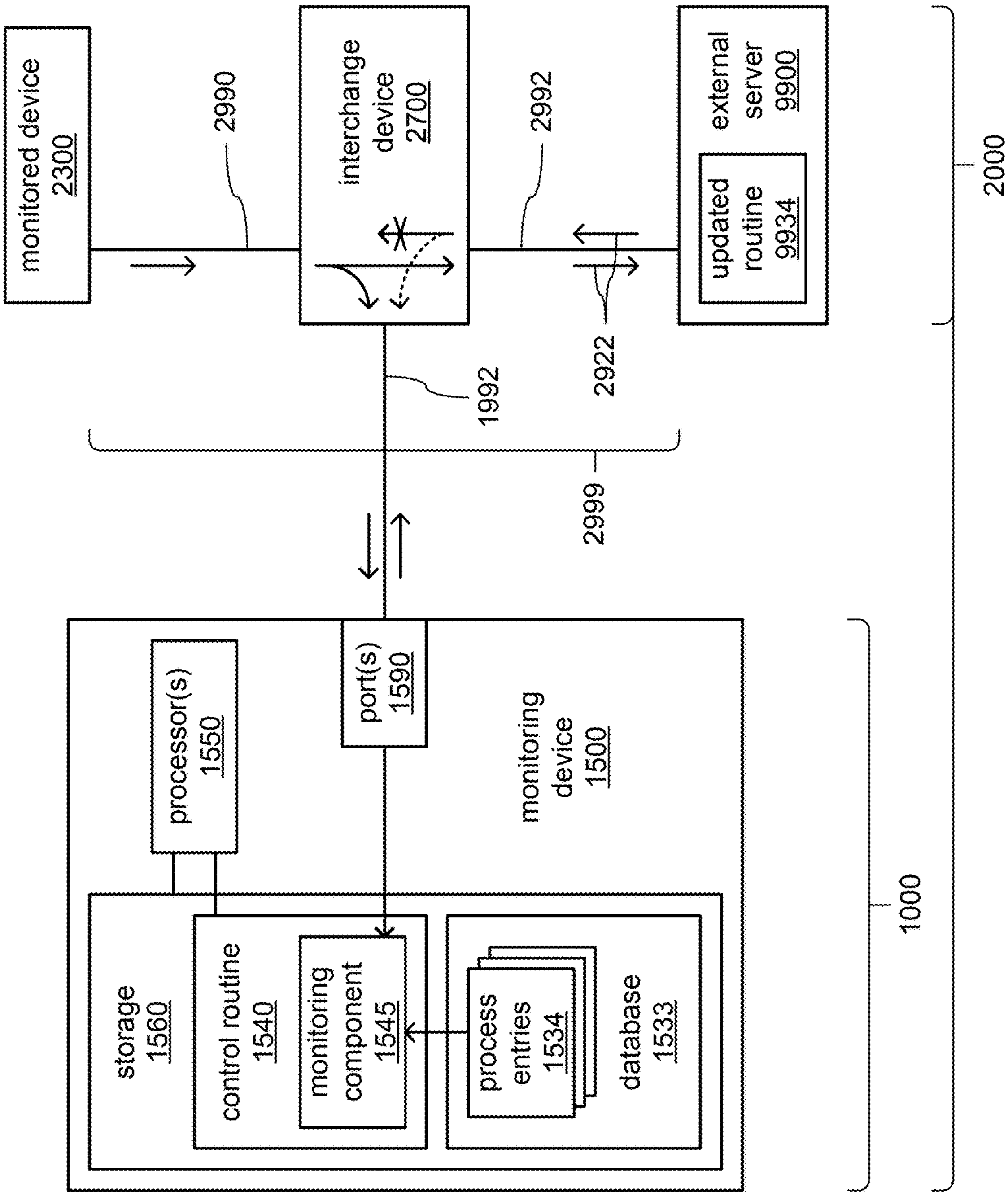

monitored device 2300
2990
interchange device 2700
2992
external server 9900
updated routine 9934
2922
1992
2999
2000
processor(s) 1550
port(s) 1590
monitoring device 1500
storage 1560
control routine 1540
monitoring component 1545
process entries 1534
database 1533
1000

FIG. 12A

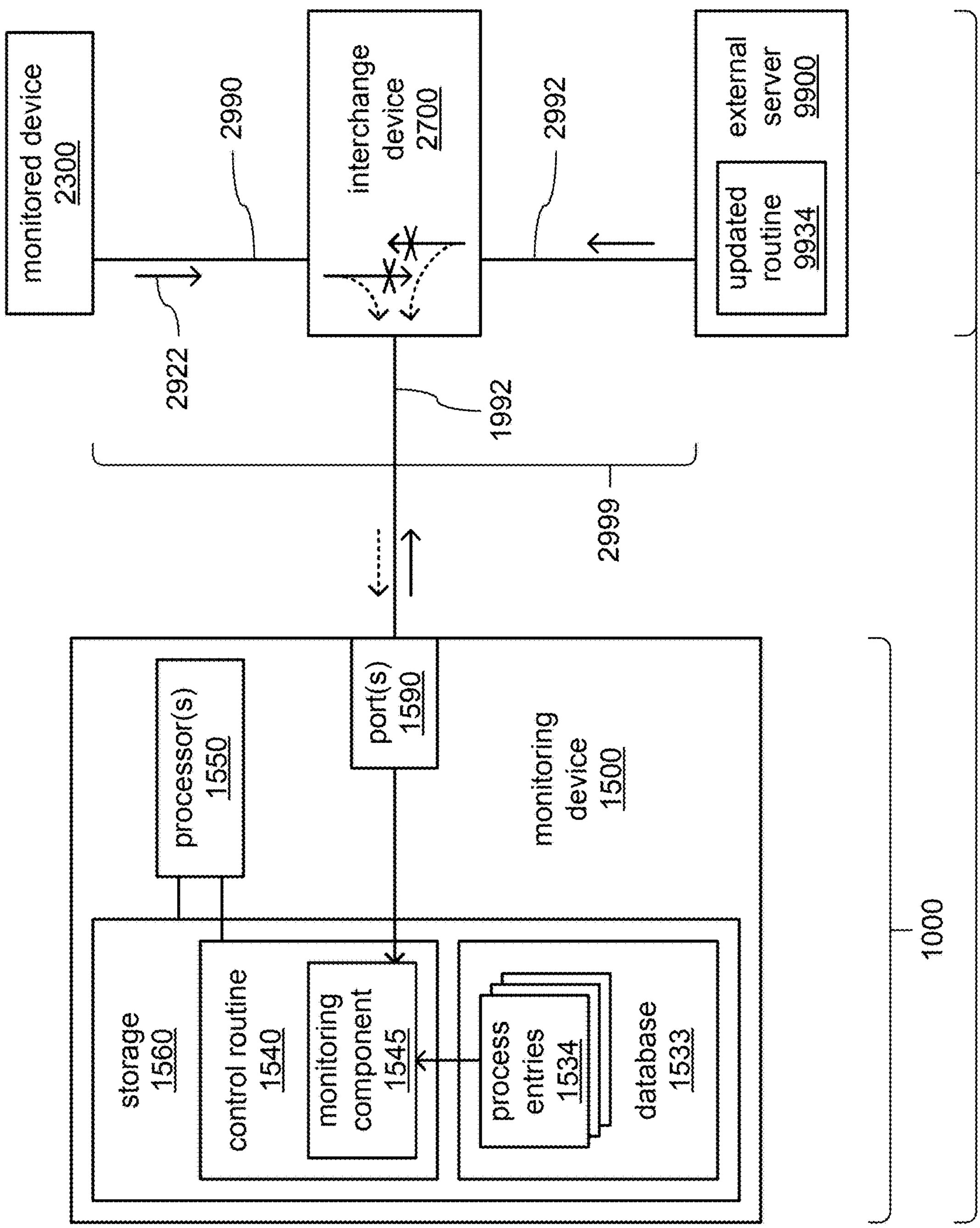

monitored device
2300
2990
interchange device
2700
2992
external server
9900
updated routine
9934
2000
2922
1992
2999
port(s)
1590
processor(s)
1550
monitoring device
1500
storage
1560
control routine
1540
monitoring component
1545
process entries
1534
database
1533
1000

FIG. 12B

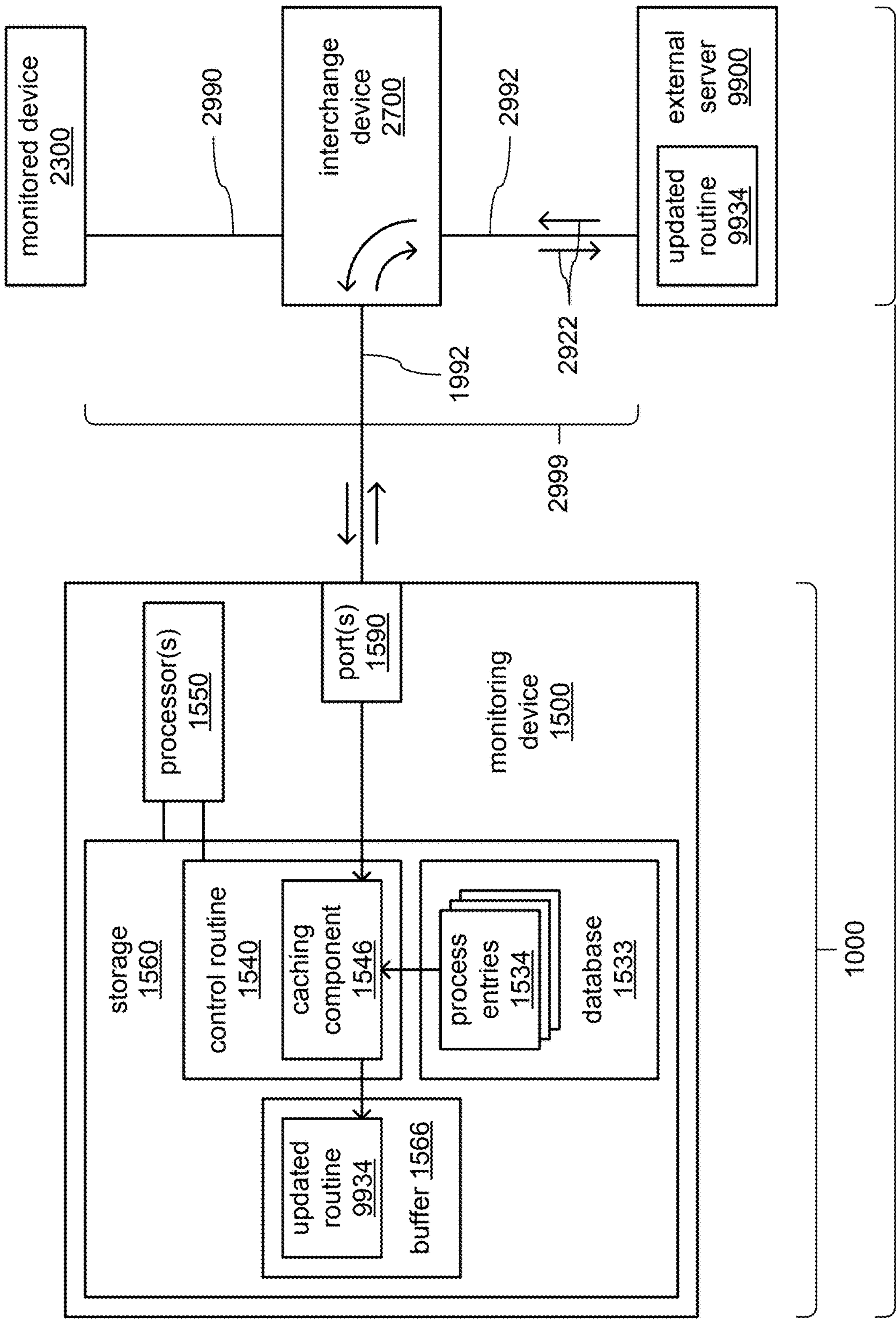
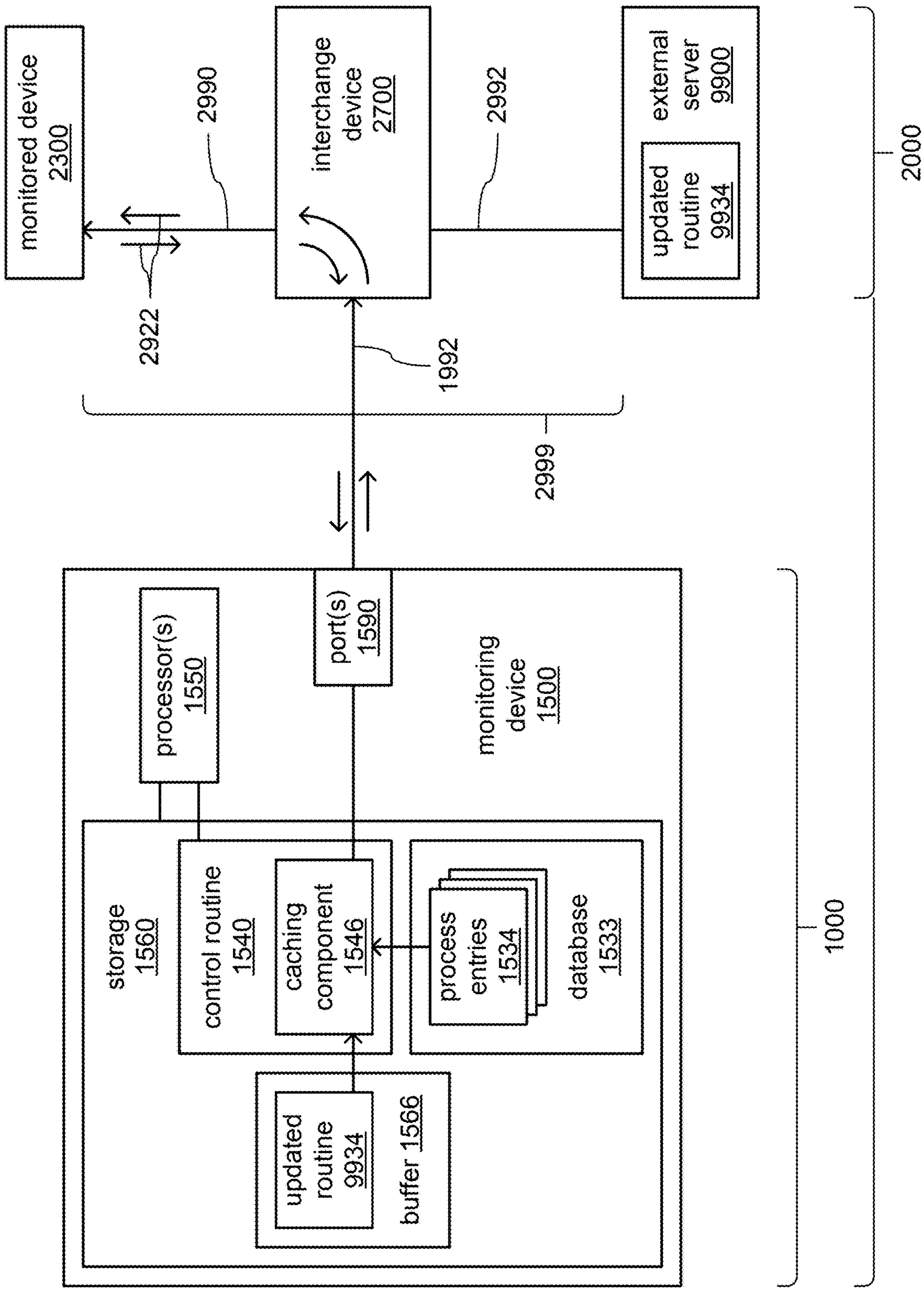

monitored device 2300
2990
interchange device 2700
2992
external server 9900
updated routine 9934
2000
2922
1992
2999
port(s) 1590
processor(s) 1550
monitoring device 1500
storage 1560
control routine 1540
caching component 1546
process entries 1534
database 1533
updated routine 9934
buffer 1566
1000

FIG. 13A



monitored device 2300
2990
interchange device 2700
2992
external server 9900
updated routine 9934
2000
2922
1992
2999
storage 1560
control routine 1540
caching component 1546
process entries 1534
database 1533
updated routine 9934
buffer 1566
processor(s) 1550
port(s) 1590
monitoring device 1500
1000

FIG. 13B

SYSTEM AND METHOD FOR ENHANCING NETWORK RELIABILITY AND COUNTERACTING CYBERSECURITY BREACH VIA SOFTWARE UPDATE

RELATED APPLICATIONS

[0001] This application is a continuation-in-part of U.S. patent application Ser. No. 18/243,165 entitled "EMULATION OF NETWORK TRAFFIC TO PREPARE A MONITORING SYSTEM FOR COUNTERACTING EFFECTS OF IMPROPER NETWORK TRAFFIC" filed Sep. 7, 2023 by Brandon Rains and Paul Williams; the disclosure of which is incorporated herein by reference for all purposes. In turn, U.S. patent application Ser. No. 18/243,165 is a continuation-in-part of U.S. patent application Ser. No. 18/206,003 entitled "SYSTEM AND METHOD FOR COUNTERACTING EFFECTS OF IMPROPER NETWORK TRAFFIC" filed Jun. 5, 2023 by Paul Williams; and is also a continuation-in-part of U.S. patent application Ser. No. 18/206,008 entitled "SYSTEM AND METHOD FOR ENHANCING COMPUTER NETWORK RELIABILITY BY COUNTERING DISRUPTIONS IN NETWORK COMMUNICATIONS" also filed Jun. 5, 2023 by Paul Williams; the disclosures of each of which are incorporated herein by reference for all purposes. In turn, each of U.S. patent application Ser. Nos. 18/206,003 and 18/206,008 is a continuation-in-part of U.S. patent application Ser. No. 17/106,060 entitled "METHOD AND SYSTEM OF DEDUCING STATE LOGIC DATA WITHIN A DISTRIBUTED NETWORK" filed Nov. 27, 2020 by Paul Williams (since issued as U.S. Pat. No. 11,711,382 on Jul. 25, 2023); the disclosure of which is incorporated herein by reference for all purposes.

[0002] This application also claims the benefit of the priority date of U.S. Provisional Application 63/700,302 entitled "SYSTEM AND METHOD FOR ENHANCING NETWORK RELIABILITY AND COUNTERACTING CYBERSECURITY BREACH VIA SOFTWARE UPDATE" filed Sep. 27, 2024 by Paul Williams and Brandon Rains; the disclosure of which is incorporated herein by reference for all purposes.

[0003] Each of U.S. patent application Ser. Nos. 18/206,003 and 18/206,008 also claims the benefit of the priority date of each of U.S. Provisional Application 63/445,654 entitled "SYSTEM AND METHOD FOR COUNTERACTING EFFECTS OF CYBER SECURITY BREACH OR OTHER DISRUPTION IN NETWORK COMMUNICATIONS" filed Feb. 14, 2023 by Paul Williams; and also claims the benefit of the priority date of U.S. Provisional Application 63/445,663 entitled "SYSTEM AND METHOD FOR ENHANCING COMPUTER NETWORK RELIABILITY AND COUNTERACTING EFFECTS OF A CYBER SECURITY BREACH" filed Feb. 14, 2023 by Paul Williams; the disclosures of each of which are also incorporated herein by reference for all purposes.

[0004] U.S. patent application Ser. No. 17/106,060 also claims the benefit of the priority date of U.S. Provisional Application 62/941,576 entitled "METHOD AND SYSTEM OF DEDUCING STATE LOGIC DATA WITHIN A DISTRIBUTED NETWORK" filed Nov. 27, 2019 by Paul Williams; the disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

1. Technical Field

[0005] The present disclosure relates to the field of cybersecurity, specifically counteracting cybersecurity beaches through fake updates to software.

2. Description of the Related Art

[0006] It has become commonplace to employ computing devices and computer-based networking technologies to control industrial processes, including and not limited to, chemical processes, automated assembly lines, and the provision of various utilities, including electric power. Unfortunately, this has opened the door to a wide variety of communications failures arising from the complexities of such technologies that may affect the control of industrial processes, thereby creating a plethora of information technology and industrial process failure scenarios. This has also opened the door to cyber attacks affecting the control of industrial processes, thereby creating the relatively new concern that cybersecurity breaches of computing devices may additionally result in the compromising of industrial processes. Accordingly, a new form of malicious activity has been created in which cybersecurity breaches are committed for the very purpose of disrupting and/or otherwise compromising industrial processes. Thus, in numerous possible ways, there may be failures in the timing of communications, failures in the correctness of the contents of communications, incorrect actions of industrial processes, and/or failure for industrial processes to operate at all.

[0007] Regarding communications failures, and by way of example, a computer server in an information technology network may cease performing its normal functions with little or no warning due to a malfunctioning or an erroneously configured hardware or software component. Alternatively, and by way of another example, a cyber attack technique such as a distributed denial of service (DDOS) attack may be directed against networked computing devices that are involved in the control of industrial processes. Such attacks may so thoroughly inundate such computing devices with network traffic as to entirely prevent them from engaging in communications related to industrial processes such that necessary transmissions of operational commands are at least significantly delayed, or simply never occur.

[0008] As will be familiar to those skilled in the art, the timing of the transmission of a particular operational command from one device to another may be as important to the correct performance of an industrial process as whether such a transmission ever occurs, at all. The failure to transmit information and/or commands when expected, or the failure to transmit information and/or commands at all, may result in portions of an information technology network or industrial process being performed for too long a period of time, being commenced at too late a time, or not being performed at all. A whole host of failures may result, including and not limited to, expensive failures cascading throughout an information technology network in an organization or failures in the successful production of products or successful provision of services, damage to equipment used to produce products or provide services, creation of hazardous conditions where industrial processes are performed, injuries and/or fatalities among personnel involved in performing industrial processes, and/or reputational and/or financial damage to corporate entities and/or other entities associated with information technology network or industrial processes. Still further, where an industrial process is part of a chain of related industrial processes, such compromising of the performance of one industrial process may adversely affect the ability to perform one or more preceding industrial processes, and/or one or more subsequent industrial processes.

[0009] Regarding incorrect communications content, and by way of an example, a cybersecurity breach may occur in which a computing device involved in the control of an industrial process may be successfully compromised by malicious software such that it becomes remotely controllable for the purpose of disrupting that industrial process. It may be that such a compromised computing device is caused to cease issuing proper operational commands in a proper order with proper timings and/or parameters, and instead, it may be caused to issue improper operational commands that cause improper operation of robotic arms/actuators, conveyor motors, power relays, welding devices, valves, heating and/or cooling units, etc. Such devices may be caused to operate outside of equipment design limitations, and/or with incorrect timings that violate operational and/or safety requirements. Beyond simply impairing the correct performance of an industrial process, the results could include wasting raw materials, releasing and/or spilling hazardous materials, destroying sub-assemblies, damaging equipment and/or facilities, and/or placing personnel in physical danger.

[0010] The malicious operational commands that are so issued may be altered versions of proper operational commands such that the operational commands may actually be issued when they are expected to be issued, but may include incorrect parameters that specify one or more incorrect values, such as an incorrect device identifier, temperature, pressure, direction of movement, extent of movement, countdown time, upper or lower limit, etc. Alternatively or additionally, the malicious operational commands actually may be proper operational commands for the industrial processing being performed, but they may be issued at improper times and/or out of proper sequence. As still another alternative, it may simply be that entirely new malicious operational commands are issued without any connection to the content and/or timing of proper operational commands.

[0011] Among the increasingly frequent varieties of cybersecurity attack is the provision of fake updates to software relied upon for the basic functionality of processing systems, including firmware and/or device drivers that may be associated with lower level functionality of hardware components and/or operating systems, operating system and/or security software components, application routines and/or support libraries, etc. As will be familiar to those skilled in the art, the increasing complexity of the software used within processing systems (e.g., processing systems involved in controlling industrial processes) often necessitates the repeated updating of various software components to address mistakes found therein, to increase the efficiency and/or performance thereof, to add support for new hardware components, and/or to add new functionality.

[0012] Unfortunately, and as will also be familiar to those skilled in the art, it has become commonplace practice for such software updates to be obtained through the Internet from servers operated by a wide variety of software suppliers at many different locations around the world. With so many different software suppliers, an extremely wide variety of software download and security measures of widely varying degrees of effectiveness have become employed such that there is no such thing as a single standard procedure (or just a relatively small quantity of standard procedures) for identifying and/or obtaining software updates. Additionally, as software suppliers change over time, merge over time, are acquired over time, change management over time, etc., such procedures are likely to change, as well as the physical locations and/or network address locations of those servers.

[0013] A similarly dizzying and ever-changing variety of software-based tools are often provided by the manufacturers of computing devices for use in enabling manually-controlled or automated accessing and downloading of software updates from such servers. And such software tools are likely to also need to be updated over time, themselves, as the aforedescribed changes to such servers occur over time. As a result, each provider of software updates and/or each software tool that is so provided may employ its own unique set of procedures for reaching out to a server, checking for whether there is a software update for a particular software routine, downloading a software update, checking its integrity, etc. In turn, this leads to similarly unique sets and/or sequences of operational commands and/or data being transmitted among devices through an internal network and/or through the Internet.

[0014] This chaotic situation has proven to be an irresistible invitation for those seeking to compromise processing devices (including those involved in controlling industrial processes) by creating fake servers, and/or using other measures, to cause fake versions of software updates to be downloaded and used. In this way, various malicious measures such as "back doors" are introduced into processing devices, thereby providing malicious persons with the ability to access those processing systems. As will be familiar to those skilled in the art, such access may enable such malicious persons to access and steal confidential information, and/or to take control of an industrial process and/or other important process to cause damage, inflict downtime and/or to commit a wide variety of types of theft.

[0015] As those skilled in the art will readily recognize, it is often inevitable that some degree of software updating from external sources for those updates is necessary for the ongoing proper operation of processing devices as they perform their various functions. Thus, it is often not practical and/or not possible to avoid all risk of downloading and using fake software updates that introduce malicious code by simply preventing all software updates. It is also often not practical and/or not possible to directly manually control the accessing and downloading of each software update, as there may be far too many software updates occurring on far too frequent a basis within a particular organization and/or within a particular facility of an organization.

[0016] A similar concern exists regarding software updates that may be provided by an internal source. It is not uncommon for personnel associated with the maintenance of processing devices within an organization to create an internal repository of software updates that may be deemed to be trusted. It may be that such software updates have been, in some way, verified and/or tested to confirm their legitimacy such that they are deemed to not pose a threat when downloaded and used within other processing devices within that organization. However, such internal methods of verification, even if effective in avoiding the introduction of malicious software, often cannot prevent the introduction of coding errors and/or other non-malicious issues.

[0017] The use of cyber attack techniques in such attacks on industrial processes often begets the temptation to focus on using longstanding cybersecurity measures to counter them. Such longstanding cybersecurity measures include, and are not limited to, the use of various malicious code digital signatures to detect 1) the transmission, receipt and/or storage of particular sequences of executable instructions of malicious pieces of software, 2) the transmission of particular malicious combinations of operational commands across a network by a computing device, and/or 3) the performance of particular malicious combinations of actions by a computing device. Such approaches have been useful in attempting to prevent the infiltration and/or execution of malicious software, and/or halting further execution of malicious software. However, as will be familiar to those skilled in the art, such approaches often set up a form of "arms race" between developers of malicious software and developers of the signatures used in such detection.

[0018] Unavoidably, there is a delay between the deployment of new malicious software and/or other varieties of attacks, and the development of corresponding defensive measures (e.g., signatures) such that it is inevitable that at least some of such software and/or attacks will be successful in causing harm before being detected. As a result, such approaches usually do little to address the harm done to industrial processes in situations where malicious software or other unexpected and/or nefarious activity is not yet detected until after some amount of damage has been underway for at least some amount of time.

[0019] An additional issue is the disruption to the normal functioning of computing devices that is often caused by the introduction of various longstanding cybersecurity measures, including to the function of controlling an industrial or other process. Such measures usually entail the installation of new cyber security software on existing computing devices, thereby consuming resources of those devices. It is not unheard of for a computing device to require enhancements to processing and/or storage resources, or to require being replaced with another computing device having such enhancements in plate, to accommodate the consumption of resources by cyber security software. Indeed, as a result of such issues, it is not uncommon for the installation of cybersecurity software on computing devices to be regarded as a case where the proverbial cure is worse than the proverbial disease. Additionally, it is also not unheard of for the installation of cyber security software and/or the replacement of computing devices to cause changes in behavior in controlling a process. More specifically, there may be slight alterations to the timing and/or ordering of the transmission of operational commands and/or data that may seem innocuous, but which beget unforeseen deleterious results as a result of triggering an unforeseen and/or unknown quirk of the logic employed in controlling a process.

[0020] Further, such installations often require some degree of adjustment in distinguishing between undesired events arising from failures and/or cyber security attacks, and events that normally occur during normal functioning associated with controlling a process. Such adjustments to address missed conditions and/or false alarms can further compound disruptions, and may still further become another avenue by which unforeseen deleterious results are triggered.

[0021] The present invention addresses these and other drawbacks of the prior art by providing a unique approach to detecting and optionally mitigating the effects of such events on computing devices and/or communications among computing devices involved in the performance of industrial processes and other repetitive loop processes. The present invention also addresses these and other drawbacks of the prior art by providing a unique approach to thoroughly preparing a monitoring system to perform such detection and mitigation prior to deployment.

BRIEF SUMMARY

[0022] Techniques are described for providing a system of one or more devices that implements a method for counteracting the effects of ongoing attempts to introduce fake firmware or software into a controlled environment.

[0023] A monitoring system includes: a processor configured to perform operations including: place the monitoring system into a training mode to generate a model of a software update process; receive indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system, wherein at least one of the multiple monitored devices is configured to initiate the software update process; and from the received indications of observed transmissions of operational commands or operational information associated with the software update process, generate the model of the software update process, wherein the model includes indications of an expected order of transmissions of operational commands or operational information associated with the software update process.

[0024] A monitoring system includes a processor configured to perform operations including: place the monitoring system into an operating mode to use a model of a software update process to analyze observed transmissions of operational commands or operational information associated with the software update process; receive, from one or more interchange devices, indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system, wherein at least one of the monitored devices is configured to provide an updated routine in the software update process; and compare the received indications of observed transmissions of operational commands or operational information associated with the software update process to indications in the model of expected transmissions of operational commands or operational information associated with the software update process to determine whether a particular transmitted operational command is a proper operational command associated with the software update process or to determine whether particular transmitted operational information is proper operational information associated with the software update process.

[0025] A method of generating a model of a software update process includes: receiving, by a processor of a monitoring system, and from an interchange device of a monitored system, indications of observed transmissions, through the interchange device, of operational commands or operational information among multiple monitored devices of the monitored system; and from the received indications of observed transmissions of operational commands or operational information associated with the software update process, generating, by the processor, the model of the software update process, wherein the model includes indications of an expected order of transmissions of operational commands or operational information associated with the software update process.

[0026] A method of using a model of a software update process to analyze observed transmissions of operational commands or operational information associated with the software update process includes: receiving, by a processor of a monitoring system, and from at least one interchange device of a monitored system, indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system; and comparing, by the processor, the received indications of observed transmissions of operational commands or operational information associated with the software update process to indications in the model of expected transmissions of operational commands or operational information associated with the software update process to determine whether a particular transmitted operational command is a proper and expected operational command associated with the software update process, or to determine whether the particular transmitted operational information is proper and expected operational information associated with the software update process.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The disclosure will be better understood and when consideration is given to the drawings and the detailed description which follows. Such description makes reference to the annexed drawings wherein:

[0028] FIGS. **1**A, **1**B, **1**C, **1**D, **1**E and **1**F, together, provide block diagrams of an example embodiment of a combination of a monitored system, a monitoring system to monitor network activity of the monitored system, and an emulation system to prepare the monitoring system for use.

[0029] FIGS. **2**A, **2**B and **2**C present details of example embodiments of the monitoring system monitoring network activity within the monitored system.

[0030] FIGS. **3**A, **3**B, **3**C, **3**D and **3**E, together, present details of an example embodiment of interactions among components of the monitored system and an external system in controlling a process performed within the external system.

[0031] FIGS. **4**A, **4**B and **4**C present details of example embodiments of interactions of components of the monitored system and of an external system with external servers in update download processes.

[0032] FIGS. **5**A, **5**B and **5**C, together, provide a more detailed presentation of aspects of a gathering phase of preparing the monitoring system for use—data is gathered to serve as part of the basis for generating emulations of behaviors of components of the monitored system, of an external system and/or of a process.

[0033] FIGS. **6**A, **6**B, **6**C, **6**D, **6**E and **6**F, together, provide a more detailed presentation of aspects of a pre-training phase of preparing the monitoring system for use—the gathered data is used to define behaviors and/or timings of the emulations.

[0034] FIGS. **7**A and **7**B, together, provide a more detailed presentation of aspects of an emulation training phase of preparing the monitoring system for use—the emulations are used to train, test and/or demonstrate abilities of the monitoring system.

[0035] FIGS. **8**A, **8**B, **8**C and **8**D, together, provide an example alternate embodiment of preparing the monitoring system for use that may be employed for supplemental training of the monitoring system, and/or that may be employed where information concerning process control logic is unavailable.

[0036] FIGS. **9**A and **9**B, together, present details of an example embodiment of counteracting the effects of a lack of transmission of an expected operational command.

[0037] FIGS. **10**A, **10**B, **10**C and **10**D, together, present details of an example embodiment of counteracting the effects of a transmission of an improper operational command.

[0038] FIGS. **11**A and **11**B, together, present details of another example embodiment of counteracting the effects of a transmission of an improper operational command.

[0039] FIGS. **12**A and **12**B, together, present details of an embodiment of using the monitoring system to counteract improper network activity in an update download process.

[0040] FIGS. **13**A and **13**B, together, present details of an embodiment of using the monitoring system to control the performance of an update download process.

DETAILED DESCRIPTION

[0041] In the following detailed description, reference is made to the accompanying drawings that form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

[0042] Broadly speaking, in a form of machine learning, observations of actual network traffic associated with the performance of an industrial process and/or an update download process are used as at least one basis to generate a model of that process and its associated network activity. That model is then used to identify when that process is being performed, and to identify instances of improper network activity that does not conform to what has been learned to be expected network activity. Each such identified instance of improper network activity may then be corrected to better conform to that expected network activity and/or may then be responded to in other ways to prevent the process from going awry.

[0043] A monitoring system includes: a processor configured to perform operations including: place the monitoring system into a training mode to generate a model of a software update process; receive indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system, wherein at least one of the multiple monitored devices is configured to initiate the software update process; and from the received indications of observed transmissions of operational commands or operational information associated with the software update process, generate the model of the software update process, wherein the model includes indications of an expected order of transmissions of operational commands or operational information associated with the software update process.

[0044] A monitoring system includes a processor configured to perform operations including: place the monitoring system into an operating mode to use a model of a software update process to analyze observed transmissions of operational commands or operational information associated with the software update process; receive, from one or more interchange devices, indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system, wherein at least one of the monitored devices is configured to provide an updated routine in the software update process; and compare the received indications of observed transmissions of operational commands or operational information associated with the software update process to indications in the model of expected transmissions of operational commands or operational information associated with the software update process to determine whether a particular transmitted operational command is a proper operational command associated with the software update process or to determine whether particular transmitted operational information is proper operational information associated with the software update process.

[0045] A method of generating a model of a software update process includes: receiving, by a processor of a monitoring system, and from an interchange device of a monitored system, indications of observed transmissions, through the interchange device, of operational commands or operational information among multiple monitored devices of the monitored system; and from the received indications of observed transmissions of operational commands or operational information associated with the software update process, generating, by the processor, the model of the software update process, wherein the model includes indications of an expected order of transmissions of operational commands or operational information associated with the software update process.

[0046] A method of using a model of a software update process to analyze observed transmissions of operational commands or operational information associated with the software update process includes: receiving, by a processor of a monitoring system, and from at least one interchange device of a monitored system, indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system; and comparing, by the processor, the received indications of observed transmissions of operational commands or operational information associated with the software update process to indications in the model of expected transmissions of operational commands or operational information associated with the software update process to determine whether a particular transmitted operational command is a proper and expected operational command associated with the software update process, or to determine whether the particular transmitted operational information is proper and expected operational information associated with the software update process.

High Level Overview of Systems and Interplay Among Systems

[0047] FIGS. **1**A, **1**B, **1**C, **1**D, **1**E and **1**F, taken together, provide a high level presentation of an example of a monitoring system **1000** being prepared, and then being deployed, to monitor transmissions of operational commands and/or operational information within a monitored system **2000** for purposes of a controlling aspects of an industrial or other type of process performed within an external system **3000**. The particular external system **3000** that is so controlled may be one of multiple external systems **3000** that define an external domain **4000** by which products may be produced; utility services may be monitored, controlled and/or provided; etc. An emulation system **5000** that emulates behaviors of the monitored system **2000** and/or of the particular external system **3000** may be employed in training and/or testing the monitoring system **1000** as part of the monitoring system **1000** for its deployment. Additionally, one or more external servers **9000** may, from time to time, provide updated versions of various routines that are executed by devices of the monitored system **2000** and/or the particular external system **3000**. Such updated versions of such routines may, on occasion, change the behaviors of the monitored system **2000** and/or of the particular external system **3000** such that occasional retraining of the monitoring system **1000** may be needed.

[0048] FIG. **1**A introduces components of each of the systems **1000**, **2000**, **3000**, **4000** and **5000**; FIGS. **1**B-D, together, depict aspects of using the monitoring system **1000** to monitor and/or correct communications traffic of the monitored system **2000** after being prepared for such use; and FIGS. **1**E-F, together, depict aspects of preparing the monitoring system **1000** for such use.

[0049] Turning to FIG. **1**A, in some embodiments, the monitored system **2000** may include a variety of computing devices interconnected by an internal network **2999**. Among such computing devices may be multiple monitored devices **2300** that are involved in controlling the industrial process or other type of process that occurs within the one of the external systems **3000** that is depicted as being coupled to one of the monitored devices **2300**. As part of being used together in a cooperative manner to control that process, various operational commands or information may be transmitted through the internal network **2999** among at least a subset of those monitored devices **2300**.

[0050] For purposes of this patent application, it is important to note that the use of such terms as "monitoring" and "monitored" refer to the monitoring, by the monitoring system **1000**, of communications through the internal network **2999** and among the monitored devices **2300** of the monitored system **2000** as part of providing reliability and/or cyber security services for the monitored system **2000**. Such terms also refer to the monitoring of communications through the internal network **2999** and between the monitored devices **2300** and one or more external servers **9900** reachable through an external network **9992** (e.g., the Internet). Thus, these terms do not refer to the use of the monitored system **2000** to control of any industrial process or other type of process performed within an external system **3000**.

[0051] Also among the computing devices connected to the internal network **2999** may be one or more unmonitored devices **2100** where none of the communications therewith through the internal network **2999** are monitored by the monitoring system **1000**. It may be that at least a subset of the unmonitored device(s) **2100** does engage in communications with at least a subset of the multiple monitored devices **2300** through the internal network **2999**. However, it may also be that none of those communications are associated with controlling an industrial process or other type of process performed within one of the external systems **3000**, and therefore, it may be deemed unnecessary to monitor those communications.

[0052] As depicted, the internal network **2999** may include one or more interchange devices **2700**. Each such interchange device **2700** may be any of a variety of types of network device, including and not limited to, a router, network switch, wireless network access point, network bridge, etc. In the internal network **2999**, it may be that each transmission of operational command(s) and/or operational information among devices coupled to the internal network **2999** is transmitted through at least one interchange device **2700**.

[0053] Also as part of controlling an external system **3000**, and as depicted, at least one of the monitored devices **2300** may be coupled to that external system **3000** by one or more communications links **3999** to both monitor and control various aspects of a process that is performed with that external system **3000**. Thus, in controlling such a process, there may be transmissions between such a monitored device **2300** and such an external system **3000** through the one or more links **3999**, in addition to there being transmissions among multiple monitored devices **2300** through the internal network **2999**. Depending on the particular technologies employed by the link(s) **3999**, one or more relay devices **3700** may be incorporated into the one or more links **3999**.

[0054] In view of their central role in conveying transmissions through the internal network **2999**, it may be that monitoring transmissions of operational commands and/or operational information among the monitored devices **2300** entails monitoring network traffic passing through each of the one or more interchange devices **2700**. To do so, at least one of the one or more interchange devices **2700** may be coupled to the monitoring system **1000** via a link **1992**. Similarly, such monitoring through each of the one or more interchange devices **2700** may also be applied to the monitoring of transmissions of operational commands and/or operational information between the monitored devices **2300** and one or more external servers **9900** reachable via the external network **9992**.

[0055] As depicted, the monitoring system **1000** may include one or more monitoring devices **1500**, and each of the one or more monitoring devices **1500** may be still another computing device. It may be that the monitoring system **1000** is remotely located from the monitored system **2000** such that it may be deemed to be entirely separate therefrom. Alternatively, it may be that the monitoring system **1000** is co-located with at least a portion of the monitored system **2000**, and/or is otherwise integrated with the monitored system **2000**, such that it may be deemed to be included therein.

[0056] The preparation of the one or more monitoring devices **1500** for use in monitoring the monitored system **2000** may include three distinct phases. In a gathering phase, one or more gathering devices **5700** of the emulation system **5000** may be temporarily coupled to the interchange device (s) **2700** by one or more links **5992** to observe and capture communications activity occurring on the internal network **2999**. This may include activity among the monitored devices **2300**, and/or activity between the monitored devices **2300** and one or more external servers **9900**. Also, at least one gathering device **5700** may be additionally temporarily coupled to the relay device(s) **3700** and/or to at least one monitoring device **2300** by one or more links **5993** to observe and capture communications activity occurring on the link(s) **3999**. This may include activity between one or more of the external systems **3000** and one or more external servers **9900**, which may be routed through one or more of the monitored devices **2300**, as well as through both the internal network **2999** and at the link(s) **3999**. In so doing, the gathering device(s) **5700** may generate observation data descriptive of the activity captured from at least the internal network **2999**, if not also from the link(s) **3999**.

[0057] The communications activity that is captured may include network activity associated with an industrial process (or other type of process) that occurs within an external system **3000**, and is under the control of monitored devices **2300**. Thus, such network activity may include operational commands and operational information exchanged with an external system **3000** to the performance of a process occurring therein at a relatively finely detailed and step-by-step level, as well as operational commands and operational information exchanged between monitored devices **2300** to coordinate the control of that process at a somewhat higher level. However, the communications activity that is captured may also include network activity associated with a single monitored device **2300** or a single external system **3000** retrieving an updated version of a software routine from an external server **9900**. Thus, such network activity may include operational commands and operational information exchanged with an external server **9900** for authentication, to query whether there is an updated version of a particular software routine currently available, to coordinate a download of such an updated routine as multiple blocks of data conveyed through a series of transmission, and/or to check the integrity of such an updated routine following its download.

[0058] Following the gathering phase, in a pre-training phase, the gathering device(s) **5700** may be uncoupled from the interchange device(s) **2700**, the relay device(s) **3700** and/or the at least one monitored device **2300**, and the observation data generated by the gathering device(s) **5700** may be provided to the one or more emulation devices **5500**. In so providing the observation data to the emulation device (s) **5500**, the gathering device(s) **5700** may be temporarily coupled thereto through one or more links of a training network **5999**. The emulation device(s) **5500** may be used to combine the observation data with other data concerning various aspects of the monitored device(s) **2300**, various aspects of the external system(s) **3000**, and/or various aspects of the process(es) being performed within the monitored device(s) **2300** and/or the external system(s) **3000**. Such a combining of information may be used to generate and/or tune aspects of emulations of the monitored device(s) **2300**, the external system(s) **3000**, and/or the process(es).

[0059] Following the pre-training phase, in an emulation training phase, the emulation device(s) **5500** may be temporarily coupled to the monitoring device(s) **1500** through one or more links **5991** in a manner that is meant to mimic the manner in which the monitoring device(s) **1500** are to be coupled to the monitored system **2000** after the monitoring device(s) **1500** have been prepared for use. The emulation device(s) **5500** may then generate an emulation of behaviors of a combination of the monitored system **2000**, the external system(s) **3000**, and/or a process as part of providing the monitoring device(s) **1500** with an emulation of corresponding communications activity on the internal network **2999** for use in training and/or testing the monitoring device(s) **1500**.

[0060] Thus, through such use of emulation, the monitoring device(s) **1500** are able to be trained for use in a manner that minimizes down time and/or other interruptions in controlling the process. Further, following such use of the emulations to train the monitoring device(s) **1500** for use, the emulations may be employed to test the effectiveness of the monitoring device(s) **1500** in various simulated scenarios to demonstrate their capabilities, provide confidence in the range of their capabilities, and/or ensure that various particular needs of operators of the monitored system **2000** are met.

[0061] Still further, such emulations may be used to simulate various conditions that may additionally provide new insights into other aspects of the monitored system **2000**, of an external system **3000**, and/or of a process. In this way, flaws in the logic employed in controlling a process and/or in the state machine of a process may be revealed that may create a susceptibility to one or more forms of attack and/or failure. Thus, it may be that such emulations go beyond providing proof of the suitability of the monitoring devices **1500** for use, and additionally identify ways in which the monitored system **2000**, an external system **3000**, and/or a process may be improved.

[0062] Following the preparation of the monitoring device (s) **1500** through the aforementioned three phases, the monitoring device(s) **1500** may be uncoupled from the emulation device(s) **5500**, and then coupled to the one or more interchange device(s) **2700** of the monitored system **2000** via the one or more links **1992**. The monitoring device(s) **1500** may then commence monitoring the communications activity through the internal network **2999** for instances of improper network traffic and/or instances of a lack of proper network traffic, and may respond to selected ones of such instances in various ways that will be described in greater detail.

[0063] Thus, through such use of the monitoring device(s) **1500** (which are separate and distinct from the monitored device(s) **2300**) to detect anomalous network traffic on the internal network **2999**, the effects of a cyber attack or a malfunction on the control of an industrial or other process is able to be mitigated. Also, this is able to be accomplished without consuming processing, storage and/or other resources of the monitored devices **2300**, which might impair the control of the process.

Systems and Interplay Among Systems During Normal Use

[0064] FIGS. **1**B-D, taken together, present aspects of the use of the monitored system **2000** to control an example process **3005***x* being performed within an example external system **3000***x*, and present aspects of the use of monitoring device(s) **1500** of the monitoring system **1000** to monitor and/or actively correct the corresponding network activity on the internal network **2999**. More specifically, an example monitored device **2300***x* of the multiple monitored devices **2300** of the monitored system **2000** may be coupled to the external system **3000***x* via one or more links **3999***x*, while at least one other example monitored device **2300***a* may cooperate with the monitored device **2300***x* through the internal network **2999** to control the process **3005***x* within the external system **3000***x*. Also, at least one monitoring device **1500** of the monitoring system **1000** may be coupled to at least one interchange device **2700** of the internal network **2999** to monitor the corresponding network traffic between at least the monitored devices **2300***a* and **2300***x* for instances of improper network activity that may adversely affect such controlling of the process **3005***x*, and to take various actions to correct such improper network activity.

[0065] These same figures, taken together, also present aspects of the operation of the monitored system **2000** to update software employed by one or more of the monitored devices **2300**, and present aspects of the use of monitoring device(s) **1500** of the monitoring system **1000** to monitor and/or actively correct the corresponding network activity on the internal network **2999**. More specifically, one of the monitored devices **2300***a* or **2300***x*, and/or the external system **3000***x*, may attempt to identify and/or download an updated routine **9934** from an external server **9900** as an update to at least a portion of a control routine **2340** and/or another routine. As will be explained in greater detail, there may be a distinct update download process that controls at least some aspects of the corresponding network activity that results on at least the internal network **2999**. At least one monitoring device **1500** of the monitoring system **1000** may be coupled to at least one interchange device **2700** of the internal network **2999** to also monitor the network traffic between one of the monitored devices **2300***a* or **2300***x* and that server **9900** for instances of improper network activity that may be associated with accessing a fake version of that server **9900** and/or downloading a fake version of the updated routine **9934**.

[0066] To be clear, it should be understood that what is depicted in FIGS. **1**B-D includes using the monitoring device(s) **1500** of the monitoring system **1000** to monitor and/or correct network traffic through the internal network **2999** at a time after the monitoring device(s) **1500** have been prepared for use by performing the earlier described three phases (i.e., the gathering phase, the pre-training phase, and the emulation training phase). As previously mentioned, FIGS. **1**E-F, taken together, depict aspects of such preparation of the monitoring device(s) **1500** of the monitoring system **1000** for use. Also, a more detailed discussion of such preparation is provided still later in this present application.

[0067] Turning to FIG. **1**B, as depicted, the external system **3000***x* in which an example process **3005***x* is performed, may include multiple sensing devices **3200** to detect various conditions, and/or multiple effecting devices **3800** able to be commanded to perform various functions. Depending on the nature of the depicted process **3005***x* that is performed within the external system **3000***x*, each of the sensing devices **3200** may be any of a variety of type of sensing device based on any of a variety of technologies to sense any of a variety of conditions, including and not limited to, a temperature sensor, pressure sensor, light sensor, vibration sensor, accelerometer, gyroscope, spectrometer, chemical release sensor, particle emission detector, manually-operable control, manual data input device, air speed sensor, RADAR, LIDAR, SONAR, RPM sensor, etc. Correspondingly, depending on the nature of the process **3005***x* that is performed within the external system **3000***x*, each of the effecting devices **3800** may be any of a variety of type of effecting device based on any of a variety of technologies to effect any of a variety of actions, including and not limited to, a robotic arm, gantry crane, remotely controllable mobile platform, welding device, metal press, valve, heater, cooler, power supply, magnet or set of magnets, data storage device, display system, aerofoil or hydrofoil control surface, rudder, magnetron, radiation source, electric motor, internal combustion engine, turbine engine, etc.

[0068] Thus, in being coupled to the external system **3000***x* via the link(s) **3999***x*, the monitored device **2300***x* may be coupled to the sensing devices **3200** and/or to the effecting devices **3800** of the external system **3000***x* via the link(s) **3999***x*. As depicted, and depending on the technologies employed by the link(s) **3999***x*, it may be that the link(s) **3999***x* incorporate one or more relay devices **3700***x*. By way of example, it may be that the relay device(s) **3700***x* serve to boost signal strength over relatively long electrical and/or optical conductors of the link(s) **3999***x*; serve as transceiver (s) for one or more wireless connections on which the link(s) **3999***x* may be based; serve as converter(s) between portions of the link(s) **3999***x* that may employ different signaling, protocols, transmission media, etc.; and/or serve as isolators (e.g., optoisolators) to prevent harmful current and/or voltage spikes from being conveyed through the link(s0 **3999***x*.

[0069] The process **3005***x* that may be performed within the external system **3000***x* may be any of a variety of types of industrial process (e.g., a chemical process, the automated assembly of a product on an assembly line, the provision of electric power, etc.), or may be any of a variety of types of non-industrial process (e.g., data archival and storage, data mining, etc.). Regardless of what the process **3005***x* may be, the process **3005***x* may have multiple states where particular action(s) are to be performed and/or where particular event (s) are to occur within each state. Thus, the performance of the process **3005***x* may progress through a tree of such states with particular transitions occurring between particular states at particular times, in a particular order, and/or in response to particular conditions.

[0070] As depicted, the monitored device **2300***x* may include one or more processors **2350***x*, a storage **2360***x*, a port **2390***x* to couple the monitored device **2300***x* to the internal network **2999**, and/or an interface **2393***x* to couple the monitored device **2300***x* to the link(s) **3999***x*. The storage **2360***x*, the port **2390***x* and/or the interface **2393***x* may each be communicatively coupled to the processor(s) **2350***x* to exchange executable instructions and/or data therewith through the exchange of electrical, optical, magnetic and/or other signals through one or more buses and/or other form of interconnect within the monitored device **2300***x*. Further, the storage **2360***x* may store a control routine **2340***x* that may include instructions executable by the processor(s) **2350***x* to cause the processor(s) **2350***x* to perform various functions. In some embodiments, the storage **2360***x* may also store log data **2330***x* that may include indications of transmissions to and/or from the external system **3000***x* via the link(s) **3999***x*, and/or may include other information associated with those transmissions.

[0071] As depicted, the monitored device **2300***a* may include one or more processors **2350***a*, a storage **2360***a*, and/or a port **2390***a* to couple the monitored device **2300***a* to the internal network **2999**. The storage **2360***a* and/or the port **2390***a* may each be communicatively coupled to the processor(s) **2350***a* to exchange executable instructions and/or data therewith through the exchange of electrical, optical, magnetic and/or other signals through one or more buses and/or other form of interconnect within the monitored device **2300***a*. Further, the storage **2360***a* may store a control routine **2340***a* that may include instructions executable by the processor(s) **2350***a* to cause the processor(s) **2350***a* to perform various functions.

[0072] Turning to FIG. **1**C, as depicted, transmissions **2922** of operational commands and/or operational information may be exchanged through the internal network **2999** among such monitored devices **2300** as the depicted monitored devices **2300***a* and **2300***x*, and transmissions **2922** may be exchanged through the internal network **2999** between such monitored devices **2300** and the one or more external servers **9900**. Additionally, transmissions **3933** of operational commands and/or operational information may be exchanged through such links **3999** as the depicted link(s) **3999***x* between such monitored devices **2300** and external systems **3000** as the depicted monitored device **2300***x* and external system **3000***x*.

[0073] In various embodiments, the control routine **2340***x* may implement logic **2345***x* for controlling at least some lower level aspects of performances of the process **3005***x* within the external system **3000***x*. It may be that the control routine **2340***x* is operative on the processor(s) **2350***x* of the monitored device **2300***x* to cause the processor(s) **2350***x* to monitor the sensing devices **3200** of the external system **3000***x* to monitor aspects of the process **3005***x* performed within the external system **3000***x*, and/or to command the effecting devices **3800** of the external system **3000***x* in a manner that causes the processor(s) **2350***x* to put the external system **3000***x* into at least a subset of the multiple states of the process **3005***x*. In so doing, the effecting devices **3800** of the external system **3000***x* may be operated in a concerted manner by the monitored device **2300***x* to perform various steps of the process **3005***x*, while being guided by data received by the monitored device **2300***x* from the sensing devices **3200** of the external system **3000***x*. Stated differently, the control routine **2340***x* may be capable of causing the processor(s) **2350***x* to monitor for and/or to implement at least a subset of those multiple states of the process **3005***x*. Additionally, it may be that the control routine **2340***x* is operative on the processor(s) **2350***x* to monitor for conditions within the process **3005***x* that are outside of at least a subset of expected minimums and/or maximums, and to respond by transmitting operational information indicative of such conditions to the monitored device **2300***a*.

[0074] In various embodiments, the control routine **2340***a* may implement logic **2345***a* for also controlling at least some higher level aspects of performances of the process **3005***x* within the external system **3000***x*. It may be that the control routine **2340***a* is operative on the processor(s) **2350***a* of the monitored device **2300***a* to cause the processor(s) **2350***a* to transmit operational commands to the monitored device **2300***x* to control the performance of the process of the external system **3000***x* based, at least in part, on operational information received from the monitored device **2300***x*. Stated differently, the control routine **2340***a* may be capable of causing the processor(s) **2350***a* to command the occurrence of transitions among at least a subset of the states of the process of the external system **3000***x*, thereby causing the process to be performed. Additionally, it may be that the control routine **2340***a* is operative on the processor(s) **2350***a* to provide a user interface to an operator tasked with overseeing the performance of a process **3005***x* within an external system **3000***x*.

[0075] Thus, within the monitored devices **2300***a* and **2300***x*, the control routines **2340***a* and **2340***x*, together, may implement logic **2345***a* and **2345***x* that causes the processors **2350***a* and **2350***x*, respectively, to cooperate to control the process **3005***x*. While the monitored device **2300***x* may be coupled to the external system **3000***x* so as to be capable of such lower level functionality as directly monitoring for and/or implementing states of the process **3005***x* performed therein, it may be that such higher level functionality as the overall actual performance of the process **3005***x* is controlled by the monitored device **2300***a*. In this way, the monitored devices **2300***a* and **2300***x* may cooperate to implement the logic of the finite state machine for the process performed within the external system **3000***x*.

[0076] More precisely, it may be that the monitored device **2300***a* outputs transmissions **2922** onto the internal network **2999** that convey operational commands and/or operational information to the monitored device **2300***x* to cause the monitored device **2300***x* to implement at least a subset of the transitions between states within the external system **3000***x*. In turn, the monitored device **2300***x* may output corresponding transmissions **3933** onto the link(s) **3999***x* to various ones of the effecting devices **3800** that convey commands to act in a manner that causes such transitions. It may also be that the monitored device **2300***x* receives, via the link(s) **3999***x*, transmissions **3933** from the effecting devices **3800** confirming receipt and/or performance of such instructions to act, and/or transmissions **3933** from the sensing devices **3200** providing data concerning detected conditions. In turn, the monitored device **2300***x* may output corresponding transmissions **2922** onto the internal network **2999** that convey operational information to the monitored device **2300***a* indicative of data received from the sensing device(s) **3200** to enable the monitored device **2300***a* to determine when particular ones of such transitions between states of the process **3005***x* have occurred and/or should occur.

[0077] Further, it may be that components of an external system **3000***x*, such as sensing devices **3200** and/or effecting devices **3800** of the external system **3000***x*, may incorporate or otherwise implement various forms of relatively simple logic for locally handling various specific events. Such events may include loss of communication with the monitored device **2300***x* that would otherwise monitor and control the external system **3000***x*, such that the locally implemented logic serves as a backup form of monitoring and/or control for a limited period of time until such communication is reestablished. Alternatively or additionally, such events may include an emergency situation, such as the outbreak of a fire or other condition that triggers the locally implemented logic to independently act to quickly implement a transition to a known failsafe state, while also transmitting operational information indicative of such conditions to the monitored device **2300***x*.

[0078] Alternatively or additionally, within the monitored devices **2300***a* and/or **2300***x*, the control routines **2340***a* and/or **2340***x* may implement logic **2345***a* and/or **2345***x* that causes the processors **2350***a* and/or **2350***x*, respectively, to access one or more external servers **9900** to retrieve one or more updated routines **9934** by which portions of the control routines **2340***a* and/or **2340***x* (and/or still other software routines) may be updated. Such logic may require manual triggering by an operator of one or both of these devices interacting therewith via a user interface (UI), such as a combination of keyboard and display (not specifically shown). Alternatively or additionally, such logic may be automatically triggered at timed intervals and/or in response to other factors. Regardless of the exact manner in which such logic is triggered, the processor(s) **2350***a* and/or **2350***x* may be caused by such logic within the control routine(s) **2340***a* and/or **2340***x*, respectively, to at least attempt to engage in communications with one or more external servers **9900**. Such communications may include transmissions **2922** of operational commands and/or operational information pertinent to determining whether there are updated routine(s) **9934** for the control routine(s) **2340***a* and/or **2340***x* (and/or for other software routines), and/or to effect the downloading of updated routine(s) **9934** from the one or more external servers **9900**.

[0079] Not unlike the example process **3005***x* occurring within the example external system **3000***x*, it may be that the process of identifying and/or downloading updated routine (s) **9934** from external server(s) **9900** also progresses through multiple states of a state machine. For example, there may be distinct states of searching for and/or identifying pertinent updated routine(s) **9934** based on what version of a control routine **2340** (and/or what version(s) of other software routines) is currently stored and/or executed within a corresponding monitored device **2300**. There may also be a distinct state of downloading one of such identified updated routines **9934**, and/or a distinct state of checking the integrity of such a download to confirm that such updated routine **9934** has not been corrupted during the download process through the networks **2999** and/or **9992**.

[0080] Accordingly, the control and/or monitoring of such states may be based on transmissions **2922** of operational commands and/or operational information through the networks **2999** and/or **9992**. For example, triggering the commencement of such a state of searching for and/or identifying pertinent update routine(s) **9934** may entail the transmission of various operational commands that may convey identifiers and/or version levels of various routines as parameters. The act of downloading an updated routine **9934** may entail the transmission of that updated routine **9934** as multiple blocks of operational information. The protocol used in coordinating such transmissions **2922** of blocks of operational information may entail the transmission of multiple instances of operational command(s) confirming the successful reception of each such block. The state of checking integrity may entail the transmission of operational commands confirming that all block have been transmitted and/or received, the transmission of operational commands conveying a checksum or other value by which such a confirmation may be performed, etc.

[0081] Further, within the external system **3000***x*, similar logic may be implemented to also determine whether updated routine(s) **9934** for routine(s) that may be executed within the external system **3000***x* are available, to also download such updated routine(s) **9934** that are found to be available, and/or to perform integrity checks of such downloaded ones of such routines **9934**. Thus, the external system **3000***x* may implement another process (separate and distinct from the process **3005***x*) that may implement another state machine for identifying and/or downloading such updated routine(s) **9934**. In so doing, the fact of the indirect connection of the external system **3000***x* to the internal network

2999 through the monitored device **2300**x may result in transmissions of operational commands and/or operational information as transmissions **3933** through the link(s) **3999**x, in addition to corresponding transmissions of operational commands and/or operational information as transmissions **2922** through the networks **2999** and/or **9992**.

[0082] Turning to FIG. **1**D, as depicted, each interchange device **2700** of the internal network **2999** may include one or more processors **2750**, a storage **2760**, multiple bidirectional ports **2790**, and/or a span port **2791**. The storage **2760**, the ports **2790** and/or the span port **2791** may each be communicatively coupled to the processor(s) **2750** to exchange executable instructions and/or data therewith through the exchange of electrical, optical, magnetic and/or other signals through one or more buses and/or other form of interconnect within each interchange device **2700**. Further, the storage **2760** may store a control routine **2740** that may include instructions executable by the processor(s) **2750** to cause the processor(s) **2750** to perform various functions. Alternatively or additionally, a portion of the storage **2760** may be allocated to serve as a buffer **2766**.

[0083] As previously discussed, each interchange device **2700** may be any of a variety of types of network device. Thus, in some embodiments, the depicted interchange device **2700** may be a relatively simple hub device in which execution of the control routine **2740** by the processor(s) **2750** may cause the processor(s) **2750** to, in response to receiving a transmission **2922** at one of the ports **2790**, output the very same transmission **2922** in a broadcasting manner at all other ports **2790**. Alternatively, in other embodiments, the depicted interchange device **2700** may be a relatively more sophisticated device (e.g., a network switch device) in which execution of the control routine **2740** by the processor(s) **2750** may cause the processor(s) **2790** to use internally stored address information associated with each port **2790** to more selectively relay a transmission **2922** received at one port **2750** to just one other port **2750** or to just a subset of the other ports **2750**.

[0084] Regardless of the level of sophistication of the depicted interchange device **2700**, it may be that received transmissions **2922** are temporarily stored within the buffer **2766** for a predetermined period of time and/or until there is an indication of success in being relayed onward through the internal network **2999**. This may be done to enable one or more attempts at retransmission to be performed in response to an indication of failure in an initial attempt at relaying onward through the internal network **2999**.

[0085] Further, the span port **2791** may be implemented as an output-only port that relays a copy and/or an indication of each transmission **2922** that is received at any port **2790** of the interchange device **2700**. Thus, as depicted, the span port **2791** may be coupled to a monitoring device **1500** of the monitoring system **1000** to enable copies and/or indications of some or all network traffic that passes through the interchange device **2700** to be provided to that monitoring device **1500**. As will shortly be explained in greater detail, this enables that monitoring device **1500** to detect an instance in which a transmission **2922** of a particular operational command and/or operational information among the monitored devices **2300**, or between a monitored device **2300** and an external server **9900**, was expected to occur within a particular span of time, but failed to occur. This also enables that monitoring device **1500** to detect an instance in which an improper transmission **2922** of an operational command and/or operational information among the monitored devices **2300**, or between a monitored device **2300** and an external server **9000**, occurs. As additionally depicted, the same depicted monitoring device **1500** may also be coupled to the depicted interchange device **2700** via another of the ports **2790**. As will be explained in greater detail, such an additional coupling therebetween may enable the monitoring device **1500** to respond to such an occurrence of a lack of an expected transmission **2922**, and/or such an occurrence of an improper transmission **2922**, by outputting a replacement transmission **2922**, itself.

[0086] In embodiments in which the interchange device **2700** is of a more sophisticated variety, it may be that execution of the control routine **2740** causes the processor(s) **2750** thereof to respond to commands received from such a monitoring device **1500** to limit the copies and/or indications of network traffic that are provided through the span port **2791** to transmissions **2922** of particular types and/or to transmissions **2922** associated with particular devices. In this way, it may be that the depicted interchange device **2700** is caused to cooperate with the depicted monitoring device **1500** to limit the copies and/or indications of network traffic that are output to the monitoring device **1500** to operational commands and/or operational information transmitted among monitored devices **2300**, and/or between monitored devices **2300** and specific external server(s) **9900**.

[0087] Alternatively or additionally, in embodiments in which the interchange device **2700** is of a more sophisticated variety, it may be that the execution of the control routine **2740** causes the processor(s) **2750** thereof to respond to commands received from such a monitoring device **1500** to at least temporarily retain transmissions **2922** meeting one or more specified criteria within the buffer **2766**. In this way, it may be that the depicted interchange device **2700** is caused to cooperate with the depicted monitoring device **1500** to at least temporarily delay allowing a transmission **2922** to proceed through the interchange device **2700** between monitored devices **2300**, and/or between a monitored device **2700** and an external server **9900**, thereby at least providing the monitoring device **1500** with an amount of time needed to analyze aspects of the transmission **2922** to determine whether it is a proper transmission **2922** so as to be able to determine whether to command the interchange device **2700** to allow it to continue onward to its intended destination device.

[0088] As also depicted, each such monitoring device **1500** may include one or more processors **1550**, a storage **1560**, and/or one or more ports **1590** for coupling to one or more interchange devices **2700** of the internal network **2999**. The storage **1560** and/or the port(s) **1590** may each be communicatively coupled to the processor(s) **1550** to exchange executable instructions and/or data therewith through the exchange of electrical, optical, magnetic and/or other signals through one or more buses and/or other form of interconnect within each monitoring device **1500**. Further, the storage **1560** may store a control routine **1540** that may include instructions executable by the processor(s) **1550** to cause the processor(s) **1550** to perform various functions. Alternatively or additionally, the storage **1560** may store a database **1533** of information concerning states, operational commands and/or operational information associated with one or more processes that may be performed within an external system **3000** (e.g., the process **3005**x performed within the external system **3000**x, or associated with one or more processes for downloading an update routine **9934** from an external server **9900**), as well as information concerning actions to be taken in response to situations observed on the internal network **2999** that at least appear to fall outside what is expected to occur during the performance of each of those one or more processes.

[0089] More precisely, and as will be explained in greater detail, the database **1533** may include multiple process entries **1534**, with each process entry **1534** corresponding to a single process that may be performed within an external system **3000** under the control of monitored device(s) **2300**, or to a single process for downloading an updated version of a software routine from an external server **9900**. Still more precisely, each process entry **1534** may store a model of one of such processes. Such a model may define the processing states of a process, along with aspects of network activity associated with each processing state. As will also be explained in greater detail, these models may be generated from at least observations of network activity as part of preparing the monitoring devices **1500** for use, and during such use of the monitoring devices **1500**, these models may be repeatedly referred to identify network activity associated with a particular process and/or to identify improper network activity that may be associated with an issue developing during the performance of a process.

[0090] With the monitoring device(s) **1500** so coupled to the interchange device(s) **2700**, the monitoring for improper network traffic through the internal network **2999** may include, and not be limited to: 1) monitoring for instances of a lack of transmission of a particular operational command or particular operational information when expected within a particular span of time; and/or 2) monitoring for instances of a transmission of an improper or invalid operational command, or of improper or invalid operational information or data.

[0091] Regarding the lack of a transmission **2922** of a particular operational command or particular operational information occurring when expected within a particular span of time, such an instance of failure of transmission may arise for any of a variety of reasons. Among those reasons may be any of a variety of hardware and/or software malfunctions that may have occurred within a particular monitored device **2300** or external server **9900** such that it is no longer functioning sufficiently to transmit the operational command or information when expected. Alternatively or additionally, among those reasons may be a cyber attack in which a particular monitored device **2300** or external server **9900** that was to transmit the operational command or information through the internal network **2999** has succumbed to malicious software or other form of internal cyber attack that prevents it from doing so, and/or has been inundated with network activity through the internal network **2999** such that it is prevented from transmitting the expected operational command or operational information.

[0092] Again, there may be particular transmissions **2922** of operational commands and/or operational information through the internal network **2999** that are associated with various transitions between states of such a process within an external system **3000**. By way of example, it may be that a particular operational command or operational information is meant to be transmitted through the internal network **2999** within a particular span of time in response to the occurrence of particular conditions, such as a transition into a particular state. Or, by way of another example, it may be that a particular operational command or operational information is meant to be transmitted through the internal network **2999** within a particular span of time, and/or as part of a particular sequence of operational commands and/or operation information, to cause a transition into a particular state. Thus, a failure of one of those transmissions **2922** to occur when expected within a particular span of time may result in a failure of occurrence of a transition between states that would otherwise normally take place, and/or may result in an errant transition to an incorrect state.

[0093] The monitoring device(s) **1500** of the monitoring system **1000** may be configured to respond to such a failure of occurrence of the transmission of a particular operational command or information through the internal network **2999** by taking any of a variety of actions. By way of example, in some embodiments, the monitoring device(s) **1500** may simply provide an alert to designated personnel of such a failure (e.g., providing an audible and/or visual alert, and/or transmitting an electronic alert message, such as a text message or phone call). Alternatively or additionally, in other embodiments, the monitoring device(s) **1500** may store, in a log or other similar data structure maintained within the storage **2760**, an indication of such a failure. Also alternatively or additionally, in still other embodiments, the monitoring device(s) **1500** may, through the interchange device(s) **2700**, transmit that particular operational command or operational information through the internal network **2999**, itself, such that the particular operational command or operational information is still provided to whichever monitored device **2300** or external server **9900** is supposed to receive it, thereby enabling the associated industrial process, download process, or other type of process to continue without interruption.

[0094] Regarding the transmission of an improper or invalid operational command, such an instance of an improper transmission **2922** may also arise for any of a variety of reasons. Among those reasons may be an honest operator mistake, or any of a variety of hardware and/or software malfunctions that may have occurred within the a particular monitored device **2300** or external server **9900** such that it is no longer functioning sufficiently to select correct operational commands and/or operational information to be transmitted. Alternatively or additionally, among those reasons may be that the monitored device **2300** or external server **9900** that transmits the improper operational command or operational information has succumbed to malicious software or other form of cyber attack that either, itself, causes the particular monitored device **2300** or external server **9900** to transmit the improper operational command or operational information, or that enables the monitored device **2300** or external server **9900** to be remotely commanded to do so. It should also be noted that such an instance of an improper transmission **2922** may arise elsewhere within and/or proximate to the internal network **2999**, such as interfering electrical activity caused by a lightning strike, etc. such that the contents of a proper transmission **2922** may have been corrupted, thereby creating the improper transmission **2922**.

[0095] Again, there may be particular transmissions **2922** of operational commands and/or operational information through the internal network **2999** that are associated with various transitions between states of such a process within an external system **3000**, or that are associated with various transitions between states of a process of downloading an updated routine **9934**. Thus, a transmission **2922** of an improper operational command to take an improper action, or to take an action with improper parameter(s), may result in a transition to another state that occurs out of proper sequence, or may result in a transition to an improper state that is entirely outside of such a tree of proper states.

[0096] The monitoring device(s) **1500** of the monitoring system **1000** may be configured to respond to the transmission of such an improper operational command through the internal network **2999** by taking any of a variety of actions. By way of example, in some embodiments, the monitoring device(s) **1500** may simply provide an alert to designated personnel of such a transmission (e.g., providing an audible and/or visual alert, and/or transmitting an electronic alert message, such as a text message or phone call). Alternatively or additionally, in other embodiments, the monitoring device(s) **1500** may store, in a log or other similar data structure maintained within the storage **2760**, an indication of such a transmission **2922**. Also alternatively or additionally, in still other embodiments, the monitoring device(s) **1500** may, through the interchange device(s) **2700**, block the transmission **2922** of such an improper operational command such that it is not received at whichever monitored device **2300** or external server **9900** is the intended destination device, and/or cause the transmission **2922** of a proper operational command that replaces and/or countermands the improper operational command.

[0097] More precisely, in some embodiments, it may be that the monitoring device(s) **1500** operate the interchange device(s) **2700** to cause the interchange device(s) **2700** to intercept each operational command that is transmitted by a monitored device **2300** or an external server **9900** that meets various pre-selected parameters (e.g., type of operational command, protocol used, destination, etc.). Each such operational command may initially be relayed to the monitoring device(s) **1500** for analysis, instead of being relayed onward to its intended destination device. The analysis that is performed may be based on machine learning where correct transmissions **2922** of operational commands associated with controlling industrial processes (e.g., the process **3005**x) and/or associated with downloading updated routines **9934** through the internal network **2999** have been observed over time to build up a knowledge base of what transmissions **2922** of operational commands are expected to occur and when for each process. Where the analysis results in a determination that the operational command is proper, then the interchange device(s) **2700** may be directed to proceed with relaying the operational command onward towards its intended destination. However, where the analysis results in a determination that the operational command is improper or invalid, then, at a minimum, the interchange device(s) **2700** may be directed to refrain from relaying the operational command onward towards its intended destination device.

[0098] Still further action may be taken by the monitoring device(s) **1500** in response to an improper operational command based on various factors. By way of example, where a transmitted improper operational command is the type of operational command that is expected, is transmitted at the expected time, and to the expected destination device, but includes invalid parameter(s), then the monitoring device(s) **1500** may transmit a corrected version of that improper operational command, through interchange device(s) **2700**, and onward to the same destination device, so as to cause the proper operational command to be received at that destination device at the expected time.

[0099] Alternatively, in other embodiments, it may be that the interchange device(s) **2700** do not, in any way, intercept the operational commands that are transmitted therethrough. Instead, for each operational command that is transmitted by a monitored device **2300** or an external server **9900** that meets various pre-selected parameters (e.g., type of operational command, protocol used, destination, etc.), a copy is relayed to the monitoring device(s) **1500** for analysis as it is also relayed onward to its intended destination. Again, the analysis that is performed may be based on machine learning and/or a comparison against the database **1533** with its models providing indications of expected operational commands. Where the analysis results in a determination that the operational command or information transmission is proper, then no further action may be taken. However, where the analysis results in a determination that the operational command or information transmission is improper, then further action may be taken by the monitoring device(s) **1500**, such as sending an alert or transmitting additional operational command(s), through interchange device(s) **2700**, and onward to the same destination device. The additional operational command(s) may be generated based on an analysis of the improper operational command to countermand the action specified in the improper operational command.

Systems and Interplay Among Systems During Preparation for Normal Use

[0100] FIGS. **1**E-F, taken together, present aspects of the preparation of the monitoring system **1000** for use in monitoring and/or correcting network traffic occurring on the internal network **2999** of the monitored system **2000** that are associated with controlling the process **3005**x being performed within the external system **3000**x, and/or associated with downloading updated routines **9934** from one or more external servers **9900**. More specifically, information gathered about the monitored system **2000**, the external system **3000**x, and/or the one or more external servers **9900** is provided to one or more emulation devices **5500** of the emulation system **5000** to enable the provision of a set of emulations of different portions of the system **2000**, the system **3000**x, and/or the one or more external servers **9900**. In so doing, an emulation of network activity through at least the internal network **2999** may also be provided, and may be used to train and/or test the one or more monitoring devices **1500** as part of preparing those monitoring device(s) **1500** for such use.

[0101] Turning to FIG. **1**E, during the gathering phase, one or more gathering devices **5700** may be temporarily coupled, via at least one link **5992**, to span port(s) **2791** and/or to tap(s) of interchange device(s) **2700** of the internal network **2999** of the monitored system **2000**. Through the link(s) **5992**, and in a manner similar to what was described in reference to FIG. **1**D for the monitoring device(s) **1500**, the gathering device(s) **5700** may receive indications of transmissions **2922** occurring on the internal network **2999**. Also in a manner similar to what was described in reference to FIG. **1**D for the monitoring device(s) **1500**, the indications of transmissions **2922** received by the gathering device(s) **5700** through the link(s) **5992** may be limited to transmissions **2922** that are associated with controlling processes performed within one or more external systems **3000** (e.g., the process **3005***x* that is performed within the external system **3000***x*). Alternatively or additionally, the transmissions **2922** may be limited to those that are associated with downloading updated routines **9934** for the control routines **2340** executed within the monitored devices(s) **2300** (e.g., the monitored devices **2300***a* and/or **2300***x*), and for routines that may be executed within the one or more external systems **3000** (e.g., the external system **3000***x*).

[0102] Additionally, in some embodiments, it may be that the gathering device(s) **5700** are also temporarily coupled, via at least one link **5993**, to either the relay device(s) **3700***x* (if present) or to the monitored device **2300***x*. As previously discussed, based on the type(s) of communications technology employed by the link(s) **3999***x*, it may be that the link(s) **3999***x* incorporate the relay device(s) **3700***x* to boost signals by which the transmissions **3933** are conveyed, to convert between differing protocols and/or differing transmission media used by the link(s) **3999***x*, and/or to provide protective isolation. Further, it may be that the relay device(s) **3700***x* also incorporate span port(s) and/or a wire tap(s) to which the gathering device(s) **5700** may be temporarily coupled via the at least one link **5993** in a manner similar to the span port(s) **2791** and/or wire tap(s) of the interchange device(s) **2700**.

[0103] Alternatively, in other embodiments in which the link(s) **3999***x* do not incorporate the relay device(s) **3700***x*, it may be deemed desirable to avoid attempting to couple the gathering device(s) **5700** to the link(s) **3999***x*, at all. As those skilled in the art will readily recognize, it may be deemed desirable to not risk causing disruptions to the ability of the link(s) **3999***x* to convey transmissions **3933** therethrough, and to employ an alternate approach to obtaining information concerning those transmissions **3933**, if an alternate approach is available. Such an alternate approach may be to couple the gathering device(s) **5700** to the monitored device **2300***x* to obtain a copy of the log data **2330***x* that may be generated therein, which may include various details of transmissions **3933** to and from the external system **3000***x*.

[0104] As those skilled in the art will readily recognize, each of the monitored devices **2300**, including the monitored devices **2300***a* and **2300***x*, may be any of a wide variety of types of computing device based on any of a variety of implementations of any of a variety of computing architectures. Among such a wide assortment of options for implementing the monitored device **2300***x*, a subset of those options may be configured and/or configurable to generate the log data **2330***x* to include any of a variety of types of information concerning any of variety of types of event that may occur while controlling the process **3005***x* within the external system **3000***x*, and/or while cooperating with the external system **3000***x* to retrieve updated routines **9934** for the use within the external system **3000***x*. In embodiments in which the log data **2330***x* is generated to include indications of each transmission **3933** through the link(s) **3999***x*, it may be possible to use the log data **2330***x* in lieu of direct observations of those transmissions **3933** by connecting to the link(s) **3999***x*.

[0105] By way of example regarding controlling the process **3005***x*, in some of such embodiments, it may be that the log data **2330***x* is limited to recording instances of anomalies and/or improper conditions that might occur during individual performances of the process **3005***x*, such as instances of exceeding ranges of temperature, pressure, speed, capacity, timing, etc. Alternatively, it may be that the log data **2330***x* is far more verbose such that indications of each transmission **3933** that is sent or received by the monitored device **2300***x* in connection with the process **3005***x* may be recorded therein, along with timestamps thereof.

[0106] By way of another example regarding downloading updated routines **9934**, in some embodiments for use within the external system **3000***x*, it may be that the log data **2330***x* is limited to recording instances of anomalies and/or improper conditions that might occur during an instance of downloading an updated routine **9934**, such as instances of anomalies in communicating with an external server **9900**, instances of interruption and/or corruption of the download, etc. Alternatively, it may be that the log data **2330***x* is far more verbose such that indications of each transmission **3933** that is sent or received by the monitored device **2300***x* in connection with downloading an updated routine **9934** may be recorded therein, along with timestamps thereof.

[0107] Regardless of the exact manner in which information concerning transmissions **3933** through the link(s) **3999***x* is captured or otherwise obtained (if that is possible) in addition to information concerning transmissions **2922** through the internal network **2999**, the captured information concerning transmissions **2922** and/or **3933** may be stored within at least one gathering device **5700** as the observation data **5730**. As will shortly be explained, following this gathering phase in which the observation data **5730** is so generated to include information for such transmissions **2922** and/or **3933** occurring over a pre-determined period of time, the observation data **5730** may then be provided to the one or more emulating devices **5500** of the emulation system **5000** as an input to the pre-training phase.

[0108] As depicted, each gathering device **5700** may include one or more processors **5750**, a storage **5760**, and/or one or more ports **5790** for being temporarily coupled to at least interchange device(s) **2700** of the internal network **2999**. The storage **5760** and/or the port(s) **5790** may each be communicatively coupled to the processor(s) **5750** to exchange executable instructions and/or data therewith through the exchange of electrical, optical, magnetic and/or other signals through one or more buses and/or other form of interconnect within each monitoring device **1500**. Further, the storage **5760** may store the observation data **5730** and/or a control routine **5740** that may include instructions executable by the processor(s) **5750** to cause the processor (s) **5750** to perform various functions.

[0109] More specifically, in executing the control routine **5740**, the processor(s) **5750** of the gathering device(s) **5700** may be caused to monitor the port(s) **5790** for the receipt of indications of transmissions **2922** occurring on the internal network **2999**. Additionally, in some embodiments, the processor(s) **5750** may be caused to monitor the port(s) **5790** for the receipt of either indications of transmissions **3933** occurring through the link(s) **3999***x*, or the log data **2330***x*. In response, the processor(s) **5750** may be caused to combine such received information concerning transmissions **2922** and/or **3933** (e.g., the contents of each transmission **2922** and/or **3933**, and/or when each transmission occurred) to generate the observation data **5730**.

[0110] Referring to FIG. **1**A, in addition to FIG. **1**E, it should be noted that, in an alternate embodiment, the emulation system **5000** may not include the gathering device (s) **5700**, and it may be that the monitoring device(s) **1500** are employed to perform the functions of the gathering device(s) **5700**. More precisely, in such an alternate embodiment, and at a time prior to the monitoring device(s) **1500** being prepared for use in monitoring and/or correcting network traffic on the internal network **2999**, it may be that the monitoring device(s) **1500** are able to be operated in a gathering mode during the gathering phase to receive indications of transmissions **2922** occurring on the internal network **2999**, and/or transmissions **3933** occurring through the link(s) **3999**, and to generate the observation data **5730** therefrom in a manner similar to what has been described in reference to the gathering device(s) **5700**. In such an alternate embodiment, the monitoring device(s) **1500** may then be temporarily coupled to the emulation device(s) **5500** to both provide the observation data **5730** thereto as part of completing the gathering phase, and to be trained and/or tested as part of the emulation training phase, as has been described.

[0111] In another alternate embodiment, it may be that one or more of the monitored devices **2300** (e.g., one or both of the monitored devices **2300***a* and **2300***x*) are employed to perform the functions of the gathering device(s) **5700**. More precisely, processor(s) **2350** within each such monitored device **2300** may caused, by execution of a portion of the control routine **2340**, to generate and store at least a portion of the observation data **5730** to include indications of network traffic through the internal network **2999** that it has output and/or that it has received. Such portions of the observation data **5730** may then be subsequently combined to form the completed observation data **5730**, which may then be provided to the emulation device(s) **5500**.

[0112] In still another embodiment, it may be that the interchange device(s) **2700** of the internal network **2999** are employed to perform the functions of the gathering device(s) **5700**. More precisely, processor(s) **2750** within each such interchange device **2700** may caused, by execution of a portion of the control routine **2740**, to generate and store at least a portion of the observation data **5730** to include indications of network traffic that passes therethrough. Again, such portions of the observation data **5730** may then be subsequently combined to form the completed observation data **5730**, which may then be provided to the emulation device(s) **5500**.

[0113] Turning to FIG. **1**F, following the gathering phase, and during the pre-training phase, the gathering device(s) **5700** that had been temporarily coupled to at least the monitored system **2000** and/or to the link(s) **3999***x* to gather information and/or to generate the observation data **5730**, may be uncoupled therefrom, and may then be temporarily coupled, to one or more emulation device(s) **5500** via the training network **5999**. In this way, the emulation device(s) **5500** may be provided with at least the observation data **5730** generated by the gathering device(s) **5700**.

[0114] Also, the emulation device(s) **5550** may be provided with configuration data **5530** that includes one or more instances of device data **2303** concerning characteristics of one or more monitored devices **2300** (e.g., the monitored devices **2300***a* and **2300***x*), the instance of the control routine **2340** executed within each, the logic **2345** implemented within each such instance of the control routine **2340** to perform one or more processes (e.g., a process for downloading an updated routine **9934**), and/or the state machine (s) implemented thereby for each such process. Alternatively or additionally, the configuration data **5530** may include one or more instances of system data **3003** concerning characteristics of one or more external systems **3000** (e.g., the external system **3000***x*), the logic implemented within each such external system **3000** to perform one or more processes (e.g., the process **3005***x*, and/or a process for downloading an updated routine **9934**), and/or the state machine(s) implemented thereby for each such process. It should be noted that in various different embodiments, information concerning the characteristics of one or more of such items may not be available, and as a result, some of such information may need to be derived from the observation data **5730** to enable the provision of at least some emulations, as will be explained in greater detail.

[0115] As depicted, each such emulation device **5500** may include one or more processors **5550**, a storage **5560**, and/or one or more ports **5590** for coupling to the training network **5999** and/or to links **5991**. The storage **5560** and/or the port(s) **5590** may each be communicatively coupled to the processor(s) **5550** to exchange executable instructions and/or data therewith through the exchange of electrical, optical, magnetic and/or other signals through one or more buses and/or other form of interconnect within each emulation device **5500**. Further, the storage **5560** may store a control routine **5540** that may include instructions executable by the processor(s) **5550** to cause the processor(s) **5550** to perform various functions. Alternatively or additionally, the storage **5560** may store the observation data **5730**, the configuration data **5530** and/or a database **5533**. Also alternatively or additionally, portions of the storage **5560** may be allocated to serve as virtual machines (VMs) or containers **5566** to support the provision of emulations.

[0116] As will be explained in greater detail, during the pre-training phase, execution of the instructions of the control routine **5540** may cause processor(s) **5550** of the emulation device(s) **5500** to use the information in the observation data **5730** concerning observed transmissions **2922** and/or **3933**, together with information in the configuration data **5530** to generate, within each VM or container **5566**, an emulation of a separate device or other component that is present within the monitored system **2000**, the external system **3000***x*, and/or one or more external servers **9900**. These may include separate emulations for each of the monitored devices **2300***a* and **2300***x*, emulation(s) of the process **3005***x* and/or the external system **3000***x* in which the process **3005***x* is performed, and/or separate emulations for each of one or more external servers **9900**. These VMs or containers **5566** may be linked to enable the separate emulations within each VM or container **5566** to interact in a cooperative manner to provide an emulation of the monitored system **2000**, the external system **3000***x*, and/or one or more external servers **9900** interacting with each other. In so doing, an emulation may also be provided of at least the communications activity occurring through the internal network **2999**, if not also through the link(s) **3999***x*.

[0117] As will also be explained in greater detail, following the pre-training phase, and during the emulation training phase, these emulations may be used in training the monitoring device(s) **1500** to distinguish between proper network activity on the internal network **2999** that is associated with one or more processes being performed, and improper network activity on the internal network **2999** that is at least suggestive of a malfunction and/or cyber attack where the monitoring device(s) **1500** should take some form of action. As will be explained in greater detail, it may be that the monitoring device(s) **1500** are also prepared to distinguish network activity on the internal network **2999** that is indicative of an error condition where the monitoring device(s) **1500** should not take action, and instead, allow other devices or external systems (e.g., one or more monitored devices **2300** and/or external systems **3000**) to take action.

[0118] In commencing the emulation training phase, the monitoring device(s) **1500** may be temporarily coupled, via at least one link **5991**, to port(s) **5590** of the emulation device(s) **5500**. The processor(s) **5550** of the emulation device(s) **5500** may be caused by execution of the control routine **5540** to provide, at such port(s) **5590**, access to the emulation of at least the communication activity through the internal network **2999**. In this way, the monitoring device(s) **1500** may be subjected to conditions that are at least relatively similar to the conditions that will be encountered when coupled to at least the actual interchange device(s) **2700** of the internal network **2999**.

[0119] Referring to FIG. **1**A, in addition to FIG. **1**F, following the emulation training phase, the monitoring device(s) **1500** may be uncoupled from the emulation device (s) **5500**, and may then be coupled, via the link(s) **1992**, to the interchange device(s) **2700** of the internal network **2999** of the monitored system **2000** for use. As will be explained in greater detail, the emulation training phase is meant to at least minimize (if not eliminate) the need for further training of the monitoring device(s) **1500** upon being so coupled to the actual monitored system **2000** for use. In this way, at least the potential for disruptions of the normal operation of the monitored system **2000** to control the process **3005***x*, and/or to download update routine(s) **9934**, may be at least minimized, if not eliminated.

[0120] However, as will also be explained, it may be that, over time, changes in network traffic through the internal network **2999** may be caused by changes to various other devices coupled thereto and/or to routines that are executed therein, which may result in changes to the logic and/or state machines of various processes. As will be familiar to those skilled in the art, hardware upgrades to devices that may have been intended to simply improve their speed and/or reliability of operation can still have the unintended consequence of changing their behavior, including on the networks to which they are connected. Further, and as will also be familiar to those skilled in the art, software updates that may have been intended to simply address seemingly insignificant "software bugs" can also have such unintended consequences. Thus, over time, such "drift" in the behavior of various devices, software and/or external systems may cause some degree of re-training of the monitoring device(s) **1500** to be deemed at least desirable, if not necessary.

[0121] Returning to FIG. **1**A, although the monitored system **2000** has been discussed as being made up of multiple networked computing devices **2100**, **2300** and/or **2700**, in alternate embodiments, it may be that the monitored system **2000** is, itself, a single computing device. In such alternate embodiments, the device(s) **2100**, **2300** and/or **2700** may be components of that single computing device that are interconnected by an internal network of buses **2999**. In such alternate embodiments, it may be that each of the one or more interchange devices **2700** is an integrated circuit providing a form of crosspoint switch function for the network of buses **2999** by which commands and/or data are transmitted among the other devices **2100** and/or **2300**.

[0122] Further, in such alternate embodiments, it may additionally be that the monitoring system **1000** is implemented using one or more microcontrollers that may be physically incorporated into that single computing device of the monitored system **2000**. However, it may also be that the monitoring system **1000** is otherwise isolated from the central processing units (CPUs) thereof of that single computing device.

[0123] Still further, and regardless of whether the monitored system **2000** is made up of multiple networked computing devices, or is, itself, a computing device, it may be that the monitoring system **1000** is implemented via public or private cloud-based network computing resources.

Further Detail of Network Communications and Interchange Devices

[0124] FIGS. **2**A, **2**B and **2**C, taken together, present various aspects of example implementations of the internal network **2999**, including example implementations of the interchange device(s) **2700**.

[0125] Turning to FIG. **2**A, and regarding the internal network **2999**, each of at least a subset of the ports **2790** of an interchange device **2700** of the one or more interchange devices **2700** may be coupled by a separate link **2990** to a separate unmonitored device **2100** or monitored device **2300**. As those skilled in the art will readily recognize, this may create a form of hub-and-spoke topology, or other electrically similar topology, in which an interchange device **2700** may be at the center of a set of point-to-point connections to a corresponding set of multiple devices **2100/2300**. Regarding the external network **9992** (e.g., the Internet), at least one of the ports **2790** may be coupled, via a separate link **2992** and a relay device **2900** (e.g., a modem, a media converter, etc.), to the external network **9992**, which in turn, may provide access to the one or more external servers **9900** from which various updated routines **9934** may be downloaded.

[0126] Turning briefly to FIG. **2**B, along with FIG. **2**A, in larger embodiments of the monitored system **2000**, it may be that multiple interchange devices **2700** are coupled together to form a larger version of such a hub-and-spoke topology.

[0127] Returning to FIG. **2**A, for each monitored system **2000**, the span port **2791** and/or wire tap(s) of at least a single interchange device **2700** may be coupled, via a separate link **1992**, to at least a single monitoring device **1500** of the monitoring system **1000**. As the devices **2100** and/or **2300** of the monitored system **2000** engage in communications thereamong, and/or as the devices **2300** and external server(s) **9900** engage in communications therebetween, through the internal network **2999**, copies and/or indications of at least operational commands and/or operational information that are transmitted in those communications may be relayed to the one or more monitoring devices **1500** via the span port(s) **2791** and/or wire tap(s), and via the links **1992**. In some embodiments, it may be that such a single monitoring device **1500** is additionally or alternatively coupled, by another link **1992**, to a bi-directional port **2790** of such a single interchange device **2700** to enable the single monitoring device **1500** to control one or more the interchange devices **2700**, and/or to transmit an operational command or operational information to a monitored device **2300** or external server **9900** therethrough.

[0128] Turning again to FIG. **2**B, along with FIG. **2**A, where there are multiple interchange devices **2700** incorporated into an embodiment of the monitored system **2000**, it may be that each one of the multiple interchange devices

2700 incorporates a separate network switch span port 2791. The span port 2791 of each of those multiple interchange devices 2700 may then be separately coupled, by a separate link 1992, to one or more monitoring devices 1500 of the monitoring system 1000 of that embodiment, thereby enabling each one of those multiple interchange devices 2700 to directly relay copies and/or indications of at least operational commands and/or operational information that are transmitted among the monitored devices 2300, and/or between the monitored devices 2300 and the external server(s) 9900. Alternatively, it may be that just a single one of those multiple interchange devices 2700 is so coupled to a single monitoring device 1500, and that single interchange device 2700 may relay copies and/or indications of such operational commands and/or operational information conveyed through any of the multiple interchange devices 2700 through that single span port 2791 and single link 1992.

[0129] Regardless of the quantity of interchange devices 2700 and/or the exact manner in which those interchange device(s) 2700 are coupled to one or more monitoring devices 1500, as previously discussed, the interchange device(s) 2700 may include the ability to be programmed to specify a particular subset of transmissions 2922 through the internal network 2999 for which copies and/or indications thereof are relayed to the monitoring device(s) 1500. Such a subset may be specified by identifiers of devices involved, types of transmission, types of protocol used, size of what is transmitted, time of day or day of week of transmissions, etc. Indeed, in some embodiments, it may be the use of identifiers of devices in defining such a subset that effectively defines which devices within the monitored system 2000 are the monitored devices 2300, versus which are the unmonitored devices 2100.

[0130] In embodiments in which the monitored system 2000 is made up of a set of networked computing devices, each link 1992, 2990 and/or 2992 may be implemented using any of a variety of wireless and/or cabling-based network technologies, including and not limited to, Bluetooth, Wi-Fi, cellular signaling, twisted-pair electrical cabling, coaxial electrical cabling, and fiber optic cabling. Such wireless and/or cabling-based technologies may adhere to any of a wide variety of specifications, including and not limited to, Ethernet and/or TCP/IP. In embodiments in which the monitored system 2000 is made up of a single computing device, each link 2990 and/or 1992 may be implemented using any of a variety of widely used and accepted internal bus specifications, including and not limited to, PCI-Express bus, and I^2C bus.

[0131] As previously discussed, the interchange device(s) 2700 incorporated into a monitored system 2000 may vary in their sophistication. Again, each interchange device 2700 may be relatively simple, such as a network hub that essentially broadcasts each transmission 2922 that it receives at one port 2790 to all of its other ports 2790. Alternatively, each interchange device 2700 may be relatively sophisticated such that it may employ various levels of protocol analysis to implement a switching algorithm by which each transmission 2922 that it receives at one port 2790 is directed to one other specific port 2790 (unless the transmission 2922 that is received is specifically intended to be broadcast). As those skilled in the art will readily recognize, still more sophisticated implementations of an interchange device 2700 are possible that may incorporate even more complex functions that may entail the storage and/or logging of transmissions, content-based analysis of transmissions 2922, etc.

[0132] Thus, embodiments of the monitored system 2000 are possible in which an interchange device 2700 may be too rudimentary in its capabilities to be able to provide a span port 2791 as was depicted in the interchange devices 2700 shown in FIGS. 2A-B. Alternatively or additionally, it may be that there are insufficient ports. In such situations, one or more taps may be added.

[0133] Turning to FIG. 2C, in some embodiments, it may be that a monitored device 2300 includes multiple components 2301 where transmissions 2922 of operational commands or operational information thereto and/or therefrom are at least able to be observed through the link 2990 by which that monitored device 2300 is coupled to an interchange device 2700. More specifically, it may be that a single monitored device 2300 incorporates multiple components 2301 that are each able to be separately addressed and communicated with through that link 2990 in a manner almost akin to being entirely separate devices. Alternatively or additionally, it may be that transmissions 2922 of operational commands or operational information among multiple components 2301 within a single monitored device 2300 are also reflected on that link 2990.

[0134] Thus, and by way of example, where the depicted monitored component 2301A of the depicted monitored device 2300 transmits an operational command or operational information to another depicted monitored component 2301B within the same monitored device 2300, a copy of that transmission 2922 may also be transmitted onto the depicted link 2990, thereby enabling the depicted interchange device 2700 to relay a copy and/or indication of that transmission 2922 onward to the depicted monitoring device 1500. As still another alternative, it may be that such a transmission 2922 between the depicted monitored components 2301A and 2301B is performed by the monitored component 2301A transmitting the operational command or operational information out to the depicted interchange device 2700, followed by the interchange device 2700 relaying that operational command or operational information back along the same link 2990 to the monitored component 2301B.

[0135] Regardless of the exact manner in which transmissions 2922 of operational commands or operational information are relayed to the monitoring device(s) 1500 of the monitoring system 1000, as will shortly be explained in greater detail, the copies and/or indications of operational commands or operational information that are so relayed may be compared to information stored within the database 1533 concerning what transmissions 2922 of operational commands and/or operational information are expected to occur (and when) as part of identifying instances in which observed transmissions 2922 of operational commands or operational information deviate from what is expected.

Correlations Among Transmissions and Process States—External Process Control

[0136] FIGS. 3A, 3B, 3C, 3D and 3E, taken together, present various aspects of the manner in which transmissions 2922 and 3933, a process state machine (i.e., the example process 3005*x*), and logic employed within monitored devices (i.e., the logic 2345*a* and 2345*x* of the monitored devices **2300***a* and **2300***x*, respectively) correlate and interact to control a performance of a process (i.e., the example process **3005***x*).

[0137] Turning to FIG. **3**A, as previously discussed, and as depicted, a process performed within one of the external systems **3000** may be defined as having a tree of multiple states that is traversed during its performance, such as the depicted process states **3006***p***1**, **3006***p***2**, etc., of the depicted process **3005***x* performed within the depicted external system **3000***x*. Also, such an external system as the depicted external system **3000***x* may include one or more sensing devices **3200** to monitor various aspects of the process **3005***x* performed therein, and/or one or more effecting devices **3800** to effect various aspects of a performance of the process **3005***x* therein.

[0138] As also previously discussed, one or more transmissions **2922** through the internal network **2999**, and/or one or more corresponding transmissions **3933** through the link(s) **3999***x*, may be associated with the beginning of a performance of the process **3005***x*. More specifically, in various embodiments, there may be one or more transmissions **2922***np* and/or **3933***np* that trigger the commencement of the process **3005***x*. By way of example, there may be transmission(s) **2922***np* that convey operational command(s) between monitored devices **2300** (e.g., the depicted monitored devices **2300***a* and **2300***x*) through the internal network **2999** to prepare for beginning the process **3005***x*, and/or to actually begin the process **3005***x*. Such operational command(s) to the monitored device **2300***x* may, in turn, cause one or more effecting devices **3800** to be commanded with corresponding transmission(s) **3933***np* through the link (s) **3999***x* to perform operations that implement such preparations, and/or that actually begin the process **3005***x*, thereby causing a transition from a non-process state **3006***n* to the depicted first process state **3006***p***1** of the process **3005***x*.

[0139] Alternatively, it may be that the process **3005***x* is responsive to external inputs such that the beginning of a performance of the process **3005***x* may be triggered in a manner that is not based on the occurrence of transmissions. Thus, from the perspective of the internal network **2999** and the link(s) **3999***x*, it may be that the first transmission associated with a performance of the process **3005***x* is a transmission indicating that the process **3005***x* has already begun. By way of example, there may first be transmission (s) **3933***np* through the link(s) **3999***x* that convey operational information that preparations for beginning the process **3005***x* have been completed, and/or that the process **3005***x* has begun such that the transition from the depicted non-process state **3006***n* to the first process state **3006***p***1** has already occurred. Such operational information to the monitored device **2300***x* may, in turn, cause corresponding transmission(s) **2922***np* between the monitored devices **2300***a* and **2300***x* to relay such operational information onward to the monitored device **2300***a*. Such operational information may include data collected by one or more sensing devices **3200** at a time immediately preceding, during and/or immediately after the transition to the process state **3006***p***1**.

[0140] It should be noted that, during the depicted non-process state **3006***n* prior to the process **3005***x* being triggered to begin, there may be one or more transmissions that are not associated within performing the process **3005***x*, at all, such as the depicted one or more transmissions **2922***n* and/or **3933***n*. Such transmissions **2292***n* and/or **3933***n* may be associated with maintaining the external system **3000***x* in the depicted non-process state **3006***n*, such as a distinct "off" state, a "sleep" state, or a "standby" state. As will be familiar to those skilled in the art, such a non-process state **3006***n* may be configured to minimize the consumption of energy, to maintain device(s) in a known inactive state, and/or to maintain substance(s) in a known safe storage state. Such a minimal consumption of energy may be directed toward maintaining a cache of pre-loaded data in readiness for a future performance of the process **3005***x*, and/or such a safe storage state may preserve substances in a condition for use in a future performance of the process **3005***x*.

[0141] By way of example, there may be transmission(s) **2922***n* that convey operational command(s) between monitored devices **2300***a* and **2300***x* through the internal network **2999** to effect and/or maintain the non-process state **3006***n*. Such operational command(s) may, in turn, cause one or more effecting devices **3800** to be commanded via transmission(s) **3933***n* through the link(s) **3999***x* to perform operations that implement and/or maintain various aspects of the non-process state **3006***n*. Alternatively or additionally, there may be transmission(s) **3933***n* through the link(s) **3999***x*, and corresponding transmissions **2922***n* through the internal network **2999** that convey operational information to the monitored device **2300***a* concerning aspects of maintaining the non-process state **3006***n*. Such operational information may include data collected by one or more sensing devices **3200** about the ongoing preservation of data, and/or about the ongoing preservation of substance(s) in storage.

[0142] For purposes of monitoring transmissions that pass through the internal network **2999** as an approach to monitoring the performance of a process occurring within an external system **3000**, such as the example process **3005***x* occurring within the external system **3000***x*, it may be that a particular transmission **2922***n* associated with the non-process state **3006***n*, a particular transmission **2922***np* associated with the transition from the non-process state **3006***n* to the process state **3006***p***1**, or a combination of multiple transmissions **2922***n* and/or **2922***np*, is selected to serve as an indication, that is observable on the internal network **2999**, that the process **3005***x* has begun. Thus, by monitoring transmissions **2922** occurring on the internal network **2999** (through one or more interchange devices **2700**), a monitoring device **1500** may identify instances in which such particular transmissions **2922***n* and/or **2922***np* (or such a combination thereof) have occurred on the internal network **2999**, and use such instances as an indication of when to begin monitoring the internal network **2999** for transmissions **2922** associated with the process **3005***x*.

[0143] Just as some transmissions **2922** and/or **3933** may be associated with the commencement of a performance of a process, others may be associated with the performance progressing between process states. Returning to the example process **3005***x*, there may be various transmissions **3933***pp* through the link(s) **3999***x*, and/or transmissions **2922***pp* through the internal network **2999**, that may be associated with transitioning between the process states **3006***p***1** and **3006***p***2**. More precisely, there may be transmission(s) **2922***pp* and/or **3933***pp* that are associated with the process state **3006***p***1** ending (with a presumption of the next process state beginning), and/or there may be transmission (s) **2922***pp* and/or **3933***pp* that are associated with the process state **3006***p***2** beginning (with a presumption of the preceding process state having ended). Somewhat similar to the aforedescribed commencement of the process **3005***x*, this may arise from the fact that some transmission(s) **2922**_pp_ and/or **3933**_pp_ may trigger the ending of the process state **3006**_p_**1**, some may be caused to occur by the ending of the process state **3006**_p_**1**, some may trigger the beginning of the process state **3006**_p_**2**, and/or some may be caused to occur by the beginning of the process state **3006**_p_**2**. Also in a manner somewhat similar to the commencement of the process **3005**_x_, the transmission(s) **2922**_pp_ and/or **3933**_pp_ that trigger the beginning or the ending of a process state may do so by conveying operational command(s) that cause effecting device(s) **3800** to be commanded to perform operations that effectuate such a beginning or ending. Correspondingly, transmissions that are caused to occur by the beginning of a process state, or are caused to occur by the ending of a process state, may convey operational information that includes data collected by one or more sensing devices **3200** associated with such a beginning or ending.

[0144] Additionally, during a process state (e.g., during one of the depicted process states **3006**_p_**1** or **3006**_p_**2**), there may be transmissions **2922**_p_ that convey operational commands and/or operational information between the monitored devices **2300**_a_ and **2300**_x_, and/or corresponding transmissions **3933**_p_ between the monitored device **2300**_x_ and either or both of the sensing device(s) **3200** and/or the effecting device(s) **3800**. Again, such transmissions may convey operational information that may include data collected by sensing devices **3200** that is indicative of various measurements associated with a portion of the process **3005**_x_ that occurs during a process state. Alternatively or additionally, such transmissions may convey operational commands that cause effecting device(s) **3800** to perform various operations during a process state.

[0145] As previously discussed, for the example process **3005**_x_ performed within the external system **3000**_x_, the monitored devices **2300**_a_ and **2300**_x_, together, may implement the logic for controlling the progression of those performances through the various process states thereof. As part of exerting such control, the logic implemented within each of the monitored devices **2300**_a_ and **2300**_x_ functions in a manner that is at least partially responsive to received inputs. While such inputs may include transmissions **2922** and/or **3933** originating from within the monitored system **2000** and/or the external system **3000**_x_, others of such inputs may be external inputs that originate entirely outside of both systems **2000** and **3000**_x_. By way of example, the monitored device **2300**_a_ may provide a user interface (not specifically shown) that may be operable to enable manual control of when a performance of the process **3005**_x_ commences and/or other aspects thereof. By way of another example, components and/or devices within the external system **3000**_x_ may be subject to external influences (e.g., availability of various external resources, weather conditions, etc.) that may constrain, enable and/or disable various aspects of the process **3005**_x_. Further, regardless of what each such external input may be, the timing of each such external input may at least partially determine what effect it has on a performance of the process **3005**_x_.

[0146] FIG. 3B depicts aspects of various causal relationships among various aspects of a performance of a process, such as the depicted process **3005**_x_. More specifically, there may be causal relationships between transmissions **3933** and at least a subset of the process states **3006**_p_. As those skilled in the art will readily recognize, such causal relationships may dictate what transmissions **3933** are to occur on the network **2999**, the type of transmission (e.g., a transmission conveying operational command(s), or a transmission conveying operational information), aspects of the content of each transmission (e.g., which operational command is conveyed, what parameters accompany each command, and/or what data values are included in operational information that is conveyed), and/or the timing of each transmission (e.g., when each transmission is expected to occur on the network **2999**).

[0147] As previously discussed, there may be one or more transmissions **3933**_np_ that may trigger the beginning of a performance of the process **3005**_x_, and various aspects of such transmission(s) **3933**_np_ may be dictated by various requirements associated with the process **3005**_x_. By way of example, it may be that the protocol used to control the process **3005**_x_ dictates that at least one particular operational command be transmitted to triggering the commencement of the process **3005**_x_. This may dictate that a transmission **3933**_np_ of a type that conveys an operational command is required, and that the particular operational command specified by the protocol is the one that is to be conveyed. The same protocol may also dictate one or more of the parameters that are to be included with the particular operational command in that transmission **3933**_np_.

[0148] Again, the transmissions **3933**_n_ that occur during the depicted non-process state **3006**_n_ may be associated with monitoring and/or maintaining aspects of the non-process state **3006**_n_, and may not be associated with the process **3005**_x_. However, it may also be that there is a causality between the occurrence of such a transmission **3933**_n_ during the non-process state **3006**_n_, and a later transmission **3933**_np_ that triggers the commencement of the process **3005**_x_. By way of example, it may be that operational information that is conveyed in one or more of the transmissions **3933**_n_ includes data indicative of measurements taken during the non-process state **3006**_n_ that in some way influences a parameter of an operational command that is later conveyed in a transmission **3933**_np_ that triggers the commencement of the process **3005**_x_. As those skilled in the art will readily recognize, such a situation may arise where a measurement of an aspect of a device and/or of a substance that is taken during the non-process state **3006**_n_ may influence an aspects of how the process **3005**_x_ is to begin, such as a temperature reading taken during the non-process state **3006**_n_ that affects of a parameter for controlling heating or cooling within an operational command for triggering the commencement of the process **3005**_x_. As will be discussed further, there may be logic employed in the control of the process **3005**_x_ that is used to derive a value for such a parameter based on such input as an earlier-collected measurement.

[0149] As also previously discussed, there may be one or more transmissions **3933**_np_ that are caused to occur by the commencement of the process **3005**_x_. By way of example, it may be that the protocol used to control the process **3005**_x_ dictates that operational information be conveyed that includes an indication of success or failure in commencing performance of the process **3005**_x_, along an indication of the type of failure in situations where failure occurs. This may dictate that a transmission **3933**_np_ of a type that conveys operational information is required, as well as dictating what operational information is to be included. The same protocol may also dictate one or more aspects of the formatting of the data values that are used to represent that operational information in that transmission **3933**_np_. As will be discussed further, there may be logic employed in the control of the process **3005***x* that is used to identify such a failure, and thereby determine the particular indication of type of failure that may be included in such operational information.

[0150] In a manner similar to the transmission(s) **3933***np* associated with the commencement of a process **3003**, there may be one or more transmissions **3933***pp* that may trigger a transition between process states **3006**, such as the depicted transition from the process state **3006***p***1** to the process state **3006***p***2** of the depicted process **3005***x*. This may include transmission(s) **3933***pp* that separately trigger the ending of one process state **3006***p*, and/or transmission (s) **3933***pp* that separately trigger the beginning of the next process state **3006***p*, as well as transmission(s) **3933***pp* that may serve both purposes. Also in a similar manner, various aspects of such transmission(s) **3933***pp* may be dictated by various requirements associated with the process **3005***x*, and/or dictated by various requirements of the particular process states **3006** between which the transition occurs. Again, such aspects may include requirements for the transmission(s) **3933***pp* based on protocols uses. Also, and as will be discussed further, there may be logic employed in the control of the process **3005***x* that is used to determine what operational commands and/or what operational information is to be transmitted, used to derive values for parameters of operational commands that may be transmitted, and/or used to derive data values included in operational information that may be transmitted.

[0151] Again, the transmissions **3933***p* that occur during each of process state **3006***p* may be associated with monitoring and/or controlling operations that are performed during each of the process states **3006***p* as part of performing the process **3005***x*. As depicted, there may be causality between such transmissions **3933***p* and preceding transmissions **3933***np* for the commencement of the process **3005***x* and/or preceding transmissions **3933***pp* for a preceding transition between process states **3006***p*. Alternatively or additionally, there may be causality between such transmissions **3933***p* and subsequent transmissions **3933***pp* for a subsequent transition between process states **3006***p*. Also alternatively or additionally, there may be causality among such transmissions **3933***p* associated with a single process state **3006***p*, or among multiple process states **3006***p*. Each of such causalities may include influences that previously transmitted operational commands, parameters of previously transmitted operational commands, and/or data values in previously transmitted operational information, may exert on subsequently transmitted operational commands, parameters of subsequently transmitted operational commands, and/or data values in subsequently transmitted operational information. Again, there may be logic employed in the control of the process **3005***x* that is used to determine what operational commands and/or what operational information is to be transmitted, used to derive values for parameters of operational commands that may be transmitted, and/or used to derive data values included in operational information that may be transmitted.

[0152] FIG. **3**C depicts aspects of various timing relationships among various aspects of a performance of a process, such as the depicted process **3005***x*. More specifically, there may be timing relationships between transmissions **3933** and at least a subset of the process states **3006***p*. The earlier discussed causal relationships may at least partially dictate aspects of the timing of at least a subset of the transmissions **3933***n*, **3933***np*, **3933***p* and/or **3933***pp*.

[0153] More specifically, the start times (Tstart), the stop times (Tstop) and/or the transmission duration times (Txmt) for at least a subset of these transmissions may be at least partially determined by whether each of such transmissions is causes or is caused by the commencement of the process **3005***x*, and/or causes or is caused by a transition between process states **3006***p*. Thus, although FIG. **3**C depicts an example transmission **3933***np* as occurring at least partially simultaneously with the transition from the non-process state **3006***n* and to the process state **3006***p***1**, the entirety of this same example transmission **3933***np* could occur entirely before that transition or entirely afterward. Similarly, although an example transmission **3933***pp* is depicted as occurring at least partially simultaneously with the transition from the process state **3006***p***1** and to another process state **3006***p*, the entirety of this same example transmission **3933***pp* could occur entirely before that transition or entirely afterward.

[0154] Also depicted is an example transmission **3933***n* that occurs entirely within the non-process state time period (Tnps) of the non-process state **3006***n*, and an example transmission **3933***p* that occurs entirely within the process state time period (Tps) of the process state **3006***p***1**.

[0155] However, despite the role that causality in connection with such transitions may play in at least partially dictating timing of various ones of these transmissions **3933***n*, **3933***np*, **3933***p* and/or **3933***pp*, other factors unrelated to such transitions may also play a role. By way of example, during the portion of the performance of the process **3005***x* that occurs during the depicted process state **3006***p***1**, a circumstance may arise that triggers the occurrence of a transmission **3933***p* during that process state that conveys operational information indicative of a milestone in the process **3005***x* having been reached, or of an anomalous event having been detected by one of the sensing devices **3200** (e.g., a high temperature reading, or of a lack of imminent lack of available data storage space). While such a transmission **3933***p* may occur entirely within the Tps of the process state **3006***p***1**, its occurrence may be entirely based on logic used to trigger such notification transmissions, and may not actually be dictated by any direct constraint relative to either the start or end of that time period beyond the fact that such logic may only be used during the process state **3006***p***1**.

[0156] FIGS. **3**D and **3**E present differing examples of responses by the logic **2345***a* and **2345***x* implemented within the monitored devices **2300***a* and **2300***x*, respectively, to various combinations of internal and external inputs. As each of these two figures makes clear, and as will be explained in greater detail, there are time delays inherent in implementing such logic to also be taken into account in providing emulations of computing devices that implement such logic, such as the monitored devices **2300***a* and **2300***x*.

[0157] Turning to FIG. **3**D, the monitored device **2300***a* may implement a UI (not specifically shown) by which an operator of the monitored device **2300***a* may manually input a command to cause the process **3005***x* being performed within the external system **3000***x* to progress from process state **3006***p***1** to process state **3006***p***2**. By way of example, such an operator may enter a command to perform an action such as heat a tank filled with water in the external system **3000***x*.

[0158] It is important to note that the timing of such manual input by the operator may be extremely difficult, if not impossible, to predict. This, in turn, may make the timing of when the first transmission in the depicted series of transmissions similarly difficult to predict. However, the timings of at least some of the following transmissions relative to each other may be far more easily predicted, as will be explained in greater detail.

[0159] In response to receiving such manually entered input from the operator, the logic **2345***a* implemented within the monitoring device **2300***a* may cause the monitored device **2300***a* to output a transmission **2922**cmd onto the internal network **2999** that conveys an operational command to the monitored device **2300***x* to do the commanded action (e.g., heat a tank). As will be explained in greater detail, such a response to such manual input may be an example of a causal relationship implemented as part of the logic **2345***a* within the monitored device **2300***a* for controlling the process **3005***x*.

[0160] In response to receiving the transmission **2922**cmd, the monitored device **2300***x* may relay the operational command to do the commanded action onward in a transmission **3933**cmd**1** to the effecting device **3800** of the external system **3000***x* via the one or more communication links **3999***x*. As depicted, there is a time delay within the monitored device **2300***x* between the time of receipt of the transmission **2922**cmd and the time of output of the transmission **3933**cmd**1**.

[0161] As will be explained in greater detail, this relaying of the operational command, through the monitored device **2300***x*, and from the transmission **2922**cmd to the transmission **3933**cmd**1**, is an example of a causal relationship implemented as part of the logic **2345***x* within the monitored device **2300***x* for controlling the process **3005***x*. Stated differently, the logic **2345***x* implemented within the monitored device **2300***x* may include logic to cause such relaying of an operational command of this type through the monitored device **2300***x*, and onward to the external system **3005***x*.

[0162] In response to receiving the transmission **3933**cmd**1**, the effecting device **3800** may act on the operational command therein to do the commanded action (e.g., turning on a heating device to heat a tank). The effecting device **3800** may output, onto the links **3999***x* and back to the monitored device **2300***x*, a transmission **3933**conf**1** conveying operational information that confirms that the operational command is being acted upon by doing the commanded action.

[0163] In response to receiving the transmission **3933**conf**1**, the monitored device **2300***x* may relay the operational information confirming that the operational command is being acted on onward in a transmission **2922**conf**1** to the monitored device **2300***a* via the internal network **2999**. As depicted, there is again a time delay within the monitored device **2300***x* between the time of receipt of the transmission **3933**conf**1** and the time of output of the transmission **2922**conf**1**. Via the UI, the operator may be informed of this received confirmation.

[0164] Within the external system **3000***x*, one or more sensing devices **3200** may output a succession of transmissions indicative of the results, over time, of the commanded action being performed. More specifically, as the tank is being heated, each transmission in a series of transmissions **3933**sen**1** through **3933**senx may convey operational information concerning the increasing temperature of the tank to the monitored device **2300***x* via the communication link(s) **3999***x*. The transmission **3933**senx may convey operational information indicating that the tank has reached a particular temperature that is associated with a transition to the process state **3006***p***2**.

[0165] The amount of time that it takes to heat the tank to the particular temperature may be subject to external inputs that make such an amount of time relatively difficult to predict. More specifically, if the tank is part of an apparatus that is exposed to the elements, then the amount of time required to heat the tank to the particular temperature may be affected by the weather. Thus, the quantity of transmissions within the series of transmissions **3933**sen**1** to **3933**senx may be difficult, if not impossible, to predict, and this may make the timing of when the transmission **3933**senx is output onto the link(s) **3999***x* similarly difficult to predict.

[0166] In response to receiving the transmission **3933**senx, the monitored device **2300***x* may output a transmission **3933**cmd**2** conveying an operational command to the same effecting device **3800** to stop acting on the earlier operational command in the earlier transmission **3933**cmd**1**. Thus, the effecting device **3800** may be commanded to stop heating the tank. Again, there is a time delay within the monitored device **2300***x* between the time of receipt of the transmission **3933**senx and the time of output of the transmission **3933**cmd**2**.

[0167] In response to receiving the transmission **3933**cmd**2**, the effecting device **3800** may act on the operational command therein to cease doing the commanded action (e.g., cease heating the tank). The effecting device **3800** may output, onto the link(s) **3999***x* and back to the monitored device **2300***x*, a transmission **3933**conf**2** conveying operational information that confirms that the earlier operational command to heat the tank is now no longer being acted upon.

[0168] In response to receiving the transmission **3933**conf**2**, the monitored device **2300***x* may relay the operational information confirming that the original operational command to heat the tank has ceased to be acted upon onward in a transmission **2922**conf**2** to the monitored device **2300***a* via the internal network **2999**. Again, there is a time delay within the monitored device **2300***x* between the time of receipt of the transmission **3933**conf**2** and the time of output of the transmission **2922**conf**2**. Via the UI, the operator may be informed of this received confirmation.

[0169] Turning to FIG. **3**E, during a performance of the process **3005***x*, a condition may develop that necessitates stopping the process **3005***x* as a safety measure. By way of example, it may be that a temperature of a device within the external system **3000***x* exceeds a threshold, as detected by sensing device **3200**, and the logic **2345***a* and **2345***x* employed by the monitored devices **2300***a* and **2300***x*, respectively, may respond by automatically stopping the process **3005***x* and providing an alert to an operator.

[0170] In a manner similar to the timing of manual input in the above example of FIG. **3**D, the timing of when such a condition as overheating may occur may be at least extremely difficult, if not impossible, to predict. This, in turn, may make the timing of when a transmission responsive to such an event is output similarly difficult to predict. However, it should again be noted that the timings of at least some of the transmissions relative to each other may be far more easily predicted, as will be explained in greater detail.

[0171] During a process state **3033***p* of the process **3005***x*, one or more sensing devices **3200** of the external system **3000***x* may output a series of transmissions **3933**sen**1** through **3933**senx onto the communication link(s) **3999** and to the monitored device **2300***x*. Each of these transmissions in this series may convey operational information concerning the status of one or more aspects of the process **3005***x*, which may include a temperature of a device of the external system **3005***x*.

[0172] As depicted, the process **3005***x* may go awry such that a transition is made from the depicted process state **3033***p* in which the process **3005***x* may be proceeding normally, and to an improper state **3006***i* in which the process **3005***x* is no longer proceeding normally. For purposes of this example, it may be that the temperature of that same device of the external system **3005***x* has risen beyond a particular threshold.

[0173] In response to receiving the transmission **3933**senx, the logic **2345***x* implemented within the monitored device **2300***x* for controlling the process **3005***x* may output two transmissions. One of the two transmissions may be a transmission **2922**alert output onto the internal network **2999** to convey operational information to the monitored device **2300***a* that includes an alert that a condition such as an excessive temperature level of a device has arisen. Logic **2345***a* within the monitored device **2300***a* for controlling the process **3005***x* may present the alert to an operator via a UI (not specifically shown). The other of the two transmission may be a transmission **3933**cmd that is output onto the link(s) **3999***x* to convey an operational command to an effecting device **3800** within the external system **3000***x* to perform an action that stops the process **3005***x*.

[0174] As depicted, the output of each of the transmissions **2922**alert and **3933**cmd occurs with some amount of delay following receipt of the transmission **3933**senx. As also depicted, the delay in the output of the transmission **3933**cmd may be somewhat longer as a result of a need for the logic **2345***x* implemented within the monitored device **2300***x* to determine whether the process **3005***x* is to be stopped, and if so, what measure(s) to take to cause the process **3005***x* to be stopped.

[0175] In response to receiving the transmission **3933**cmd, an effecting device **3800** of the external system **3000***x* may act on the operational command therein to cause the process **3005***x* to be stopped. The effecting device **3800** may output, onto the link(s) **3999***x* and back to the monitored device **2300***x*, a transmission **3933**conf conveying operational information that confirms that the operational command to stop the process **3005***x* is now being acted upon.

[0176] In response to receiving the transmission **3933**conf, the monitored device **2300***x* may relay the operational information confirming that the process **3005***x* is being stopped onward in a transmission **2922**conf to the monitored device **2300***a* via the internal network **2999**. Again, there is a time delay within the monitored device **2300***x* between the time of receipt of the transmission **3933**conf and the time of output of the transmission **2922**conf. Via the UI, the operator may be informed of this received confirmation.

Correlations Among Transmissions and Process States—Update Download Control

[0177] Each of FIGS. **4**A, **4**B and **4**C present aspects of a separate example of the manner in which transmissions **2922** and **3933**, an update download state machine, and/or logic employed within different monitored devices or external system correlate and interact to control a performance of an update download process.

[0178] Referring to all three of FIGS. **4**A-C, as previously discussed, just as the earlier example process **3005***x* performed within the external system **3000***x* may progress through a set of process states (e.g., a tree process states of a state machine), the same may also be true of an update download process **3005***ud* that may also be performed within the example external system **3000***x*, and/or may also be true of an update download process **2305***ud* that may be performed within the example monitored device(s) **2300***a* and/or **2300***x*.

[0179] In a manner also similar to the earlier example process **3005***x*, one or more transmissions **2922** through the internal network **2999**, and/or one or more corresponding transmissions **3933** through the link(s) **3999***x*, may be associated with the beginning of a performance of the process **3005***x*. More specifically, in various embodiments, there may be one or more transmissions **2922** and/or **3933** that trigger the commencement of the update download process **2305***ud* and/or **3005***ud*. By way of example, there may be various transmission(s) **2922** that convey operational command(s) among the monitored devices **2300***a* and/or **2300***x*, and/or that convey operation command(s) between those monitored devices and the external system **3000***x* that are meant to trigger start of an update download process **2305***ud* or **3005***ud*, and/or that are meant to cause a cessation of one or more other processes in preparation for such an update download process.

[0180] As those familiar with the updating of low level firmware, operating system routines, libraries that support the execution of application routines, and/or the application routines, will readily recognize, it is often necessary to cause the performances of one or more other processes to cease before engaging in such an update to avoid any of a variety of unpredictable behaviors that may be caused by an updating process. By way of example, it may be that an update process causes one or more hardware components and/or software routines to become inaccessible such that other process(es) may become uncontrollable, unpredictably change to an improper state, remain stuck in a particular process state for longer than is proper, etc. Hardware components that may become inaccessible may include storage devices, interrupt controllers, network interfaces, etc. Alternatively or additionally, software links between differing software routines may become broken such that necessary cooperation between those routines may be entirely lost. Thus, in situations where there is a seemingly well-founded belief that a particular update download process is able to be performed within a particular and/or system without a need to cease the performance(s) of other process(es), it may be regarded as a best practice to do so in any case.

[0181] As an alternative to one or more transmissions **2922** and/or **3933** triggering the commencement of an update download process **2305** or **3005**, and also in a manner similar to the earlier example process **3005***x*, it may be that the commencement of an update download process is responsive to external inputs such that the beginning of a performance of the process **3005***x* may be triggered in a manner that is not based on the occurrence of such transmissions. Thus, from the perspective of the internal network **2999** and the link(s) **3999***x*, it may be that the first transmission **2922** or **3933** associated with a performance of an update download process **2305***ud* or **3005***ud* is a transmission indicating that such a process has already begun. By way of example, there may first be transmission(s) **2922** and/or **3933** that convey operational information that preparations for beginning an update download process have been completed, and/or that such a process has begun such that the transition to a first process state associated with such a process has already occurred.

[0182] As previously discussed in reference to the earlier example process **3005***x*, it may be that a monitoring device **2300** and/or an external system **3000** provides a user interface (UI) by which an operator thereof may manually trigger a download of an updated routine from an external server **9900**. Alternatively or additionally, there may be a recurring moment or interval of time (e.g., a particular hour each night, a particular day of each week, a particular date of each month, etc.) at which at least a check for the availability of such downloads is made (which may be immediately followed by the occurrence of such downloads). Thus, the trigger for downloading an updated routine may, at least in part, be a timing component within a monitoring device **2300** and/or an external system **3000**.

[0183] Thus, in a manner that is further similar to the earlier example process **3005***x*, in monitoring network activity on the internal network **2999**, it may be one or more particular transmissions **2922** associated with a non-process state, one or more particular transmissions **2922** associated with a first process state of an update download process **2305***ud* or **3005***ud*, or a combination thereof, that serves as the indicator used by the monitoring device(s) **1500** that an update download process **2305***ud* or **3005***ud* has begun. Alternatively or additionally, it may be a combination of one or more transmissions **2922** associated with the cessation of one or more other unrelated processes, and one or more transmissions **2922** associated with the commencement of an update download process that serves as the indicator of that commencement of that update download process. Also alternatively or additionally, it may be that a combination of one or more particular transmissions **2922**, and particular timing (e.g., a recurring interval of time, a time of each day, a day of each week, a date in each month, etc.) that serves as the indicator that commencement of that update download process.

[0184] Additionally similar to the earlier example process **3005***x*, it may be particular transmissions **2922** that serve as indicators of transitions between process states of such an update download process. Again, each of such transmissions **2922** associated with such a transition may be a transmission **2922** that is connected to activity occurring at the end of a previous process state, that is connected to activity occurring at the beginning of a next process state, or that is connected to explicitly triggering and/or explicitly indicating a transition therebetween.

[0185] However, despite such similarities that may exist in correlations between process states and observable transmissions **2922** and/or **3933**, the update download processes **2305***ud* and/or **3005***ud* may differ from the earlier example process **3005***x* in that the control of the update download processes **2305***ud* and/or **3005***ud* may not involve so much cooperation through the internal network **2999**. More specifically, and as previously discussed, control of the example process **3005***x* performed within the external system **3000***x* involved cooperation among at least the two monitored devices **2300***a* and **2300***x* through the internal network **2999**. In contrast, the complexity of the control required to simply download an updated routine may be far less, and indeed, may be made up largely of repeated transmissions of a very limited variety of operational commands that serve as little more than an ongoing handshake. Thus, the logic and/or state machine that may be implemented within one of the monitored devices **2300***a* or **2300***x*, or that may be implemented within the external system **3000***x*, to receive an updated routine may be relatively simple in comparison with the multiple logic(s) and/or state machine(s) implemented to control the earlier example process **3005***x* using the combination of monitored devices **2300***a* and **2300***x*.

[0186] Each of FIGS. **4**A-C present a differing example of actions and responses to actions implemented within the monitored devices **2300***a* and **2300***x*, and the external system **3000***x* associated with downloading an updated routine from an external server **9900**. Also, as each of these figures makes clear, there are time delays inherent in implementing logic and/or a state machine that are to be taken into account in providing emulations thereof.

[0187] FIG. **4**A presents an example of an update download process **3005***ud* implemented as a state machine that is performed within the external system **3000***x* to download an updated routine from an external server **9900**.

[0188] As depicted, there may be various transmissions **2922***n* and/or **3933***n* that occur at a time before the update download process **3005***ud* begins within the external system **3000***x* with the entry into the first process state **3006***p***1** of the update download process **3005***ud*. More specifically, and as depicted, it may be that no process was being performed within the external system **3000***x* during a non-process state **3006***n* in the time leading up to the commencement of the update download process **3005***ud*.

[0189] As depicted, the update download process **3005***ud* may commence (starting with the first process state **3006***p***1**) without a trigger from a transmission **2922** or **3933** of any type. Instead, it may be that the first process state **3006***p***1** is triggered to begin by another external input, such as the previously discussed commanded beginning by an operator through a UI of the external system **3000***x*, or the previously discussed timed trigger to begin. Regardless of the exact manner in which the first process state **3006***p***1** of the update download process **3005***ud* is caused to begin, it may be that a function of the process state **30006***p***1** is to cause the cessation of performances of one or more other processes within the external system **3000***x*. Again, this may be done as a best practice to prevent a situation in which there is unpredictable behavior in the performance of a process that might cause damage to property and/or injury to personnel. Thus, during the process state **3006***p***1**, the external system **3000***x* may be caused to output, onto the link(s) **3999***x* and to the monitored device **2300***x*, a transmission **3933**cmd**1** conveying an operational command to cause a cessation in performance of one or more other processes that may be performed within the external system **3000***x* under the control of the monitored devices **2300***a* and **2300***x*.

[0190] As previously discussed in connection with the example process **3005***x*, there may be cooperation among at least the monitored devices **2300***a* and **2300***x* that provides the control over the example process **3005**x and/or other processes that may be performed within the external system **3000**x. Thus, such a transmission **3933**cmd**1** through the link(s) **3999**x, and directed to the device(s) that exercise control over processes performed within the external system **3000**x may be necessary to cause performance(s) of such processes to stop.

[0191] In response to receiving the transmission **3933**cmd**1** at the monitored device **2300**x, the logic **2345**x implemented therein may cause the monitored device **2300**x to relay the operational command onward to the monitored device **2300**a in a transmission **2922**cmd**1** thereto via the internal network **2999**. The logic **2345**x may also respond by at least ceasing to cause the monitored device **2300**x to output, onto the link(s) **3999**x and to the external system **3000**x, transmissions **3933** that would cause one or more processes (other than the update download process **3005**ud) to continue to be performed within the external system **3000**x.

[0192] The logic **2345**x may further respond to the receipt of the transmission **3933**cmd**1** by causing the monitored device **2300**x to output, onto the link(s) **3999**x and to the external system **3000**x, one or more further transmissions **3933** (not specifically shown) to actively take one or more steps to cause the cessation of performance(s) of one or more processes that might currently be underway within the external system **3000**x. Alternatively or additionally, the logic **2345**x may further respond by causing the monitored device **2300**x to output, onto the link(s) **3999**x and to the external system **3000**x, one or more further transmissions **3933** (not specifically shown) to cause various components of the external system **3000**x to be placed into a known stable state. As previously discussed, this may entail a variety of actions, including and not limited to, powering down various components, stopping the movement of various components, moving various components to a known stable condition, setting conditions of various components to enable data and/or various substances to be maintained at known stable conditions, etc. The various known stable states into which various components may be placed may be specifically selected to enable one or more other processes **3005** to be relatively easily resumed after the update download process **3005**ud is completed.

[0193] In response to receiving the transmission **2922**cmd**1** at the monitored device **2300**a, the logic **2345**a implemented therein may at least cease to cause the monitored device **2300**a to output, onto the internal network **2999** and to the monitored device **2300**x, further transmissions **2922** that would cause the monitored device **2300**x to output further transmissions **3933** to the external system **3000**x that would cause one or more processes (other than the update download process **3005**ud) to continue to be performed within the external system **3000**x. Upon causing at least such a cessation of such transmissions, the logic **2345**a may also respond by causing the monitored device **2300**a to output, onto the internal network **2999** and back to the monitored device **2300**x, a transmission **2922**conf**1** conveying operational information indicating that the command to cause the cessation of performance of one or more other processes within the external system **3000**x has been received and acted upon at the monitored device **2300**a.

[0194] In response to receiving the transmission **2922**conf**1** at the monitored device **2300**x, and upon causing at least the cessation of transmissions to the external system **3000**x that would cause one or more processes (other than the update download processes **3005**ud) to continue to be performed therein, the logic **2345**x may cause the monitored device **2300**x to output, onto the link(s) **3999**x and back to the external system **3000**x, a transmission **3933**conf**1** conveying operational information indicating that the command to cause the cessation of performance of one or more other processes within the external system **3000**x has been received and acted upon at both of the monitored devices **2300**a and **2300**x.

[0195] As depicted, and as previously discussed in connection with FIGS. **3**B-C, delays may be incurred, not just in action being taken in response to a received operational command, but also in simply relaying an operational command. More specifically, and by way of example, the simple act of relaying the operational command received via the transmission **3933**cmd**1**, through the logic **2345**x, and onward in the transmission **2922**cm**1** may take an appreciable amount of time that may be of significance in distinguishing between proper and improper network activity.

[0196] The receipt of the transmission **3933**conf**1** at the external system **3000**x may serve as the trigger for transitioning from the process state **3006**p**1**, and to the process state **3006**p**2**. Regardless of the exact manner in which the process state **3006**p**2** of the update download process **3005**ud is caused to begin, it may be that a function of the process state **30006**p**2** is to check whether an updated routine of a sought after type is currently available from the external server **9900**. To perform such a check, the external system **3000**x may be caused to output, onto the link(s) **3999**x and to the monitored device **2300**x, a transmission **33933**query**1** conveying an operational command for the external server **9900** to provide an indication of whether such an updated routine is currently available.

[0197] In response to receiving the transmission **3933**query**1** at the monitored device **2300**x, the logic **2345**x implemented therein may cause the monitored device **2300**x to relay the operational command onward to the external server **9900** in a transmission **2922**query**1** thereto via at least the internal network **2999**. As previously discussed, such external servers **9900** may be accessible through a connection between the internal network **2999** and an external network **9992** (see FIG. **1**A), such as the Internet. In various embodiments, the external system **3000**x may have been previously provided with an IP address or other identifier that may be used in the transmission **3933**query**1** to specify the particular external server **9900** as the intended destination of the transmission **3933**query**1**. In relaying the operational command received in the transmission **3933**query**1**, the logic **2345**x may also relay that same IP address or other identifier onward in the transmission **2922**query**1**.

[0198] In response to receiving the transmission **2922**query**1** at the external server **9900**, logic **9954** that may be implemented therein may cause the external server **9900** to respond with a transmission **2922**resp**1** that conveys operational information that answers to the check of whether such an updated routine is currently available. As depicted, such an answer may be a simple binary yes or no. The received transmission **2922**query**1** may have included an indication of an identifier (e.g., an IP address or other identifier) of a device to which the responding transmission **2922**resp**1** may be directed so as to cause the responding transmission **2922**resp**1** to ultimately be received at the monitored device **2300**x.

[0199] In response to receiving the transmission **2922**resp**1** at the monitored device **2300***x*, the logic **2345***x* implemented therein may cause the monitored device **2300***x* to relay its operational information onward to the external system **3000***x* in a transmission **3933**resp**1** thereto via the link(s) **3999***x*. Again, as depicted, various delays may be incurred at various steps in the round trip of transmissions **3933**query**1**, **2922**query**1**, **2922**resp**1** and **3933**resp**1**.

[0200] The receipt of the transmission **3933**resp**1** at the external system **3000***x* may serve as the trigger for transitioning from the process state **3006***p***2**, and to the process state **3006***p***3**. Regardless of the exact manner in which the process state **3006***p***3** of the update download process **3005***ud* is caused to begin, it may be that a function of the process state **30006***p***3** is to trigger the beginning of the actual download of an updated routine from the external server **9900**, if the transmission **3933**resp**1** conveys an indication that there is such an updated routine currently available. Presuming that there is such an updated routine currently available, the external system **3000***x* may be caused to output, onto the link(s) **3999***x* and to the monitored device **2300***x*, a transmission **3933**cmd**2** conveying an operational command to the external server **9900** for the download to begin.

[0201] As depicted, that operational command may be relayed onward to the external server **9900** by the monitored device **2300***x* as a transmission **2922**cmd**2** output onto the internal network **2999**, and onward to external server **9900**. As further depicted, in response to receiving the transmission **2922**cmd**2**, the logic **9954** of the external server **9900** may commence outputting further transmissions (not specifically shown) that convey portions of an updated routine as portions of operational data.

[0202] FIG. **4**B presents an example of an update download process **2305***ud* implemented as a state machine that is performed within the monitored device **2300***x* to download an updated routine from an external server **9900**. Thus, a substantial difference between what is depicted in the previous example of FIG. **4**A versus what is depicted in this example of FIG. **4**B, is a trading of roles between the monitored device **2300***x* and the external system **3000***x*. More specifically, in FIG. **4**B, it is the monitored device **2300***x* that is receiving the download of an updated routine, and not the external system **3000***x*, as was the case in FIG. **4**A. However, despite this substantial difference, the progression of process states **2306***p***1**, **2306***p***2**, **2306***p***3**, etc., and the corresponding progression of transmissions **2922** and **3933** depicted in FIG. **4**B is similar in many ways to what was depicted in FIG. **4**A.

[0203] Again, it may be deemed to be a best practice to cease the performances of one or more processes **3005** within the external system **3000***x* to prevent unpredictable activity therein. However, in this example, the concern over occurrences of unpredictable activity arises in this example of FIG. **4**B from the fact that the monitored device **2300***x* is the one of the monitored devices **2300** that is in direct communication with the external system **3000***x*, and from the fact that the monitored device **2300***x* is involved in controlling processes **3005** that are performed within the external system **3000***x*. Thus, the risk that updating a routine executed within the monitored device **2300***x* may cause the monitored device **2300***x* to, itself, behave unpredictably may be deemed a sufficient cause for placing components of the external system **3000***x* in a known stable state before such updating occurs.

[0204] Thus, as depicted, the update download process **2305***ud* may be begin with a first process state **2306***p***1** in which transmissions **2922**cmd**1** and **3933**cmd**1** are output that convey operational commands to cause such a cessation in performance of one or more processes **3005** within the external system **3000***x*. More specifically, the transmission **3933**cmd**1** may convey operational command(s) to the external system **3000***x* to place various components in known stable states, while the transmission **2922**cmd**1** may convey operational command(s) to the monitored device **2300***a* to cease the output of transmission that would cause one or more processes to continue to be performed therein. As further depicted, as the monitored device **2300***a* and the external system **3000***x* complete taking action on those operational commands, they may output transmissions **2922**conf**1** and **3933**conf**1** conveying indications back to the monitored device **2300***x* that the operational commands conveyed in the transmissions **2922**cmd**1** and **3933**cmd**1**, respectively, have been received and acted upon.

[0205] The receipt of the transmissions **2922**conf**1** and **3933**conf**1** at the monitored device **2300***x* may serve as the trigger for transitioning from the process state **2306***p***1**, and to the process state **2306***p***2**. Similar to the process state **3006***p***2** of update download process **3005***ud*, and regardless of the exact manner in which the process state **2306***p***2** of the update download process **2305***ud* is caused to begin, it may be that a function of the process state **23006***p***2** is to check whether an updated routine of the one or more types that may be sought is currently available from the external server **9900**.

[0206] However, and as depicted, communications between the monitored device **2300***x* and the external server **9000** may be more direct than the communications between the external system **3000***x* and the external server **9000** in the example of FIG. **4**A. During the process state **2306***p***2**, transmissions **2922**query**1** and **2922**resp**1** may be output and received as part of checking whether an updated routine of a sought after type is currently available from the external server **9900**. Presuming that such an updated routine is currently available, then the actual download thereof may be triggered to begin by the output and receipt of the transmission **2922**cmd**2** conveying an operational command to the external server **9900** to begin that download.

[0207] FIG. **4**C presents another example of an update download process **2305***ud* implemented as a state machine, but which is performed within the monitored device **2300***a*, instead of the monitored device **2300***x*, to download an updated routine from an external server **9900**. Thus, a substantial difference between what is depicted in the previous example of FIG. **4**B versus what is depicted in this example of FIG. **4**C, is a trading of roles between the monitored devices **2300***a* and **2300***x*. More specifically, in FIG. **4**C, it is the monitored device **2300***a* that is receiving the download of an updated routine, and not the monitored device **2300***x*, as was the case in FIG. **4**B. However, despite this substantial difference, the progression of process states **2306***p***1**, **2306***p***2**, **2306***p***3**, etc., and the corresponding progression of transmissions **2922** and **3933** depicted in FIG. **4**C is similar in many ways to what was depicted in FIG. **4**B.

[0208] Again, it may be deemed to be a best practice to cease the performances of one or more processes **3005** within the external system **3000***x* to prevent unpredictable activity therein. In a manner somewhat similar to the example of FIG. **4**B, the concern over occurrences of unpredictable activity in this example of FIG. **4**C arises from the fact that the monitored device **2300***a* is involved in controlling processes **3005** that are performed within the external system **3000***x*. Thus, the risk that updating a routine executed within the monitored device **2300***a* may cause the monitored device **2300***a* to, itself, behave unpredictably may be deemed a sufficient cause for placing components of the external system **3000***x* in a known stable state before such updating occurs.

[0209] Thus, as depicted, the update download process **2305***ud* may be begin with a first process state **2306***p***1** in which transmissions **2922**cmd**1** and **3933**cmd**1** output that convey and relay an operational command to cause such a cessation in performance of one or more processes **3005** within the external system **3000***x*. More specifically, and in a manner that is more similar to the example of FIG. **4**A, the transmission **2922**cmd**1** may convey the operational command from the monitored device **2300***a*, and to the operational device **2300***x*, while the transmission **3933**cmd**1** may relay that operational command onward from the monitored device **2300***x*, and to the external system **3000***x*. As further depicted, upon completing the operational command, the external system **3000***x* may output a transmission **3933**conf**1** conveying an indication to the monitored device **2300***x* that the operational command relayed in the transmission **3933**cmd**1** was received and acted upon. As also further depicted, upon receiving the transmission **3933**conf**1** and upon completing the operational command, the monitored device **2300***x* may output a transmission **2922**conf**1** conveying an indication to the monitored device **2300***a* that the operation command conveyed in the transmission **2922**cmd**1** has been received and acted upon by both the monitored device **2300***x* and the external system **3000***x*.

[0210] The receipt of the transmission **2922**conf**1** at the monitored device **2300***a* may serve as the trigger for transitioning from the process state **2306***p***1**, and to the process state **2306***p***2**. Again, regardless of the exact manner in which the process state **2306***p***2** of the update download process **2305***ud* is caused to begin, it may be that a function of the process state **23006***p***2** is to check whether an updated routine of the one or more types that may be sought is currently available from the external server **9900**.

[0211] In a manner that is more similar to the example of FIG. **4**B, communications between the monitored device **2300***a* and the external server **9000** may be more direct than the communications between the external system **3000***x* and the external server **9000** in the example update download process **3005***ud* of FIG. **4**A. Again, during the process state **2306***p***2**, transmissions **2922**query**1** and **2922**resp**1** may be output and received as part of checking whether an updated routine of a sought after type is currently available from the external server **9900**. Presuming that such an updated routine is currently available, then the actual download thereof may be triggered to begin by the output and receipt of the transmission **2922**cmd**2** conveying an operational command to the external server **9900** to begin that download.

Gathering Data to Use in Generating Emulations—Gathering Phase

[0212] FIGS. **5**A, **5**B and **5**C, taken together, present various aspects of the gathering phase of preparing one or more monitoring devices **1500** for use in monitoring and/or correcting improper network traffic through the internal network **2999**. Again, at a time prior to the pre-training phase (to be discussed in reference to FIGS. **6**A-F), one or more gathering devices **5700** may be temporarily coupled to one or more interchange devices **2700**, and/or one or more relay devices **3700***x* of the link(s) **3999***x*, to generate the observation data **5730** for being provided to one or more emulation devices **5500**. Also, the emulation device(s) **5500** may also be provided with the configuration data **5530** descriptive of aspects of at least monitored device **2300** (e.g., the monitored devices **2300***a* and **2300***x*), external systems **3000** (e.g., the external system **3000***x*), and/or processes performed within monitored devices **2300** (e.g., the process **2305***ud*) and/or performed within external systems **3000** (e.g., the processes **3005***ud* and/or **3005***x*).

[0213] Turning to FIG. **5**A, as previously discussed, to generate such emulations of portions of the monitored system **2000** (e.g., the monitored devices **2300***a* and/or **2300***x*), the process(es) **2305** performed within one or more of the monitored devices **2300** (e.g., the process **2305***ud*), the one or more of the external systems **3000** (e.g., the external system **3000***x*), and/or the process(es) **3005** performed within one or more of the external systems **3000** (e.g., the processes **3005***ud* and/or **3005***x*), among the data required is the observation data **5730** that documents at least transmissions **2922** through the internal network **2999** among the monitored devices **2300** within which process(es) **2305** are performed and/or among the monitored devices that control the process(es) **3005** performed within the external system(s) **3000**. It may be deemed preferable that such documented transmissions **2922** include whole sets of transmissions **2922** associated with multiple complete performances those processes **2305** and/or **3005**. In this way, variations in transmission timing, parameters in transmissions of operational commands, and/or data values in transmissions of operational information that may be normal variations may be taken into account in such emulations. In turn, this may enable the monitoring device(s) **1500** to be prepared to distinguish such normal variations from anomalous variations that are at least more likely to be associated with a malfunction or a cyber attack.

[0214] As also previously discussed, it may also be deemed desirable to, if possible, cause the observation data **5730** to similarly document whole sets of corresponding transmissions **3933** that are also associated with the same multiple complete performances of those same processes **2305** and/or **3005**. The inclusion of information concerning the content and timing of the transmissions **3933** may enable the emulation of the progression of a process **2305** (e.g., the process **2305***ud*) or a process **3005** (e.g., the processes **3005***ud* and/or **3005***x*) through its process states to be more accurate.

[0215] Again, considerable importance is usually placed on not taking action that may impair the functionality of the one or more links **3999** that couple an external system **3000** to one of the monitored devices **2300**. This arises from the fact that the link(s) **3999** often directly couple the sensing devices **3200** and effecting devices **3800** that most directly effect changes in process state, including changes to a pre-selected non-process state in response to an unexpected condition that may damage equipment and/or create a hazard. Thus, attaching devices to the link(s) **3999** of an external system **3000** as part of capturing and documenting the transmissions **3933** that pass therethrough may be deemed to present an unacceptable risk. As a result, at least where such components as the relay device(s) **3700***x* are not available and/or not able to be coupled to, it may be deemed more desirable to obtain such information concerning such transmissions **3933** from log data that may be generated and maintained by the monitored device(s) **2300** that are directly involved in such communications (e.g., the depicted log data **2330***x* generated within the monitored device **2300***x* for transmissions **3933** associated with the external system **3000***x*).

[0216] Turning to FIG. **5**B, as previously discussed, each of the control routines **2340***a* and **2340***x* of the monitored devices **2300***a* and **2300***x* may implement logic **2345***a* and **2345***x*, respectively, that are each part of the logic employed to perform process(es) **2305** thereat (e.g., the process **2305***ud*) and/or employed to control performances of process(es) **3005** performed within the external system **3000***x* (e.g., the processes **3005***ud* and/or **3005***x*). Thus, the configuration data **5530** may be generated to include device data **2303***a* concerning the monitored device **2300***a*, device data **2303***x* concerning the monitored device **2300***x*, and/or system data **3003***x* concerning the external system **3000***x*.

[0217] In some embodiments, each of the device data **2303***a* and **2303***x* may include indications of various characteristics of the hardware and/or software of its corresponding one of the monitored devices **2300***a* and **2300***x*. Such characteristics may include, and are not limited to, the type and/or revision level of the processor(s); the type and quantity of volatile and/or non-volatile storage; the type and revision of operating system and/or driver software executed; the type and revision of other software executed; the type and/or revision of network interface(s); etc. The system data **3003***x* may include indications of similar characteristics of the external system **3000***x*. Alternatively or additionally, the system data **3003***x* may include indications of characteristics of the sensing devices **3200** and/or the effecting devices **3800**. The provision of such information for the monitored devices **2300***a* and/or **2300***x*, and/or for the external system **3000***x* may enable a more accurate emulation of each by enabling more accurate emulations of components thereof.

[0218] In some embodiments, each of the device data **2303***a* and **2303***x* may include a state machine definition **2304***a* and **2304***x*, respectively, that is descriptive of various aspects of the process states **2306** of process(es) **2305** (e.g., the process **2305***ud*) that may be performed within its corresponding one of the monitored device **2300***a* and **2300***x*.

[0219] Alternatively or additionally, the each of the device data **2303***a* and **2303***x* may include an alternate description of the logic **2345***a* and **2345***x*, respectively, which as previously discussed, may implement the state machines. In some of such embodiments, such an alternate description may take the form of a copies of at least portions of the control routines **2340***a* and **2340***x* that include the logic **2345***a* and **2345***x*, respectively. The system data **3003***x* may similarly include a state machine definition **3004***x* that is descriptive of various aspects of the process states **3006** of process(es) **3005** (e.g., the processes **3005***ud* and/or **3005***x*) that may be performed within the external system **3000***x*.

[0220] It should noted that embodiments are possible in which at least some of such content of the configuration data **5530** is not available. This situation may arise as a result of security and/or trade secret concerns, where information concerning details of one or more processes **2305** and/or **3005** is deemed to be too sensitive to make available. Alternatively or additionally, it may be that a description of one or more of the process **2305** and/or **3005** as a state machine has never been generated such that it simply does not exist. Thus, in such embodiments, the state machine definition(s) **2304** and/or **3004** may be derived from other information contained within the configuration data **5530** and/or from within the observation data **5730** as part of the pre-training phase of preparing the monitoring device(s) **1500** for use.

[0221] Turning to FIG. **5**C, in embodiments in which one or more state machine definitions **2304** and/or **3004** are provided, each such provided state machine definition **2304** or **3004** may include a separate entry for each process state, and/or may include an entry descriptive of various aspects of the manner in which a performance of the corresponding process **2305** or **3005** is caused to begin. FIG. **5**C presents an example of such a provided state machine definition **3004***x* for the earlier-described example process **3005***x*.

[0222] Turning to the depicted example ID entry **3007** of the example state machine definition **3004***x*, various conditions and/or events that cause and/or indicate the commencement of the process **3005***x* may be specified. By way of example, such a specification may specify one or more particular transmissions **2922** through the internal network **2999** and/or one or more particular transmissions **3933** through the link(s) **3999***x* that may trigger and/or indicate commencement of the process **3005***x*. Further, for each transmission, the transmission type (e.g., conveying an operational command or conveying operational information), parameter values for operational commands, data values for operational information may also be specified. Still further, where multiple values are possible for a parameter value or for a data value, a group of discrete values and/or a range of values may be specified in any of a variety of ways (e.g., a maximum value or upper limit, a minimum value or lower limit, a median value, a mean value, an average value, a list of discrete values, a model/formula for deriving values, a probability distribution, etc.).

[0223] In some embodiments in which upper and/or lower limits on a value are specified, such upper and/or lower limits may be selected to include values that are associated with various error conditions that have been planned for such that the state machine of the process **3005***x* and/or the logic **2345***a* and/or **2345***x* involved in controlling the processor **3005***x* have been designed to include measures to address those conditions. Indeed, in some embodiments, there may be ranges of values and/or sets of discrete values specified for parameter values and/or data values that are associated with a performance of the process **3005***x* without error conditions, and other ranges of values and/or sets of discrete values specified for parameter values and/or data values that are associated with a performance of the process **3005***x* in which one or more anticipated error conditions occur.

[0224] Turning to the depicted example state entry **3008** of the example state machine definition **3004***x*, for each state entry **3008**, various conditions and/or events that cause and/or indicate entry into the corresponding state, that cause and/or indicate exit from the corresponding state, and/or that cause and/or indicate the selection of the next state may be specified. By way of example, such a specification may specify one or more particular transmissions **2922** through the internal network **2999** and/or one or more particular transmissions **3933** through the link(s) **3999**x that may trigger and/or indicate such entry into and/or exit from the corresponding state, and/or that may cause and/or indicate the selection of the next state. Further, for each transmission, the transmission type, parameter values for operational commands, data values for operational information may also be specified. Again, where multiple values are possible for a parameter value or for a data value, a group of discrete values and/or a range of values may be specified in any of a variety of ways (e.g., a maximum value or upper limit, a minimum value or lower limit, a median value, a mean value, an average value, a list of discrete values, a model/formula for deriving values, a probability distribution, etc.). Further, each state entry **3008** may also specify upper and/or lower timing limits for the performance of different actions during the corresponding state.

Using Gathered Data to Define Aspects of Emulations—Pre-Training Phase

[0225] FIGS. **6**A, **6**B, **6**C, **6**D, **6**E and **6**F, taken together, present various aspects of the pre-training phase of preparing one or more monitoring devices **1500** for use in monitoring and/or correcting improper network traffic through the internal network **2999**. Again, at a time following the gathering phase (just discussed in reference to FIGS. **5**A-C), and prior to the emulation training phase (to be discussed in reference to FIGS. **7**A-B), one or more emulation devices **5500** may use the observation data **5730** together with the configuration data **5530** to prepare for generating emulations of portions of the monitored system **2000** (e.g., the monitored devices **2300**a and **2300**x), one or more of the external systems **3000** (e.g., the external system **3000**x), one or more processes **2305** and/or **3005** (e.g., the process **3005**x performed within the external system **3000**x). In so doing, preparations are made for using a set of such emulations together to prepare the one or more monitoring devices **1500**.

[0226] It should be noted that, for purposes of illustration by example, an example is used of preparing one or more monitoring devices **1500** to monitor and/or correct improper network traffic in connection with performing the process **3005**x within the external system **3000**x under the control of the monitored devices **2300**a and **2300**x. Thus, FIGS. **6**A-F present aspects of preparing and/or generating emulations to be used in training such monitoring device(s) **1500** (as will then be described in greater detail in FIGS. **7**A-B).

[0227] Turning to FIG. **6**A, the control routine **5540** executed by processor(s) **5550** of at least one of the emulation devices **5500** may include an interpretation component **5541** to parse and/or analyze logic employed by monitored devices **2300**, a correlation component **5542** to correlate portions of such logic and/or of state machine definitions, and an augmentation component **5543** to begin the generation of the database **5533**.

[0228] As previously discussed, each of the monitored devices **2300** (including the monitored devices **2300**a and **2300**x) may be a computing device. Thus, it is envisioned that each monitored device **2300** may incorporate any of a variety of types of processor, storage components, peripheral components, etc. As a result, any of a variety of processor instruction sets, peripheral and/or storage addressing architectures, etc. may be used. Each of the device data **2303**a and **2303**x within the configuration data **5530** may specify such aspects of its corresponding one of the monitored devices **2300**a and **2300**x thereby enabling the selection of a version of the interpretation component **5541** that may incorporate a matching instruction de-compiler, analyzer and/or interpreter to enable parsing of each of the logic **2345**a and **2345**x, respectively, as the interpretation component **5541** is executed by the processor(s) **5550** of at least one of the emulation devices **5500**.

[0229] As also previously discussed, in some embodiments, the configuration data **5530** may include a separate instance of system data **3003** for each external system **3000** (e.g., the depicted instance of system data **3003**x for the external system **3000**x). Further, each such instance of system data **3003** may include, for each process **3005** that is performed within its corresponding external system **3000**, a separate state machine definition **3004** that describes the process states **3006** of its corresponding process **3005** (e.g., the depicted state machine definition **3004**x for the process **3005**x that is performed within the external system **3000**x). However, as also previously discussed, there may be other embodiments in which it may be that a state machine definition **3004** is not provided for one or more processes **3005**.

[0230] In embodiments in which the state machine definition **3004**x for the process **3005**x is provided in the system data **3000**x, execution of the correlation component **5542** may cause the processor(s) **5550** to identify the process states **3006** of the process **3005**x that correspond to each of various portions of the logic **2345**a and **2345**x. With such correlations identified, execution of the augmentation component **5543** may cause the processor(s) **5550** to generate a process entry **5534** for the process **3005**x within the database **5533** to include the contents of that provided state machine definition **3004**x, with augmentations from the correlated portions of logic.

[0231] However, in embodiments in which the state machine definition **3004**x for the process **3005**x is not provided in the system data **3003**x, then execution of the correlation component **5542** may cause the processor(s) to correlate portions of the logic **2345**a to portions of the logic **2345**x. The processor(s) **5550** may then be caused to derive the set of process states **3006** of the process **3005**x from the now correlated combination of the logic **2345**a and **2345**x. With such correlations identified, and with the set of process states **3006** derived, execution of the augmentation component **5543** may cause the processor(s) **5550** to generate the process entry **5534** for the process **3005**x within the database **5533**.

[0232] Thus, each process entry **5534** of the database **5533** may be generated to have an internal structure somewhat similar to that of a state machine definition **3004** (e.g., the state machine definition **3004**x depicted in FIG. **5**C). More specifically, the process entry **5534** generated for the process **3005**x may have an ID entry **5537** and multiple state entries **5538** that would correspond in organization an content to the ID entry **3007** and the multiple state entries **3008**, respectively, within the state machine definition **3004**x that may or may not be provided for the process **3005**x.

[0233] However, in comparison to the ID entry **3007** of the state machine definition **3004**x for the process **3005**x, the ID entry **5537** of the corresponding process entry **5534** may include additional information extrapolated from correlated portions of the logic **2345**a and **2345**x associated with causing the commencement of a performance of the process **3005***x*. Such additional information may include, and not be limited to, upper and/or lower limits of parameter values and/or data values, and/or upper and/or lower limits for the timings of when each transmission is to occur. Such information may be descriptive of both transmissions **2922** that pass through the internal network **2999** and transmissions **3933** that pass through the link(s) **3999***x*.

[0234] Also, in comparison to the state entries **3008** of the state machine definition **3004***x* for the process **3005***x*, each state entry **5538** of the corresponding process entry **5534** may include additional information extrapolated from correlated portions of the logic **2345***a* and **2345***x* associated with the corresponding process state **3006** of the process **3005***x*. Again, such additional information may include, and not be limited to, upper and/or lower limits of parameter values and/or data values, and/or upper and/or lower limits for the timings of when each transmission is to occur. Also again, such information may be descriptive of both transmissions **2922** that pass through the internal network **2999** and transmissions **3933** that pass through the link(s) **3999***x*.

[0235] Where such extrapolations from the logic **2345***a* and **2345***x* are concerned, it should be noted that, while the state machine definition **3004***x* may have specified various constraints for timings, parameter values and/or data values based on various requirements for each state of the process **3005***x*, the logic **2345***a* and/or **2345***x* may impose additional constraints that are also required to be satisfied. As a result, the specifications of timings, parameter values and/or data values within the ID entry **5537** and the state entries **5538** within the process entry **5534** for the process **3005***x* may be more constrained than corresponding specifications provided in the state machine definition **3004***x*.

[0236] Turning to FIG. **6**B, the correlation component **5542** may again be executed to cause the processor(s) **5550** to correlate each of the observed transmissions **2922** or **3933** indicated in the observation data **5730** to one of the process states **3006** described in the process entry **5534** of the database **5533** for the process **3005***x*. The augmentation component **5543** may also again be executed to cause further augmentation of the ID entry **5537** and/or the state entries **5538** of that process entry **5534** with indications of timings, operational commands, parameter values and/or data values that were actually observed for each of the observed transmissions **2922** or **3933**.

[0237] As a result, the entries **5537** and/or **5538** within the process entry **5534** for the process **3005***x* are caused to include indications of real world observations for such information for each transmission **2922** or **3933** that is associated with each process state **3006**, in addition to the ranges and/or sets of possible timings, commands and/or values that are specified by the state machine definition **3004***x*, and/or that are extrapolated from the logic **2345***a* and **2345***x*.

[0238] In some embodiments, a statistical analysis may be performed based on the observed timings of the observed transmissions **2922** and/or **3933** that takes into account whatever variation in timings may be observed across multiple performances of the process **3005***x*. From such an analysis for each such transmission, a range of timing and/or a probability distribution of timing for when the transmission is expected to occur may be specified in the process entry **5534** for the process **3005***x*.

[0239] Turning to FIG. **6**C, the control routine **5540** may also include an emulation component **5544** and a time tuning component **5545**. Execution of the emulation component **5544** may cause the processor(s) **5550** of one of the emulation devices **5500** to instantiate an execution environment in which one or more routines of a monitored device may be executed, such as the depicted control routine **2340***a* of the monitored device **2300***a*. As depicted, such an execution environment may be a virtual machine (VM) or a container **5566**. Regardless of the exact nature and/or characteristics of the execution environment that is provided, further execution of the emulation component **5544** may cause the processor(s) **5550** to retrieve indications of characteristics of a monitored device **2300** from a corresponding instance of device data **2300** (e.g., the depicted device data **2303***a* corresponding to the monitored device **2300***a*) to guide aspects of the execution environment (e.g., the VM or container **5566**) that is provided, thereby providing an execution environment for the execution of the control routine **2340***a* that relatively closely mimics various aspects of the execution environment normally provided to the control routine **2340***a* within the monitored device **2300***a*.

[0240] More specifically, and as previously discussed in reference to FIG. **5**B, each instance of device data **2303** may include indications of various characteristics of the hardware and/or software of its corresponding monitored device **2300**. Such characteristics may include, and are not limited to, the type and/or revision level of the processor(s); the type and quantity of volatile and/or non-volatile storage; the type and revision of operating system and/or driver software executed; the type and revision of other software executed; the type and/or revision of network interface(s); etc.

[0241] With such an execution environment provided, execution of the time tuning component **5545** by the processor(s) **5550** may cause the retrieval of indications from the observation data **5730** of amounts of time between transmissions **2922** and/or **3933** that are causally linked by portions of the logic **2345** of a control routine **2340** (e.g., the logic **2345***a* of the control routine **2340***a* of the monitored device **2300***a*). Referring briefly back to FIGS. **3**B-C, as was discussed in reference to the logic **2345***x*, there may be portions of that logic that are triggered to output a transmission **2922/3933** in response to the receipt of a transmission **2922/3933**, and such a response may take place with an amount of time delay. As was also discussed (but not specifically depicted), similar responses with time delays may also be exhibited by the logic **2345***a*. As has been discussed, the observation data **5730** includes timing information for each observed transmission **2922** and **3933** that documents the amount of time that elapsed for each such delay.

[0242] Returning to FIG. **6**C, in executing the time tuning component **5545**, the clock speed, number of processing cycles per second allocated to the VM or container **5566**, and/or another parameter that exerts influence over the speed of execution of the control routine **2340***a* therein may be manipulated to cause the time delays exhibited by the logic **2345***a* therein to at least relatively closely match those documented in the observation data **5730**. In this way, a time tuning setting may be identified to cause the timing behavior of the logic **2345***a* within the VM or container **5566** to match the timing behavior of the logic **2345***a* within the monitored device **2300***a*. An indication of the time tuning setting that begets such timing behavior may then be stored within the device data **2303***a* (or elsewhere within the storage **5560** of one or more of the emulation devices **5500**) as the depicted tuning specification **5535***a*.

[0243] Turning to FIG. **6**D, a similar use of information in the device data **2303***x*, emulation, and time tuning may be performed to similarly identify a time tuning setting for the logic **2345***x* of the monitored device **2300***x*, and an indication thereof may be stored within the device data **2303***x* (or elsewhere within the storage **5560** of one or more of the emulation devices **5500**) as the depicted tuning specification **5535***x*.

[0244] With such time tuning settings having been identified for use in emulating the monitored devices **2300***a* and **2300***x*, the monitored devices **2300***a* and **2300***x* may now be emulated with relatively accurate timings.

[0245] Turning to FIGS. **6**E-F, the control routine **5540** may additionally include a testing component **5546** to perform various tests of combinations of emulations of monitored devices **2300**, processes performed within monitored devices **2300**, external systems **3000**, and/or processes performed within external systems **3000**. As depicted, and as is about to be explained in greater detail, the testing component **5546** may be employed to perform various tests of a combination of emulations of the monitored devices **2300***a* and/or and **2300***x* interacting with an emulation of the process **3005***x* performed within the external system **3000***x* based on the state machine definition **3004***x*.

[0246] In executing the emulation component **5544**, the processor(s) **5550** of at least one of the emulation devices **5500** may be caused to instantiate VMs or containers **5566** in which the execution environments of the monitored devices **2300***a* and **2300***x* may be emulated to enable the control routines **2340***a* and **2340***x*, respectively, to be executed therein. Additionally, another VM or container **5566** may be instantiated in which the state machine of the process **3005***x* may be implemented. In further executing the emulation component **5544**, communications among these VMs or containers **5566** may be enabled to allow transmissions **2922** and **3933** thereamong in a manner that mimics the communications that would normally take place on the internal network **2999** and the link(s) **3999***x*.

[0247] In executing the testing component **5546**, the processor(s) **5550** may be caused to parse the process entry **5534** within the database **5533** for the process **3005***x* to retrieve lower and/or upper limits for timings for events that trigger the output of transmissions **2922** and/or **3933**. Such events may include simulated operator inputs via a UI to the logic **2345***a*, and/or simulated measurements taken by simulated sensing devices **3200** for the process **3005***x*. Experiments may then be performed in which each of such simulated inputs are performed, each timed to meet its corresponding lower and upper limits for timing of occurrence. In this way, edge and/or corner cases based on corresponding upper and/or lower limits of transmissions **2922** and/or **3933** that are triggered by these inputs may be tested.

[0248] Further execution of the testing component **5546** may then cause the resulting test timings of those transmissions **2922** and/or **3933** to be documented, and execution of the augmentation component **5543** may then cause the indications of timings for transmissions already present in the database **5533** to be augmented with these observed test timings. In this way, the upper and/or lower limits for timings with which transmissions **2922** and/or **3933** might actually occur in edge and/or corner cases may be added to the database **5533**, alongside the timings actually observed during normal operation of the monitored system **2000** together with the process **3005***x*.

[0249] In some embodiments, the resulting combination of transmission timings that are extrapolated from the combination of the logic **2345***a* and **2345***x*; that are observed to have actually taken place during normal operation; and that are derived from such experimentation with edge and/or corner cases as just described may be combined to generate a fuller set indications of transmission timings in the process entry **5534** for the process **3000***x* in the database **5533**. As a result, for each transmission **2922** or **3933**, that process entry **5534** in the database **5533** may include indications of what timings are possible in theory, versus what timings are possible in unlikely extreme timing conditions, versus what timings have actually been observed to occur. In some of such embodiments, each such indication of timings for a transmission **2922** or **3933** may take the form of a probability distribution indicative of relative likelihoods of theoretically possible timings, vs. unlikely timings under extreme conditions, vs. timings that have actually been observed to occur.

[0250] Regardless of the exact way in which such indications of transmission timings are represented in that process entry **5534** within the database **5533**, such extensive timing information for at least transmissions **2922** through the internal network **2999** may enable greater accuracy in distinguishing between proper and improper transmissions **2922** through the internal network **2999**, as well as more certain identification of when a span of time has elapsed without an expected transmission **2922** having occurred. Alternatively or additionally, such extensive timing information for transmissions **2922** and/or **3933** may enable the provision of emulations guided by such timing information for diagnostics, testing and/or training purposes.

[0251] As depicted in FIG. **6**F, with the database **5533**, including a process entry **5534** for the process **3005***x*, having been generated as described above, execution of a generation component **5547** of the control routine **5540** may cause processor(s) **5550** of an emulation device **5500** to generate the database **1533** with a corresponding process entry **1534** therefrom. It may be that each of the process entries **1534** of the database **1533** is somewhat reduced in content from their counterpart process entries **5534** of the database **5533**. By way of example, information concerning transmissions **3933** through links **3999** (e.g., the link(s) **3999***x*) may not be included in process entries **1534** of the database **1533** as the communications traffic through link **3999** may not normally be monitored by the monitoring device(s) **1500**.

[0252] In essence, as each process entry **5534** of the database **5533** is generated and then augmented (perhaps repeatedly augmented), each process entry **5534** is caused to become a definition of a model for the process **2305** or **3005** that corresponds to it. And as is about to be described, the defined within each process entry **5534** may be further refined before being stored as a process entry **1534** in the database **1533** that is ultimately provided to each monitoring device **1500** to be used in analyzing transmissions **2922** to identify corresponding processes **2305** and/or **3005**, and to be used in taking corrective action in connection with those transmissions **2922**, as will be further discussed in still greater detail.

Using the Emulations for Training—Emulation Training Phase

[0253] FIGS. **7**A and **7**B, taken together, present various aspects of the training phase of preparing one or more monitoring devices **1500** for use in monitoring, stopping and/or correcting improper network traffic through the internal network **2999**. Again, at a time following the pre-training phase (just discussed in reference to FIGS. **6**A-F), and prior to such actual use of the monitoring device(s) **1500** in monitoring, stopping and/or correcting improper network traffic through the internal network **2999**, the monitoring device(s) **1500** may be temporarily coupled to the emulation device(s) **5500** to be trained and/or tested using the emulated network traffic among at least the emulations of various monitored devices **2300** (e.g., emulations of the monitored devices **2300***a* and **2300***x*).

[0254] Turning to FIG. **7**A, as depicted, the information within a process entry **5534** within the database **5533** for a process **2305** or **3005** (e.g., the process **3005***x*) may be used as an input to a UI component **5548** of the control routine **5540** to guide the presentation of options for manual testing to an operator through a user interface (UI) that employs an input device **5520** and/or a display **5580** of an emulation device **5500**. More precisely, a process entry **5534** of the database **5533** may be used to provide an operator with options for defining one or more experiments that are based on the indications within that process entry **5534** of what parameter values, data values, timings, etc. have been determined to be possible as a result of the aforedescribed interpretation, extrapolation and experimentation operations.

[0255] Turning to FIG. **7**B, as part of such testing, the monitoring device(s) **1500** may be placed in a training mode to enable the learning of new possible transmissions **2922** that may occur as part of proper network traffic on the internal network **2999** associated with a particular process **2305** or **3005**, to enable the learning of new possible parameter values and/or data values for the contents of those transmissions **2922** that may occur through the internal network **2999**, and/or to enable the learning of new timings for those transmissions **2922**. As a result, such new information concerning those transmissions **2922** that may occur through the internal network **2999** in connection with the particular process **2305** or **3005** may be added to the specifications of transmissions, transmission contents and/or transmission timings already stored within a process entry **1534** that defines a model for that particular process **2305** or **3005** within the database **1533**.

Alternate/Supplemental Training Approach

[0256] FIGS. **8**A, **8**B, **8**C and **8**D, taken together, present various aspects of an example embodiment of an alternate approach to training a monitoring device **1500** to detect and/or respond to learned patterns of transmission through the internal network **2999**. Again, such patterns of transmission may be associated with controlling an industrial process (e.g., the process **3005***x*) and/or associated with downloading software updates (e.g., the processes **2305***ud* and/or **3005***ud*).

[0257] Turning to FIG. **8**A, in some embodiments, this alternate approach to training a monitoring device **1500** may be employed to provide supplemental training at a time after the emulation-based approach described in detail above has been employed to provide the initial training. As is about to be described, this alternate approach does not employ emulations, and allows the monitoring device **1500** to remain in place and connected to interchange device(s) **2700**. However, in other embodiments, it may be that this alternate approach to training a monitoring device **1500** may be employed for both initial and subsequent training.

[0258] As depicted, a monitoring device **1500** may be coupled to an interchange device **2700** by one or more of its ports **1590** and through the depicted link **1992**. In executing the depicted learning component **1543** of the control routine **1540**, the processor(s) **1550** may be caused to place the monitoring device **1500** in a training mode in which the monitoring device **1500** is prepared for use (or further prepared for use). In such preparations, observations are made of transmissions **2922** of operational commands and/or operational information through the internal network **2999** that are associated with a particular process **2305** or **3005**, and those observations may serve as a basis for machine learning from actual examples from when the particular process **2305** or **3005** is performed normally. As depicted, where such transmissions **2922** are associated with controlling an industrial process (e.g., the process **3005***x* performed within the external system **3000***x*), then such transmissions **2922** may be between two monitored devices **2300** (e.g., the depicted monitored devices **2300***a* and **2300***b*), and may be conveyed through corresponding links **2990***a* and **2990***b*, respectively, of the internal network **2999**. However, where such transmissions are associated with downloading an updated routine, then such transmissions **2922** may be between a monitored device **2300** (e.g., the depicted monitored device **2300***a*) and an external server **9900**, such transmissions may be conveyed through corresponding links **2990***a* and **2992**, respectively, of the internal network **2999**.

[0259] More specifically, copies and/or indications of these transmissions **2922** of operational commands and/or operational information that are associated with a particular process **2305** or **3005** are relayed by the interchange device **2700** to the monitoring device **1500** via the link **1992**. Within the monitoring device **1500**, the received copies and/or indications of such transmissions may be used to generate and/or augment information stored within a process entry **1534** of the database **1533** that corresponds to the particular process **2305** or **3005**.

[0260] Still more specifically regarding the control of an industrial process, and in embodiments where the monitoring device **1500** was earlier provided with descriptions of portions of the logic used for the process by monitored device(s) **2300** and/or by components of an external system **3000** to control the industrial process, the entry set **1531** may have already been generated at an earlier time based on such descriptions of such logic. Again, such descriptions of such logic may include descriptions of aspects of the process states **2306** or **300***g*, along with descriptions of aspects of transmissions **2922** associated with individual process states, and/or descriptions of aspects of transmissions **2922** associated with transitions between process states. As previously discussed, in such embodiments, the timings and/or data values of at least some of the transmissions, for which copies and/or indications are relayed to the monitoring device **1500**, are correlated to indications of expected transmissions **2922** within a process entry **1534** for a particular process **2305** or **3005**. The observed timings and/or data values may be used to derive models that are descriptive of variations observed in those timings and/or data values, as well as being descriptive of relative probabilities of such variations. Alternatively or additionally, the observed timings and/or data values may be used to derive and/or train models based on neural networks and/or other forms of machine learning.

[0261] Alternatively, regarding either the control of an industrial process or the process of downloading of a software update, and in other embodiments where the monitoring device **1500** was not earlier provided with descriptions of portions of the logic used for the process (e.g., not provided with a state machine definition **2304** or **3004**), the process entry **1534** for a process **2305** or **3005** may be generated from the observations made of transmissions **2922** among the particular monitored devices **2300** that are associated with that process **2305** or **3005**. More specifically, the copies and/or indications of such transmissions **2922** that are relayed to the monitoring device **1500** may be analyzed for their timings, for the types of the transmissions **2922**, for what commands and/or parameters were included in transmissions **2922** of the type used to convey operational commands, and/or for what data values were included in transmissions **2922** of the type used to convey operational information. As previously discussed, with there being no access to data concerning the process states of the process with which such transmissions are associated (e.g., no access to a state machine definition **2304** or **3004**), the analysis of at least observed variations in what types of transmissions occur and/or in what commands are transmitted may be used as a basis for deriving a set of process states of the process. Again, models based on statistical analyses and/or models based on any of a variety of machine learning technologies may be derived based on observed variations in commands and/or parameters transmitted, variations in data values transmitted, and/or variations in timings of the transmissions.

[0262] Regardless of the exact manner in which the various process entries **1534** within the database **1533** are generated and/or augmented, the processor(s) **1550** may be caused, by further execution of the learning component **1543**, to transition the monitoring device **1500** out of such a training mode upon reaching a predefined threshold, such as a threshold quantity of performances of the particular process from which observations of transmissions of operational commands or operational information are made, and/or a threshold amount of time spent in the training mode.

[0263] It should be noted that, in some embodiments, use of such a training mode may be entirely obviated by preloading the monitoring device **1500** with a database **1533** that has already been previously generated, either within the very same monitored system **2000**, or within another monitored system that is similar enough that any variations in timings of transmissions therein are relatively small such that proper operation of the monitoring device **1500** with the monitored system **2000** is not impaired. In some of such embodiments, it may be that the database **1533** was previously generated through earlier training using another monitoring device **1500** that was trained based on observing the same monitored system **2000**, or another sufficiently similar monitored system.

[0264] FIGS. **8**B, **8**C and **8**D present aspects of using this alternate approach in a situation in which descriptions of the logic used to control the process **3005***x* are not provided. Thus, in this alternate approach in such a situation entails preparing a monitoring device **1500** based on machine learning from observations of network traffic that occurs through the network **2999** during multiple performances of the process **3005***x*, but without the benefit of information correlating those observations to stages of the process **3005***x*. As a result, such preparation must be based on the timings, types, particular operation commands (including parameters) and/or particular operational information associated with the transmissions **3933** that are observed.

[0265] FIG. **8**B, when compared to FIG. **3**B, illustrates causality information that is not provided to the monitoring device **1500** as a result of not being provided with a description of the logic for controlling the process **3005***x*. In some embodiments, indications that are manually entered by an operator of what transmission(s) **3933**, or sequence of transmissions **3933**, are associated with the beginning of the process **3005***x* may be relied upon by the monitoring device **1500** to serve as an indicator for when a performance of the process **3005***x* has begun and/or is being triggered to begin. Thus, as depicted, it may still be possible for the monitoring device **1500** to correlate specific transmission(s) **3400** (e.g., one or more of the depicted transmissions **3933***n*, **3933***np* and/or **3933***p***1**) with the beginning of a performance of the process **3005***x*.

[0266] However, even with the benefit of an operator of the monitored system **2000** providing manual identification of each transmission **3933** that is associated with the process **3005***x* versus other transmissions **3933** that are not associated with the process **3005***x*, there remains no information provided to the monitoring device **1500** that correlates individual process states **3006***p* to individual transmissions **3933***p* (e.g., the specifically labeled transmissions **3933***p***1**, **3933***p***2**, and so on). As a result, identifying instances of causality between particular process states **3006***p* and the occurrence and/or content of particular transmissions **3933***p*, and/or identifying instances of causality between particular transitions between process states **3006***p* and the occurrence and/or content of particular transmissions **3933***p*, may not be possible.

[0267] FIG. **8**C, when compared to FIG. **3**C, illustrates timing information that is not provided to the monitoring device **1500** as a result of not being provided with a description of the logic for controlling the process **3005***x*. Again, indications that are manually entered by an operator of what transmission(s) **3933**, or sequence of transmissions **3933**, are associated with the beginning of the process **3005***x* may be relied upon by the monitoring device **1500** to serve as an indicator for when a performance of the process **3005***x* has begun and/or is being triggered to begin. Thus, as depicted, it may still be possible for the monitoring device **1500** to correlate the timings of specific transmission(s) **3933** (e.g., one or more of the depicted transmissions **3933***n*, **3933***np* and/or **3933***p***1**) with the beginning of a performance of the process **3005***x*.

[0268] However, even with the benefit of an operator of the monitored system **2000** providing manual identification of each transmission **3933** that is associated with the process **3005***x* versus other transmissions **3933** that are not associated with the process **3005***x*, there remains no information provided to the monitoring device **1500** that correlates the timings of the start and/or ending of individual process states **3006***p* to individual transmissions **3933***p* (e.g., the specifically labeled transmissions **3933***p***1**, **3933***p***2**, and so on). As a result, identifying specific times at which particular transmissions **3933***p* are expected to occur based on the when particular process states **3006***p* start and/or end may not be possible.

[0269] Thus, and turning to FIG. **8**D, with such a lack of access to a description of the logic for controlling the process **3003***x*, the monitoring device **1500** may rely on the aforedescribed observations of transmission type, content, parameters and/or timing as inputs to deriving models **1539***p* for each transmission **3933***p* that is to occur during a performance of the process **3005***x*. By way of example, and as depicted, individual models **1539***p***1**, **1539***p***2** and **1539***p***3** may be derived for each of the depicted transmissions **3933***p***1**, **3933***p***2** and **3933***p***3**, respectively.

[0270] Regarding timings, with no access to information concerning the logic for determining when any particular transmission **3933***p* (e.g., the depicted transmissions **3933***p***1**-*p***3**) is to be expected to occur, each of the corresponding models **1539***p* (e.g., the depicted corresponding models **1539***p***1**-*p***3**) may include a model for the time at which to begin its corresponding transmission **3933***p* that is based on the time period from when the last transmission **3933***p* ended (i.e., Tbetw). Such a model of timing may also be derived to include some degree of variation (including relative probabilities) for when to begin transmitting based on observations of such variations across multiple performances of the process **3005***x*.

[0271] Regarding what commands and/or associated parameters are to be transmitted in transmissions of operational commands, each one of such models **1539***p* (e.g., the models **1539***p***1**-*p***3**) that is associated with a transmission **3933***p* (e.g., the transmissions **3400***p***1**-*p***3**, respectively) that is of a type for transmitting an operational command may include a model for the selection of the particular command to be transmitted and/or a model for the selection of the parameter(s) to be included therewith. Such a model may take into account timings relative to one or more preceding transmissions **3933***p*, along with the content of one or more preceding transmissions **3933***p*.

[0272] Regarding what data values are to be transmitted in transmissions of operational information, each one of the models **1539***p* (e.g., the models **1539***p***1**-*p***3**) that is associated with a transmission **3933***p* (e.g., the transmissions **3933***p***1**-*p***3**, respectively) that is of a type for transmitting operational information may include a model for the derivation of data values to be transmitted therein. Again, such a model may take into account timings relative to one or more preceding transmissions **3933***p*, along with the content of one or more preceding transmissions **3933***p*.

[0273] As such models **1539***p* for each transmission **3933***p* are developed and/or refined based on observations from multiple performances of the process **3005***x*, a process entry **1534** for the process **3005***x* may be generated and stored in the database **1533**. Such a process entry set **1534** may have an organizational structure similar to what was previously described in connection with FIG. **7**E.

[0274] Again, with the lack of provision of information concerning the logic for controlling the process **3005***x*, there may be no information available concerning any aspect of the set of process states **3006***p* of the process **3005***x*. In some embodiments, the set of process states **3006***p* may be inferred from the observed transmissions **3933***p*. More specifically, from observations of the transmissions **3933***p* that occur on the network **2999** during multiple performances of the process **3005***x*, the beginnings and endings of different process states **3006***p* may be inferred to be associated with each instance in which there appears to be a point in the multiple performances at which a selection is made from among multiple observed possibilities of what transmission **3933***p* is to occur.

[0275] By way of example, where there is observed to be some variation between the transmission of one command or another among the multiple performances, or where there is observed to some variation between the transmission of an operational command and the transmission of operational information, the processor(s) **1550** may be caused by execution of the learning component **1543**, to infer that one process state **3006***p* ends at that point, and that there is a selection of what process state **3006***p* is to begin at that point.

Detection and Correction of Missing Transmission

[0276] FIGS. **9**A and **9**B, taken together, present various aspects of an example embodiment of using a monitoring device **1500** to identify an instance in which a transmission **2922** of a particular operational command or operational information does not occur when it is expected to occur, and to respond to such an instance by generating and outputting a replacement transmission **2922** of the particular operational command or operational information.

[0277] Turning to FIG. **9**A, the control routine **1540** may incorporate a monitoring component **1545** that, when executed by the processor(s) **1550**, may cause the monitoring device **1500** to enter into an operating mode in which the monitoring device **1500** is used to detect and address instances of network activity on the internal network **2999** that does not conform to what is expected.

[0278] More specifically, FIG. **9**A depicts an example instance in which a transmission **2922** of an operational command or operational information associated with controlling a particular process **3005** performed within an external system **3000** does not occur when expected. More specifically, the depicted monitored device **2300**A fails to transmit a particular operational command to the depicted monitored device **2300**B (through the depicted interchange device **2700**, and links **2990**A and **2990**B of the internal network **2999**) during a span of time in which that transmission **2922** of that operational command or operational information was expected to take place, according to the information about the expected transmission **2922** that is stored within a process entry **1534** of the database **1533** that corresponds to the particular process **3005**. In continuing to execute the monitoring component **1545**, the processor(s) **1550** may be caused to detect this lack of such a transmission **2922** of the particular operational command or operational information. More precisely, the processor(s) **1550** may be caused to detect the lack of receiving an indication from the interchange device **2700** that such a transmission **2922** has occurred on the internal network **2999**.

[0279] As previously discussed, and as will be familiar to those skilled in the art, the failure of the monitored device **2300**A to transmit a particular operational command or operational information when expected (or at all) may be caused by any of a variety of conditions. Again, the cause may be any of a variety of hardware and/or software malfunctions that may typically befall a computing device. As still another possibility, the monitored device **2300**A may be in the process of being serviced, replaced and/or upgraded under circumstances in which the need for the monitored device **2300**A to transmit the particular operational command or operational information during an expected span of time has somehow not been accommodated. Alternatively, the cause may be some form of cyber attack that has compromised the monitored device **2300**A, itself, or at least has compromised the ability of the monitored device **2300**A to access the link **2990**A and/or to use the link **2990**A to transmit the particular operational command or operational information through the internal network **2999**.

[0280] Turning to FIG. 9B, the control routine **1540** may incorporate a correction component **1547** that, when executed by the processor(s) **1550**, may cause the processor (s) **1550** to take action to address the lack of occurrence of this expected transmission **2922** on the internal network **2999**. More specifically, in response to this lack of transmission of the particular operational command or operational information, the processor(s) **1550** may be caused to use information stored within the corresponding process entry **1534** concerning this particular transmission to, itself, generate and transmit the operational command or operational information to the monitored device **2300**B (through the depicted interchange device **2700** and the depicted links **1992** and **2990**B). In effect, the monitoring device **1500** is caused to take the place of the monitored device **2300**A for purposes of transmitting the particular operational command or operational information to the monitored device **2300**B.

[0281] As part of generating and/or transmitting the particular operational command or operational information to the monitored device **2300**B, the processor **1550** (*s*) may be caused to refer to indications stored in that corresponding process entry **1534** of the database **1533** concerning what the particular operational command or operational information to be transmitted is, and/or various protocol details to be adhered to in transmitting the particular operational command or operational information. Among such protocol details may be the need to include one or more identifiers with the operational command or operational information that may specify the destination for the transmission, that identify the iteration of the particular process **3005** that the particular operational command or operational information is directed to, etc. Also among such protocol details may be an indication of a need to generate a command sequence number that identifies the relative position of the particular operational command or operational information among other operational commands or operational information that are transmitted as part of controlling the process **3005** (this command sequence number should not be confused with the sequence numbers used in TCP/IP). By way of example, such a command sequence number may need to be generated by incrementing the command sequence number of the last operational command or operational information associated with the particular process **3005** that was observed to have been transmitted.

[0282] It should be noted that, although an example of a breakdown in electronic communications involving a failure to output an expected transmission **2922** by a single monitored device **2300**A is presented and discussed in connection with FIGS. **9**A-B, it is envisioned that there could be a breakdown in electronic communications involving a failure by multiple monitored devices **2300** and involving multiple expected transmissions **2922**. In such an eventuality, it may be that the monitoring device **1500** is caused to detect such multiple failures by such multiple monitored devices **2300** (e.g., such failures occurring in both of the monitored devices **2300**A and **2300**B), and may be further caused to act to correct each of the resulting instances of the lack of output of an expected transmissions **2922**.

Detection and Correction of Improper Transmission—With Buffer

[0283] FIGS. **10**A, **10**B, **10**C and **10**D, taken together, present various aspects of an example embodiment of using a monitoring device **1500** to identify an instance in which an improper or invalid operational command is transmitted, and to respond to such an instance by blocking that transmission **2922** and/or generating and outputting a transmission **2922** that conveys a corrected and/or countermanding operational command.

[0284] Turning to FIG. **10**A, the control routine **1540** may incorporate a monitoring component **1545** that, when executed by the processor(s) **1550**, may cause the monitoring device **1500** to enter into an operating mode in which the monitoring device **1500** is used to detect and address instances of improper network activity on the internal network **2999** in which a transmission **2922** conveys an operational command that does not conform to what is expected.

[0285] By way of example, the depicted monitored device **2300**A outputs, onto its link **2990**A with the depicted interchange device **2700**, a transmission **2922** of an operational command associated with a particular process **3005** performed within an external system **3000**. At the interchange device **2700**, the operational command may be intercepted (such that the transmission **2922** of the operational command is not allowed to reach its destination, at least not initially) by being stored within the buffer **2766** therein. A copy or indication of the operational command is then relayed by the interchange device, on the depicted link **1992**, to the depicted monitoring device **1500** for analysis. Within the monitoring device **1500**, further execution of the monitoring component **1545** by the processor(s) **1550** may cause the processor(s) **1550** to refer to a process entry **1534** of the database **1533** that corresponds to the particular process **3005**. Referring to such a process entry **1534** may be done as part of analyzing the operational command to determine whether its type, its content (including any parameter values therein), the timing of its transmission, its order of transmission relative to other transmitted operational commands and/or to transmitted operational information, and/or still other aspects thereof, conform to what is expected for the particular process **3005**.

[0286] Turning to FIG. **10**B, if the processor(s) **1550** determine that the transmission **2922** received from the monitored device **2300**A (and being held at the interchange device **2700**) conveys an operational command of proper type, content, transmission timing, transmission order relative to other commands, etc., then the monitoring device **1500** may transmit an instruction along the link **1992** to the interchange device **2700** to proceed with relaying the transmission **2922**, with its operational command, onward to its intended destination device (i.e., the depicted monitored device **2300**B), thereby allowing the operational command to be received at its intended destination device.

[0287] However, and turning to FIG. **10**C, if the processor (s) **1550** determine that the transmission **2922** received from the monitored device **2300**A (and being held within the buffer **2766** of the interchange device **2700**) conveys an operational command that is improper in type, content, transmission timing, transmission order relative to other commands, etc., then the processor(s) **1550** may be caused to generate alert(s) of various types and/or to operate the port **1590** to transmit instruction(s) along the link **1992** to instruct the interchange device **2700** to refrain from relaying that transmission **2922** onward to its intended destination device (i.e., the depicted monitored device **2300**B), thereby preventing the reception of that operational command at its intended destination device.

[0288] Turning to FIG. **10**D, as previously discussed, in some embodiments, and under various pre-selected circumstances, it may be that, in addition to instructing the interchange device **2700** to refrain from relaying the transmission **2922** conveying the improper operational command onward to its intended destination device, the processor(s) **1550** of the monitoring device **1500** may be further caused to generate and output a substitute transmission **2922** that conveys a substitute proper operational command. More precisely, and as depicted, the control routine **1540** may incorporate a correction component **1547**. In being executed by the processor(s) **1550**, and in response to circumstances in which the improper operational command was transmitted at a time at which it was expected that a proper operational command was to be transmitted, the processor(s) **1550** may be caused to generate a substitute proper operational command that is of the expected type and that includes expected parameter (s). Stated differently, the processor(s) **1550** may be caused to use information stored within the process entry **1534** of the database **1533** that corresponds to the process **3005** to, itself, generate the proper operational command that should have been transmitted. The processor **1550** may then operate the port **1590** to output a transmission **2922** onto the link **1992** that conveys the generated proper operational command to the interchange device **2700**, along with instructions to relay it to the expected destination device (i.e., the monitored device **2300**B). In effect, the monitoring device **1500** is caused to take the place of the monitored device **2300**A for purposes of transmitting the proper operational command to the monitored device **2300**B.

[0289] As part of generating and/or transmitting such a substitute proper operational command to the monitored device **2300**B, the processor **1550** (*s*) may be caused to refer to indications stored in the corresponding process entry **1534** of the database **1533** concerning various protocol details to be adhered to in transmitting it. Among such protocol details may be the need to include one or more identifiers with the command that may specify the destination device for the transmission, that identify the iteration of the process **3003** that the particular command is directed to, etc. Also among such protocol details may be an indication of a need to generate a command sequence number that identifies the relative position of the proper operational command among other operational commands that are transmitted as part of controlling the industrial process (this command sequence number should not be confused with the sequence numbers used in TCP/IP). By way of example, such a command sequence number may need to be generated by incrementing the command sequence number of the last operational command associated with the process **3005** that was observed to have been transmitted.

Detection and Correction of Improper Transmission—Without Buffer

[0290] FIGS. **11**A and **11**B, taken together, present various aspects of an example embodiment of using a monitoring device **1500** to identify an instance in which an improper operational command is transmitted, and to respond to such an instance by generating and outputting a transmission **2922** that conveys a corrected and/or countermanding operational command.

[0291] Turning to FIG. **11**A, in a manner similar to what was discussed above in reference to FIG. **10**A, the processor (s) **1550** of the depicted monitoring device **1500** may be caused, by execution of the monitoring component **1545** of the control routine **1540**, to place the monitoring device **1500** into an operating mode in which the monitoring device **1500** is used to detect and address instances of improper network activity on the internal network **2999** in which a transmission **2922** conveys an operational command that does not conform to what is expected.

[0292] By way of example, and similar to what was discussed above in reference to FIG. **10**A, the depicted monitored device **2300**A outputs, onto its link **2990**A with the depicted interchange device **2700**, a transmission **2922** of an operational command associated with a particular process **3005** performed within an external system **3000**. However, unlike the interchange device **2700** of FIGS. **10**A-D, the interchange device **2700** of FIGS. **11**A-B may not incorporate the ability to intercept and temporarily store the transmission **2922** as part enabling the operational command that it conveys to be conditionally allowed to be relayed onward to its intended destination device. Instead, the interchange device **2700** of FIGS. **11**A-B may simply proceed with relaying the transmission **2922** onward to its intended destination device (i.e., the monitored device **2300**B).

[0293] However, as the interchange device **2700** so relays the transmission **2922** onward to the monitored device **2300**B via its corresponding link **2990**B, a copy and/or indication of the transmission **2922**, including the operational command therein, is relayed along the depicted link **1992** to the monitoring device **1500** for analysis. As has been discussed, within the monitoring device **1500**, the processor (s) **1550** may be caused by caused by execution of the monitoring component **1545** to refer to the process entry **1534** of the database **1533** that corresponds to the particular process **3005** as part of analyzing the transmission **2922**, including the operational command therein, to determine whether the type of the transmission **2922**, its content (including any parameter values of the operational command), the timing of the transmission **2922**, its order of being transmitted relative to other transmissions **2922**, and/or still other aspects, conform to what is expected for the particular process **3005**. If the processor(s) **1550** determine that the transmission **2922** is proper (e.g., conveys an operational command of proper type and parameters, with proper transmission timings, in proper order relative to other transmissions, etc.), then the processor(s) **1550** may be caused to take no further action regarding the transmission **2922**, since it has already been relayed to the monitored device **2300**B.

[0294] However, and turning to FIG. **11**B, if the processor (s) **1550** determine that the transmission **2922** conveys an operational command that is improper in type, content, transmission timing, transmission sequence order relative to other commands, etc., then the processor(s) **1550** may be caused to generate and transmit one or more operational commands to countermand the improper operational command and/or to provide an expected proper operational command.

[0295] More specifically, the processor(s) **1550** of the monitoring device **1500** may be further caused by execution of the correction component **1547** to generate at least one countermanding operational command that serves to stop and/or reverse whatever action or non-action was ordered to be taken by the improper operational command was conveyed in the transmission **2922**. In so doing, the processor **1550** may be caused to use both the contents of the improper operational command and information about expected transmissions of expected operational commands for the particular process **3005** stored within the corresponding process entry **1534** of the database **1533** to generate such countermanding operational command(s). The processor **1550** may then be caused to operate the port **1590** to output at least one transmission **2922** of the at least one countermanding operational command onto the link **1992** to the interchange device **2700** along with instructions to relay the at least one transmission **2922** to the same destination device as the improper operational command (i.e., to the monitored device **2300**B).

[0296] Also more specifically, and either in addition to or in lieu of the generation and transmission of at least one countermanding operational command, where the improper operational command was transmitted at a time at which it was expected that a proper operational command was to be transmitted, the processor **1550** may generate a substitute proper operational command that is of the expected type and that includes expected parameter(s). Again, in so doing, the processor(s) **1550** may be caused to use information stored within the corresponding process entry **1534** of the database **1533** about expected transmissions of operational commands for the particular process **3005** to generate such a proper operational command. Again, the processor(s) **1550** may then be caused to operate the port **1590** to output a transmission **2922** conveying the generated proper operational command onto the link **1992** to the interchange device **2700** along with instructions to relay it to the same destination device as the improper operational command (i.e., to the monitored device **2300**B). In effect, and in a manner similar to what was discussed in reference to FIG. **10**D, the monitoring device **1500** is caused to take the place of the monitored device **2300**A for purposes of transmitting the proper operational command to the monitored device **2300**B.

[0297] It should be noted that, in various situations, it may be that the generation and transmission of the proper command also serves the purposes of countermanding the improper command operational command, such that the generation and transmission of separate and distinct countermanding operational command(s) is unnecessary. Again, as part of generating and transmitting either or both of distinct countermanding operational command(s) and a proper operational command, the processor **1550** may be caused to refer to indications of various protocol details stored in the corresponding process entry set **1534** of the database **1533** that corresponds to the particular process **3005**.

Identification and Simpler Responses to Update Download Process

[0298] FIGS. **12**A and **12**B, taken together, present various aspects of an example embodiment of detecting and responding to the commencement of a process of downloading a software update for a monitored device **2300** (e.g., an update download routine **2305***ud*) based on a pattern of transmissions **2922** occurring on the internal network **2999**. FIG. **12**A depicts aspects of the detection and an example initial response, and FIG. **12**B depicts aspects of an example response of preventing such a download from at least being completed.

[0299] Turning to FIG. **12**A, in executing a monitoring component **1545** of the control routine **1540**, processor(s) **1550** of a monitoring device **1500** may cooperate with an interchange device **2700** through a link **1992** to monitor activity occurring on the internal network **2999** for a pattern of transmissions **2922** that is able to be matched to one of the processes described within one of the process entries **1534** of the database **1533**. As depicted, a process for the downloading of an updated routine **9934** by a monitored device **2300** may be so identified as having begun with a particular first transmission **2922** that is associated with that process from a monitored device **2300** that is directed toward an external server **9900**.

[0300] It is envisioned as possible that the very first transmission **2922** associated with a particular process may have characteristics, such as a type of operational command, a parameter, a particular identifier of a destination device, etc. that may be sufficiently unique that just the very first transmission **2922** is needed to provide an indication that the particular process has commenced. Thus, where such a unique first transmission **2922** is observed by the monitoring device **1500**, it may not be necessary to wait to observe even a second transmission **2922** (which may be in reply to the first transmission) before the particular process is able to be identified, and a response to its commencement is able to be determined and implemented. Indeed, FIG. **12**A depicts the implementation of a particular response in which the relaying of the very first transmission **2922** is allowed to proceed through the interchange device **2700**, but a second transmission **2922** in response thereto is then blocked from being relayed through the interchange device **2700**.

[0301] However, it is also envisioned as possible that a relatively small quantity of multiple transmissions **2922** occurring in a particular sequence and/or with a particular combination of content may need to be observed to provide a sufficiently reliable indication that a particular process has commenced. Thus, in such a situation, there may be multiple transmissions **2922** that are allowed to proceed in both directions between the monitored device **2300** and the external server **9900** before the commencement of the particular process for downloading an updated routine **9934** for execution within the monitored device **2300** is recognized (e.g., an update download process **2305***ud*).

[0302] Regarding the possible responses to the commencement of the process of downloading the depicted updated routine **9934**, in some embodiments, the process may be allowed to proceed without interruption as long as comparisons of the transmissions **2922** of operational commands and/or operational information that occur conform to what has been learned to be expected by the monitoring device **1500**. More specifically, the transmissions **2922** identified as associated with this update download process may be compared to indications in the corresponding process entry **1534** of what transmissions **2922** are expected to occur, what the content of each transmission **2922** is expected to be, what the order of those transmissions **2922** is expected to be, and/or what the timing of each of those transmissions **2922** is expected to be. In such an embodiment, it may be that the detection of even a single transmission that does not exhibit expected characteristics may serve as a basis for blocking further transmissions **2922** that are identified as associated with the update download process. Such a blockage of such subsequent transmissions **2922** may be intended to force the process to be halted, and thereby force the process to be commenced again from the beginning if it is to be carried out successfully. Such a forced restart from the beginning may be deemed an effective way to deal with the possibility of corruption of the updated routine **9934** during the downloading of it.

[0303] As an alternative to simply blocking further transmissions **2922** in this update download process in response to such a non-conforming transmission **2922**, the processor (s) **1550** may, instead, be caused to use the information within the corresponding process entry **1534** for this process to correct the non-conforming transmission **2922**. More specifically, in a manner similar to what was earlier described in connection with FIGS. **10**A-D, a transmission **2922** that is non-conforming in its content may be blocked from being relayed through the interchange device **2700**, while a replacement transmission **2922** with conforming content may be generated within the monitoring device **1500**, and provided to the interchange device **2700** to relay onward in place of the non-conforming transmission. Alternatively or additionally, in a manner similar to what was earlier described in connection with FIGS. **9**A-B, a transmission **2922** that is expected to occur at a particular time, but which does not occur when expected, may be responded to by the monitoring device **1500** generating the missing transmission **2922**, itself, and providing it to the interchange device **2700** to be relayed to the destination device to which the missing transmission would have been directed. Such corrective efforts may be made to address particular circumstances in which the affected transmissions **2922** are determinative of whether a download involves a particular external server **9900** that is deemed to be trustworthy, rather than an external server **9900** that is unknown or that is known to be an untrustworthy (e.g., fake) external server. Such corrections may be performed to address situations in which an identifier of an external server **9900** may have been mistyped by an operator of the monitored device **2300**. Alternatively, the downloading of software updates may be deemed to be sufficiently sensitive that it is deemed to be undesirable to make corrections in an update download process, regardless of whether such a correction is to an innocent mistake, rather than to a deliberate effort to download a fake software update from a fake external server.

[0304] In other embodiments, the process of downloading the depicted updated routine **9934** may be blocked immediately upon being identified as having commenced. Thus, once enough transmissions **2922** have occurred to enable the commencement of the update download process to be identified, the process may be immediately blocked or otherwise negated to prevent its completion, and to require that the update download process be commenced again from the beginning if it is to be carried out successfully. In such embodiments, it may be that a notification is transmitted to particular personnel seeking a response as to whether the process should be allowed to take place. Thus, in such embodiments, it may be that this update download process continues to be blocked each time it is identified as having commenced until such approval is received from such personnel.

[0305] FIG. **12**B depicts such an ongoing complete blockage of transmissions in either direction that are associated with a download process that is being blocked.

Buffering and Mimicry Responses to Update Download Process

[0306] FIGS. **13**A and **13**B, taken together, present various aspects of another example embodiment of responding to the commencement of a process of downloading a software update for a monitored device **2300** based on a pattern of transmissions **2922** occurring on the internal network **2999**. Each of FIGS. **13**A and **13**B depicts aspects of an example subsequent response to the detection of the commencement of such a download.

[0307] Turning to FIG. **13**A, in executing a caching component **1546** of the control routine **1540**, processor(s) **1550** of the monitoring device **1500** may cooperate with the interchange device **2700** through the link **1992** to independently complete an earlier attempted software update that entails the downloading of an updated routine **9934** from an external server **9900**. As discussed above, it may be that one or more earlier attempts by the monitored device **2300** to download the updated routine **9934** were blocked while approval to proceed was awaited from personnel.

[0308] Following the receipt of such approval, and at a time when the monitored device **2300** may have ceased attempting to perform the update download process, processor(s) **1550** may be caused to use learned information concerning transmissions **2922** associated with such an update download process to engage in communications with that external server **9900** that mimic the communications that the monitored device **2300** would have engaged in as part of performing the download earlier, if that earlier download had been permitted to proceed. In so doing, the processor(s) **1550** may retrieve indications of learned characteristics of the transmissions **2922** associated with this update download process from its corresponding one of the process entries **1534** within the database **1533**. In this way, the monitoring device **1500** may store a copy of the updated routine **9934** in preparation for providing the updated routine **9934** to the monitored device **2300** at a later time.

[0309] It should be noted that such operations may be limited to being performed with software updates for which a license or other form of permission has been granted for such storage thereof. In some of such embodiments, the opportunity of the storage of the updated routine **9934** within a buffer **1566** of the monitoring device **1500** may be used to analyze the updated routine **9934** to determine its authenticity and/or to identify any malicious code therein.

[0310] Turning to FIG. **13**B, at a later time, when the monitored device **2300** may again attempt to download the updated routine **9934**, the processor(s) **1550** may cooperate with the interchange device **2700** to similarly mimic the learned behavior of the external server **9934** to cause the monitored device **2300** to download the updated routine **9934** from the monitoring device **1500**, and not the directly from the external server **9900**. Again, the processor(s) **1550** may retrieve indications of learned characteristics of the transmissions **2922** associated with this update download process from its corresponding one of the process entries **1534** within the database **1533**.

[0311] Among the uses that may be made of the set of emulations provided by the emulation device(s) **5500**, other uses are possible including but not limited to (a) Archive Storage: a facility's support team may wish to store its network traffic captures for historical and future comparison purposes; (b) Equipment Function Validation: checking on and/or verifying the functioning of Operational Technology (OT) network equipment, Industrial Control Systems (ICS) processes, and field equipment; (c) Diagnostic Debug: debugging the functioning of OT (operational technology) network equipment, ICS processes, and field equipment to find and solve otherwise difficult to detect and/or solve problems; (d) Datamining (AI or otherwise): uncovering minute as well as major patterns and trends in production equipment command-and-control and usage that would otherwise not be readily easy to uncover; (e) Cyber Security Malware & Hacker analysis: isolating and analyzing rogue, harmful or malicious OT network traffic sent by or otherwise altered by nefarious attackers and/or their hardware/software equipment; (f) Monitoring & Reporting: there are entire classes of important and sophisticated facility reports that virtually cannot be produced any other way; (g) Cost Performance Analysis & Improvement: conducting live, in-depth technical, production, and/or performance research on an entire ICS facility full of equipment produced by different vendor makes and models of all at the same time; (h) Product Development & Improvement: ICS vendors studying the function and performance of their respective ICS hardware and software products, each for their own respective product improvement purposes and new product development purposes.

[0312] There is thus disclosed a system of one or more devices that implements a method for enhancing computer network reliability by countering disruptions in network communications involved with the update of software. The features set forth below may be combined in any of a variety of ways to create any of a variety of embodiments of such a system and/or of a method of decision making augmentation that may incorporate such a system.

[0313] A monitoring system includes: a processor configured to perform operations including: place the monitoring system into a training mode to generate a model of a software update process; receive indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system, wherein at least one of the multiple monitored devices is configured to initiate the software update process; and from the received indications of observed transmissions of operational commands or operational information associated with the software update process, generate the model of the software update process, wherein the model includes indications of an expected order of transmissions of operational commands or operational information associated with the software update process.

[0314] The monitored system may include at least one interchange device to which each monitored device of the multiple monitored devices is separately coupled; the transmissions of operational commands or operational information among the multiple monitored devices may be conveyed through the one or more interchange devices; and the monitoring system may be coupled to the at least one interchange device to receive the indications of observed transmissions of operational commands or operational information among the multiple monitored devices.

[0315] The model may include a finite state model that includes indications of multiple states of the software update process and indications of valid transitions among the multiple states.

[0316] A monitoring system includes a processor configured to perform operations including: place the monitoring system into an operating mode to use a model of a software update process to analyze observed transmissions of operational commands or operational information associated with the software update process; receive, from one or more interchange devices, indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system, wherein at least one of the monitored devices is configured to provide an updated routine in the software update process; and compare the received indications of observed transmissions of operational commands or operational information associated with the software update process to indications in the model of expected transmissions of operational commands or operational information associated with the software update process to determine whether a particular transmitted operational command is a proper operational command associated with the software update process or to determine whether particular transmitted operational information is proper operational information associated with the software update process.

[0317] The monitored system may include at least one interchange device to which each monitored device of the multiple monitored device is separately coupled; the transmissions of operational commands or operational information among the multiple monitored devices may be conveyed through the one or more interchange devices; and the monitoring system may be coupled to the at least one interchange device to receive the indications of observed transmissions of operational commands or operational information among the multiple monitored devices.

[0318] The at least one interchange device may intercept and store the particular transmitted operational command or operational information; and the processor may be further configured to respond to a determination that the particular transmitted operational command is a proper operational command associated with the software update process, or to a determination that the particular transmitted operational information is proper operational information associated with the software update process, by instructing the at least one interchange device to release the particular transmitted operational command or operational information from storage and allow the particular transmitted operational command or operational information to be relayed to its intended destination.

[0319] The at least one interchange device may intercept and store the particular transmitted operational command, and the processor may be further configured to respond to a determination that the particular transmitted operational command is an improper operational command associated with the software update process by performing operations including: instruct the at least one interchange device to refrain from relaying the particular transmitted operational command to its intended destination; generate a substitute proper operational command including a type of operational command based on an expected order of operational commands specified in the model, and including an expected parameter value based on the model; and transmit the substitute proper operational command to the at least one interchange device to be relayed to an expected destination specified in the model.

[0320] The processor may be further configured to respond to a determination that the particular transmitted operational command is an improper operational command associated with the software update process, or to a determination that the particular transmitted operational information is improper operational information associated with the software update process, by logging or providing an alert to particular personnel, or attempting one or more corrective actions.

[0321] The model may include a finite state model that includes indications of multiple states of the software update process and indications of valid transitions among the multiple states.

[0322] A method of generating a model of a software update process includes: receiving, by a processor of a monitoring system, and from an interchange device of a monitored system, indications of observed transmissions, through the interchange device, of operational commands or operational information among multiple monitored devices of the monitored system; and from the received indications of observed transmissions of operational commands or operational information associated with the software update process, generating, by the processor, the model of the software update process, wherein the model includes indications of an expected order of transmissions of operational commands or operational information associated with the software update process.

[0323] The model may include a finite state model that includes indications of multiple states of the software update process and indications of valid transitions among the multiple states.

[0324] A method of using a model of a software update process to analyze observed transmissions of operational commands or operational information associated with the software update process includes: receiving, by a processor of a monitoring system, and from at least one interchange device of a monitored system, indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system; and comparing, by the processor, the received indications of observed transmissions of operational commands or operational information associated with the software update process to indications in the model of expected transmissions of operational commands or operational information associated with the software update process to determine whether a particular transmitted operational command is a proper and expected operational command associated with the software update process, or to determine whether the particular transmitted operational information is proper and expected operational information associated with the software update process.

[0325] The at least one interchange device may intercept and store the particular transmitted operational command or operational information; and the method may further include responding, by the processor, to a determination that the particular transmitted operational command is a proper and expected operational command associated with the software update process by instructing the at least one interchange device to allow the particular transmitted operational command to be relayed to its intended destination, or responding, by the processor, to a determination that the particular transmitted operational information is proper and expected operational information associated with the software update process by instructing the at least one interchange device to allow the particular transmitted operational information to be relayed to its intended destination.

[0326] The at least one interchange device may intercept and store the particular transmitted operational command, and the method may further include responding, by the processor, to a determination that the particular transmitted operational command is an improper operational command associated with the software update process by performing operations including: instructing the at least one interchange device to refrain from relaying the particular transmitted operational command to its intended destination; generating, by the processor, a substitute proper operational command including a type of operational command based on an expected order of operational commands specified in the model, and including an expected parameter value based on the model; and transmitting the substitute proper operational command to the at least one interchange device to be relayed to an expected destination specified in the model.

[0327] The method may further include responding to a determination that the particular transmitted operational command is an improper operational command associated with the software update process, or to a determination that the particular transmitted operational information is improper operational information associated with the software update process, by logging or providing an alert to particular personnel, or attempting one or more corrective actions.

[0328] The model may include a finite state model that includes indications of multiple states of the software update process and indications of valid transitions among the multiple states.

[0329] Various other components may be included and called upon for providing for aspects of the teachings herein. For example, additional materials, combinations of materials, and/or omission of materials may be used to provide for added embodiments that are within the scope of the teachings herein.

[0330] Standards for performance, selection of materials, functionality, and other discretionary aspects are to be determined by a user, designer, manufacturer, or other similarly interested party. Any standards expressed herein are merely illustrative and are not limiting of the teachings herein.

[0331] When introducing elements of the present disclosure or the embodiment(s) thereof, the articles "a," "an," and "the" are intended to mean that there are one or more of the elements. Similarly, the adjective "another," when used to introduce an element, is intended to mean one or more elements. The terms "including" and "having" are intended to be inclusive such that there may be additional elements other than the listed elements.

[0332] While the disclosure has been described with reference to illustrative embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications will be appreciated by those skilled in the art to adapt a particular instrument, situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the claimed invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims.

We claim:

1. A monitoring system comprising a processor configured to perform operations comprising:

place the monitoring system into a training mode to generate a model of a software update process;

receive indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system, wherein at least one of the multiple monitored devices is configured to initiate the software update process; and

from the received indications of observed transmissions of operational commands or operational information associated with the software update process, generate the model of the software update process, wherein the model comprises indications of an expected order of transmissions of operational commands or operational information associated with the software update process.

2. The monitoring system of claim **1**, wherein:

the monitored system comprises at least one interchange device to which each monitored device of the multiple monitored devices is separately coupled;

the transmissions of operational commands or operational information among the multiple monitored devices are conveyed through the one or more interchange devices; and

the monitoring system is coupled to the at least one interchange device to receive the indications of observed transmissions of operational commands or operational information among the multiple monitored devices.

3. The monitoring system of claim **1**, wherein the model comprises a finite state model that comprises indications of multiple states of the software update process and indications of valid transitions among the multiple states.

4. A monitoring system comprising a processor configured to perform operations comprising:

place the monitoring system into an operating mode to use a model of a software update process to analyze observed transmissions of operational commands or operational information associated with the software update process;

receive, from one or more interchange devices, indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system, wherein at least one of the monitored devices is configured to provide an updated routine in the software update process; and

compare the received indications of observed transmissions of operational commands or operational information associated with the software update process to indications in the model of expected transmissions of operational commands or operational information associated with the software update process to determine whether a particular transmitted operational command is a proper operational command associated with the software update process or to determine whether particular transmitted operational information is proper operational information associated with the software update process.

5. The monitoring system of claim **4**, wherein:

the monitored system comprises at least one interchange device to which each monitored device of the multiple monitored device is separately coupled;

the transmissions of operational commands or operational information among the multiple monitored devices are conveyed through the one or more interchange devices; and

the monitoring system is coupled to the at least one interchange device to receive the indications of observed transmissions of operational commands or operational information among the multiple monitored devices.

6. The monitoring system of claim **4**, wherein:

the at least one interchange device intercepts and stores the particular transmitted operational command or operational information; and

the processor is further configured to respond to a determination that the particular transmitted operational command is a proper operational command associated with the software update process, or to a determination that the particular transmitted operational information is proper operational information associated with the software update process, by instructing the at least one interchange device to release the particular transmitted operational command or operational information from storage and allow the particular transmitted operational command or operational information to be relayed to its intended destination.

7. The monitoring system of claim **4**, wherein:

the at least one interchange device intercepts and stores the particular transmitted operational command; and

the processor is further configured to respond to a determination that the particular transmitted operational command is an improper operational command associated with the software update process by performing operations comprising:

instruct the at least one interchange device to refrain from relaying the particular transmitted operational command to its intended destination or to otherwise block, stop of countermand the transmitted operational command;

optionally generate a substitute proper operational command comprising a type of operational command based on an expected order of operational commands specified in the model, and comprising an expected parameter value based on the model; and

transmit the substitute proper operational command to the at least one interchange device to be relayed to an expected destination specified in the model.

8. The monitoring system of claim **4**, wherein the processor is further configured to respond to a determination that the particular transmitted operational command is an improper operational command associated with the software update process, or to a determination that the particular transmitted operational information is improper operational information associated with the software update process, by logging or providing an alert to particular personnel, or attempting one or more corrective and/or preventative actions.

9. The monitoring system of claim **4**, wherein the model comprises a finite state model that comprises indications of multiple states of the software update process and indications of valid transitions among the multiple states.

10. A method of generating a model of a software update process comprising:

receiving, by a processor of a monitoring system, and from an interchange device of a monitored system, indications of observed transmissions, through the interchange device, of operational commands or operational information among multiple monitored devices of the monitored system; and

from the received indications of observed transmissions of operational commands or operational information associated with the software update process, generating, by the processor, the model of the software update process, wherein the model comprises indications of an expected order of transmissions of operational commands or operational information associated with the software update process.

11. The method of claim **10**, wherein the model comprises a finite state model that comprises indications of multiple states of the software update process and indications of valid transitions among the multiple states.

12. A method of using a model of a software update process to analyze observed transmissions of operational commands or operational information associated with the software update process, the method comprising:

receiving, by a processor of a monitoring system, and from at least one interchange device of a monitored system, indications of observed transmissions of operational commands or operational information among multiple monitored devices of a monitored system; and

comparing, by the processor, the received indications of observed transmissions of operational commands or operational information associated with the software update process to indications in the model of expected transmissions of operational commands or operational information associated with the software update process to determine whether a particular transmitted operational command is a proper and expected operational command associated with the software update process, or to determine whether the particular transmitted operational information is proper and expected operational information associated with the software update process.

13. The method of claim **12**, wherein:

the at least one interchange device intercepts and stores the particular transmitted operational command or operational information; and

the method further comprises responding, by the processor, to a determination that the particular transmitted operational command is a proper and expected operational command associated with the software update process by instructing the at least one interchange device to allow the particular transmitted operational command to be relayed to its intended destination, or responding, by the processor, to a determination that the particular transmitted operational information is proper and expected operational information associated with the software update process by instructing the at least one interchange device to allow the particular transmitted operational information to be relayed to its intended destination.

14. The method of claim **12**, wherein:

the at least one interchange device intercepts and stores the particular transmitted operational command; and

the method further comprises responding, by the processor, to a determination that the particular transmitted operational command is an improper operational command associated with the software update process by performing operations comprising:

instructing the at least one interchange device to refrain from relaying the particular transmitted operational command to its intended destination;

optionally generating, by the processor, a substitute proper operational command comprising a type of operational command based on an expected order of operational commands specified in the model, and comprising an expected parameter value based on the model; and

transmitting the substitute proper operational command to the at least one interchange device to be relayed to an expected destination specified in the model.

15. The method of claim **12**, further comprising responding to a determination that the particular transmitted operational command is an improper operational command associated with the software update process, or to a determination that the particular transmitted operational information is improper operational information associated with the software update process, by logging or providing an alert to particular personnel, or attempting one or more corrective or preventative actions.

16. The method of claim **12**, wherein the model comprises a finite state model that comprises indications of multiple states of the software update process and indications of valid transitions among the multiple states.

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